

PARAMETRIC OPTIMIZATION OF HEAT SINKS USING CFD AND OPTIMIZATION TECHNIQUES

Final Year Project Report

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the requirements for the
Degree of Bachelor of Technology

In

Mechanical Engineering

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CERTIFICATE

This is to certify that **Mr. Ashutosh Mohapatra, Mr. Taran Poojari, and Mr. Mohammed Wafeeq Mohammad Salim Kazi**, the students of **Bachelor of Technology in Mechanical Engineering**, have completed the Final Year Project Report entitled, “**Parametric Optimization of Heat Sinks using CFD and Optimization Techniques**” to our satisfaction.

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ABSTRACT

In modern electronic and industrial systems, the increasing power density of devices demands highly efficient thermal management solutions to maintain operational stability and prevent performance degradation. Among the available techniques, air-cooled heat sinks are widely employed due to their simplicity, low cost, and ease of integration. However, conventional designs often fall short under high heat flux conditions, necessitating advanced geometrical and parametric optimization for enhanced thermal performance.

This study presents a comprehensive investigation into the thermal and hydraulic performance of heat sinks under forced convection. The work begins with the numerical validation of an existing research study on heat sinks using Computational Fluid Dynamics (CFD). Key performance metrics, including heat transfer coefficient, Nusselt number, and pressure drop, were validated against published data to establish the reliability of the simulation framework.

Following successful validation, a set of simulations was conducted to observe how the change in fin shape, along with variations in parameters such as height, diameter, and inlet velocity, affected the thermal and flow characteristics. These extended cases helped explore the influence of fin geometry on heat transfer behavior, and the corresponding results have been analyzed and discussed in subsequent sections.

The outcomes of this work provide valuable design insights for optimizing heat sink geometries in active air-cooling systems, especially for compact high-power electronic devices. The validated CFD model and extended analysis serve as a foundation for future studies involving advanced geometries and performance trade-offs in thermal management applications.

Artificial Neural Networks (ANN) were employed to predict the thermal and hydraulic performance of heat sinks based on geometric and flow parameters. Genetic Algorithms (GA) were used for optimization purposes.

[Keywords: Heat sink, Pin-fin geometry, CFD simulation, Forced convection, Genetic Algorithm, Parametric variation, ANN, Electronic cooling systems]

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CHAPTER 1: INTRODUCTION

INTRODUCTION

Heat sinks are essential thermal management components designed to dissipate unwanted heat from high-power electronic or mechanical systems. Acting as passive heat exchangers, they transfer thermal energy away from heat-generating components to a surrounding fluid medium—typically air or liquid—to maintain safe operating temperatures. Efficient heat dissipation is critical to ensure system reliability, prevent thermal runaway, and extend the operational lifespan of components.

The performance of a heat sink relies on three primary modes of heat transfer: conduction, convection, and radiation. Initially, heat is conducted from the hot component into the heat sink material—usually a thermally conductive metal such as aluminum or copper. From there, it is convected into the surrounding fluid, while a smaller portion may also be radiated away. In most applications, conduction and convection dominate.

In thermodynamic terms, a heat sink serves as a thermal reservoir, maintaining a relatively constant temperature while absorbing excess heat from the system. Devices such as CPUs, power transistors, LED modules, and battery packs depend on heat sinks to prevent thermal buildup that could otherwise impair performance or cause irreversible damage.^[1]

Given the growing demand for compact, efficient thermal solutions in high-density electronics, the parametric optimization of heat sinks has become increasingly important. This involves the systematic analysis and tuning of key geometrical and operational parameters—such as fin height, spacing, base thickness, material conductivity, and flow conditions—to improve thermal performance while minimizing weight, pressure drop, and footprint. Optimization strategies include computational fluid dynamics (CFD), artificial intelligence, and evolutionary algorithms such as genetic optimization.

This project investigates the thermal behavior of pin-fin and rectangular heat sinks under forced convection, with the goal of identifying geometries that enhance performance. The work begins with an overview of heat sink design challenges and objectives, followed by an extensive literature review that explores the performance characteristics of various geometries and the role of CFD-based optimization in thermal management. The design and simulation methodologies are then detailed, including CAD modeling, meshing strategies, and boundary

condition setups, carried out using ANSYS Fluent for both pin-fin and rectangular fin configurations.

To extend the analysis beyond conventional methods, predictive modeling techniques such as Artificial Neural Networks (ANN) and Extreme Learning Machines (ELM) are introduced to estimate thermal performance parameters. These are followed by optimization using genetic algorithms to identify the most efficient design configurations. The results of these simulations and predictive models are discussed in depth, offering insight into trade-offs between heat transfer efficiency and fluid dynamic losses. Finally, the study concludes with key findings and outlines future work, suggesting possible directions for refining the models and enhancing heat sink design strategies.

TYPES OF HEAT SINKS BASED ON FLUID FLOW

1. PASSIVE HEAT SINKS

Passive heat sinks operate through natural convection, where the movement of air is driven solely by buoyancy forces arising from temperature differences. As the surface of the heat sink heats up, the surrounding air becomes less dense and rises, while cooler air flows in to replace it. This circulation transfers heat from the sink to the ambient environment without requiring any mechanical assistance. The key advantage of passive systems lies in their simplicity and reliability—they have no moving parts, require no external power supply, and generate no noise. However, their thermal performance is limited, making them suitable primarily for low-power or thermally tolerant applications where heat dissipation demands are relatively low. ^[1]

2. ACTIVE HEAT SINKS

Active heat sinks utilize forced convection to improve thermal dissipation. In these systems, external mechanisms such as fans, blowers, or pumps are employed to increase the flow of air or fluid over the surface of the heat sink. This enhances the heat transfer rate by maintaining a higher temperature gradient between the heat sink and the cooling medium. A typical example is the cooling fan in a personal computer, which activates to cool the processor when its temperature rises. Active cooling is essential in high-power or high-density systems, such as CPUs, power amplifiers, or industrial drives, where passive methods are insufficient. While active heat sinks offer superior cooling performance, they introduce complexity, noise, power consumption, and potential maintenance issues due to the presence of moving components. ^[1]

3. HYBRID HEAT SINKS

Hybrid heat sinks combine both passive and active cooling strategies in a single system. These configurations typically operate in passive mode under low thermal loads and switch to active cooling when higher temperatures are detected. This dynamic response is usually managed by integrated temperature sensors and control circuits. Hybrid systems provide a balanced solution—offering quiet and energy-efficient operation under normal conditions while delivering enhanced cooling when needed. Such systems are well-suited for devices with variable heat generation, such as smart electronics, battery systems, or thermally adaptive embedded platforms. Although less common than purely passive or active types, hybrid heat sinks are gaining traction in advanced thermal management designs due to their efficiency, adaptability, and extended component lifespan.

TYPES OF HEAT SINKS BASED ON STRUCTURE.

1. PIN FIN HEAT SINKS

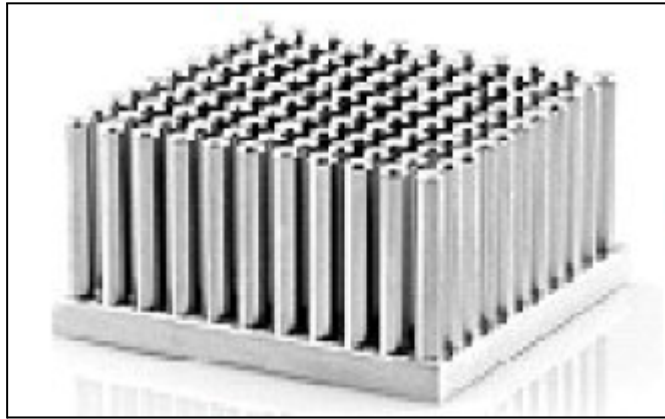


Figure 1.1: Pin fin Heat sink.^[3]

Pin fin heat sinks consist of an array of cylindrical, square, or elliptical pins that extend vertically from the base. They are particularly effective in multi-directional airflow environments and are widely used in forced convection scenarios due to their enhanced surface area and turbulence generation. It provides high heat transfer due to increased surface area and turbulence. But causes greater pressure drop compared to plate fin types.^[2]

2. RECTANGULAR PLATE FIN HEAT SINKS

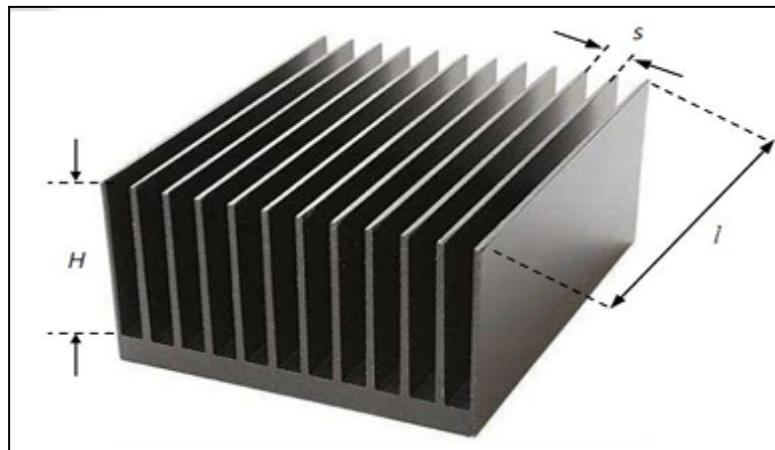


Figure 1.2: Rectangular Heat sink.^[4]

These are the most commonly used heat sinks in industry. Rectangular fins arranged in parallel maximize directional airflow efficiency, especially in forced convection setups. They're easy to manufacture and offer a good trade-off between performance and simplicity. It has simple geometry, easy to fabricate, predictable performance. But is less effective in natural convection or multidirectional flow.^[4]

3. TAPERED HEAT SINKS

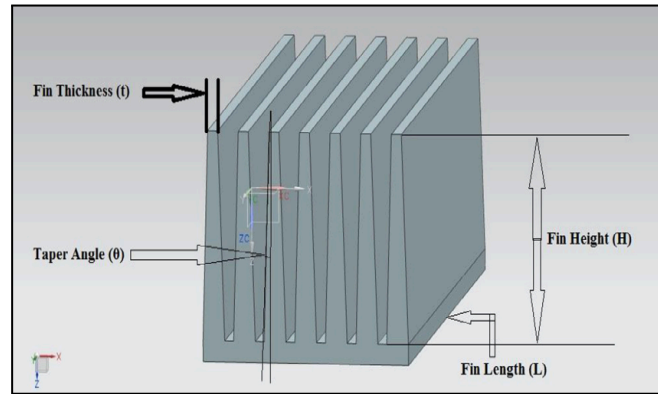


Figure 1.3: Tapered Heat sink.^[5]

Tapered heat sinks feature fins that gradually decrease in cross-sectional area from base to tip. This design enhances the local heat transfer coefficient near the fin tip, where airflow velocity is higher, and also contributes to weight reduction. Tapered fins are especially useful in lightweight applications such as aerospace or portable electronics.^[5]

4. PERFORATED HEAT SINKS

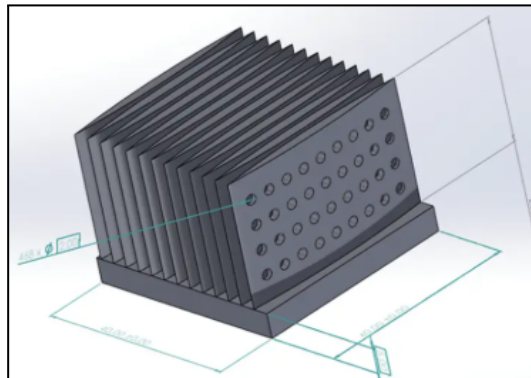


Figure 1.4: Perforated Heat sink.^[6]

Porous heat sinks, such as those made from metal foams or sintered materials, offer an extremely high surface area-to-volume ratio. These structures allow coolant (air or liquid) to permeate through the matrix, promoting enhanced convective and conductive heat transfer. Despite their superior thermal efficiency, they suffer from high pressure drop and are more challenging to clean or maintain.^[6]

5. POROUS HEAT SINKS

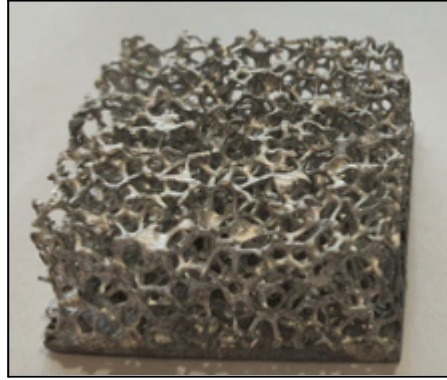


Figure 1.5: Porous Heat sink.^[7]

Porous heat sinks, such as those made from metal foams or sintered materials, offer an extremely high surface area-to-volume ratio. These structures allow coolant (air or liquid) to permeate through the matrix, promoting enhanced convective and conductive heat transfer. Despite their superior thermal efficiency, they suffer from high pressure drop and are more challenging to clean or maintain. ^[7]

6. FLARED HEAT SINKS

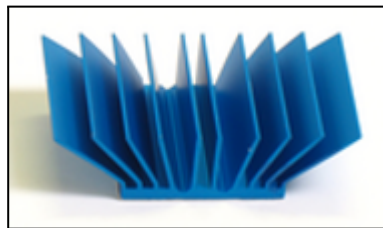


Figure 1.6: Flared Heat sink.^[8]

Flared fin heat sinks employ fins that are angled outward from the base, rather than being perpendicular. This flaring geometry improves the entry and exit of airflow, especially under natural convection, by reducing flow resistance and increasing exposure of the fin base to ambient air. ^[8]

CHAPTER 2: LITERATURE REVIEW

2.1: FUNDAMENTALS OF HEAT TRANSFER

The performance of a heat sink relies on its ability to remove unwanted heat generated by an electronic or mechanical device. This heat is transferred through various modes — primarily **conduction** and **convection**, and to a lesser degree **radiation**, depending on the operating conditions.

The efficiency and design of a heat sink are intrinsically tied to these mechanisms, which must be thoroughly understood to optimize its thermal resistance and ensure effective cooling of electronic components.

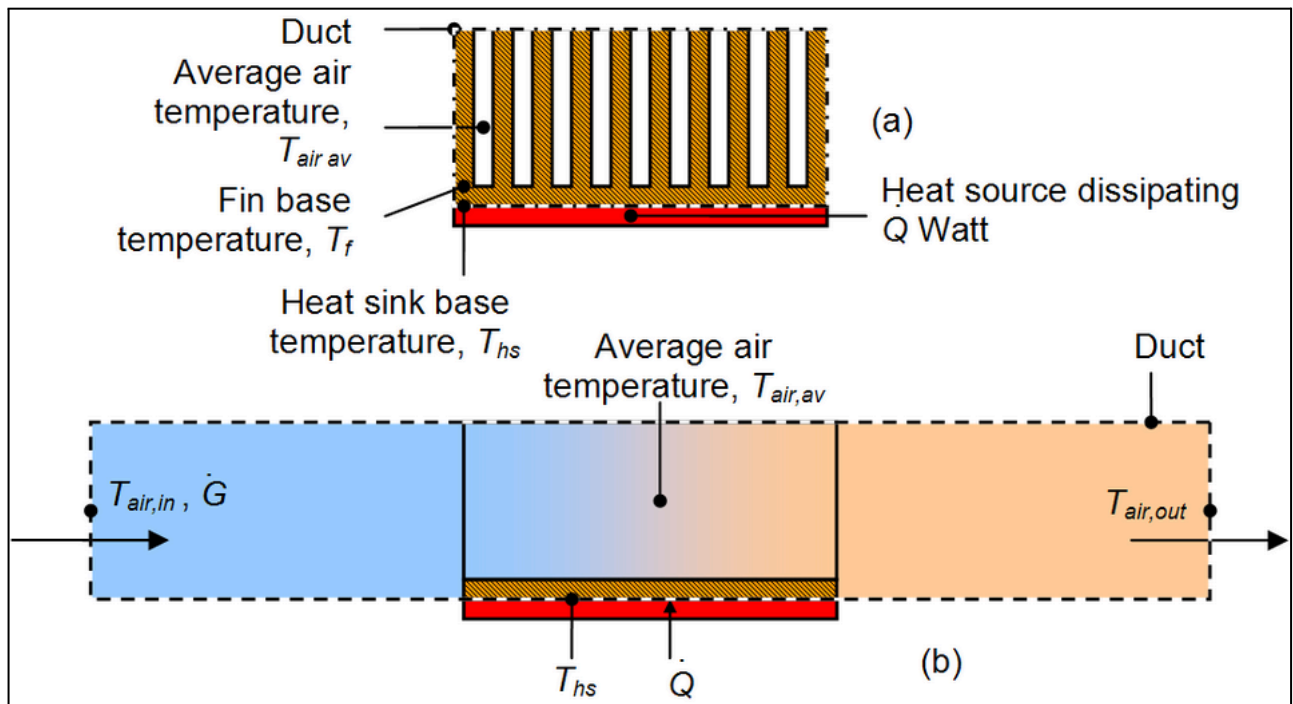


Figure 2.1: Basic heat transfer in heat sink^[9]

2.1.1 HEAT CONDUCTION

Heat conduction is the mode of energy transfer that occurs within the solid structure of the heat sink, i.e., from the contact surface (usually the device or chip) through the base and into the fins. It is mathematically governed by **Fourier's Law of Heat Conduction**:

$$q = -k.A.\frac{dT}{dx}$$

Where,

- q = Heat transfer Rate (W)
- k = Thermal conductivity of the material (W/m·K)

- A: cross-sectional area perpendicular to heat flow (m²)
- $\frac{dT}{dx}$: temperature gradient across the length (K/m)

The negative sign indicates heat flows from high to low temperature.

Efficient conduction is critical in the base region of the heat sink, as any resistance here will limit heat from reaching the extended surfaces (fins), thereby creating localized hot spots. This highlights the importance of selecting materials with high thermal conductivity.

Copper has a thermal conductivity of approximately 400 W/m·K, while aluminum (specifically 6061 or 6063 alloys) ranges from 150–200 W/m·K. Despite copper's superior performance, aluminum is preferred in many applications due to its lower cost, weight, and manufacturability.

Poor conduction within the base or fins leads to thermal resistance, which reduces the effectiveness of the overall system. In compact heat sinks, minimizing internal conduction paths is critical to avoiding hotspots and uneven heat distribution across fins.^[10]

2.1.2 CONVECTION HEAT TRANSFER

Convection is the primary mode of heat dissipation from the heat sink to the surrounding environment. The forced convection process is generally more effective than natural convection and is often induced using fans or airflow ducts in high-performance applications.

Newton's Law of Cooling is used to describe convective heat transfer:

$$q = h.A.(T_s - T_\infty)$$

Where,

- h: convective heat transfer coefficient (W/m²·K)
- A: exposed surface area (m²)
- T_s: surface temperature of the heat sink (K)
- T_∞: bulk fluid temperature (ambient air) (K)

The value of h is affected by:

- Flow velocity and turbulence intensity
- Fluid properties (viscosity, thermal conductivity)
- Fin surface geometry, roughness, and orientation
- Reynolds number and Prandtl number of the flow

For heat sink fins, the external surface area plays a critical role in determining convective efficiency. Fins increase this surface area significantly, thereby enhancing the overall heat rejection rate. However, the actual effectiveness is influenced by how efficiently air can flow between and around the fins.^[11]

The Nusselt number (Nu) is a dimensionless quantity that helps evaluate convective performance:

$$\text{Nu} = \frac{h.L}{k}$$

Where,

- L is a characteristic length (e.g., fin height),
- k is the thermal conductivity of the cooling fluid.

Empirical correlations exist to relate the Nusselt number to the Reynolds number (Re) and Prandtl number (Pr) depending on the geometry and boundary conditions of the flow.^[12]

2.1.3 RADIATION

In most forced air-cooled heat sinks operating at moderate temperatures, radiation plays a minor role compared to conduction and convection. However, at elevated temperatures, or in low-convection environments, radiation can contribute significantly to total heat dissipation.^[13]

The radiative heat transfer is governed by **Stefan–Boltzmann’s Law**:

$$q = \varepsilon.\sigma.A.(T_{\text{surface}}^4 - T_{\text{surroundings}}^4)$$

Where,

- ε : emissivity of the surface
- σ : Stefan–Boltzmann constant ($5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$)
- $T_{\text{surface}}, T_{\text{surrounding}}$: surface and surroundings temperature (K)

2.2: ENERGY BALANCE AND THERMAL RESISTANCE

2.2.1 HEAT EXTRACTED BY FLOWING AIR

When a fluid like air flows over the surface of a heat sink, it removes thermal energy through convection. The total energy removed by the air can be calculated using an energy balance equation:^[13]

$$q = m.C_p.(T_{\text{out}} - T_{\text{in}})$$

Where,

- Q : heat transfer rate (W),
- m : mass flow rate of air (kg/s),
- C_p : specific heat of air (J/kg·K)
- T_{in} , T_{out} : inlet and outlet temperatures of the fluid (K).

2.2.2 LUMPED THERMAL RESISTANCE

The concept of **thermal resistance** is frequently used in heat sink design to represent the resistance to heat flow from a device junction to the ambient environment. It is defined as:

$$R_{Th} = \frac{\Delta T}{Q} = \frac{T_{source} - T_{ambient}}{Q}$$

Where,

- R_{th} : thermal resistance (°C/W)
- ΔT : Temperature difference between the heat source and surroundings (°C)
- Q : heat flow rate (W).

The objective in thermal management is to minimize R_{Th} , which is largely affected by fin geometry, airflow, and material conductivity. Interface materials like thermal paste or phase-change materials help reduce R_{Th} .^[11]

2.3: FIN EFFICIENCY

The primary function of fins is to increase the surface area available for heat dissipation. However, not all heat supplied to the fin base is transferred to the surrounding air — some is lost due to the thermal gradient along the fin's length. This leads to the concept of fin efficiency, which quantifies the fin's ability to transfer heat compared to an ideal fin with uniform temperature.

2.3.1 FIN EFFICIENCY

It is defined as the ratio of the actual heat transferred by a fin to the heat that would be transferred if the entire fin were at the same temperature as the base.

Fin efficiency (η) is given by:

$$\eta = \frac{q(actual)}{q(ideal)} = \frac{\tanh(mL)}{mL}$$

Where,

- $m = \frac{\sqrt{2h}}{\sqrt{k.t}}$
- L: fin length,
- t : fin thickness,
- k : thermal conductivity of fin material,
- h : convective heat transfer coefficient.

The ideal condition assumes that the entire fin is isothermal — i.e., there is no temperature drop from the base to the tip. This represents an infinitely long or infinitely conductive fin, which acts as a theoretical upper limit for heat transfer capability. In practice, real fins experience temperature gradients, especially if the fin is long or the material has moderate conductivity.^[13]

2.4: BYPASS AND SHROUDING

2.4.1 BYPASS

Bypass flow is a common phenomenon and occurs when a portion of the airflow diverts around the heat sink instead of passing through the fin array — particularly when the pressure resistance through the fins is high.^[14]

Consequences of bypass include:

- Reduced effective airflow over fin surfaces.
- Non-uniform temperature distribution across the heat sink.

2.4.1 SHROUDING

A shroud is a mechanical cover or duct designed to channel air directly through the heat sink fins, minimizing bypass.^[14]

Advantages of shrouding:

- Increased utilization of available fin surface area.
- Improved temperature uniformity across the heat sink.

Shrouds are especially effective when used with staggered fin arrays, as they ensure the turbulent, redirected flow fully engages with all surfaces.

2.5: MATERIAL SELECTION

The choice of material directly influences the thermal performance, manufacturability, and cost of heat sinks. The key properties required are:^[10]

- High thermal conductivity for efficient heat conduction
- Low density for weight-sensitive applications (e.g., aerospace)
- Corrosion resistance and mechanical strength.

Commonly used materials:

Material	Thermal Conductivity (W/m·K)	Density (kg/m ³)	Comments
1. copper	~400	~8900	High conductivity, heavy, costly
2. Aluminium	150–200	~2700	Lightweight, affordable
3. Graphite	150–500 (anisotropic)	~1600–2000	Advanced material, expensive

Table 2.1: Material Properties

2.6 LEARNINGS ON PIN FINS

2.6.1: FIN ARRAY CONFIGURATIONS

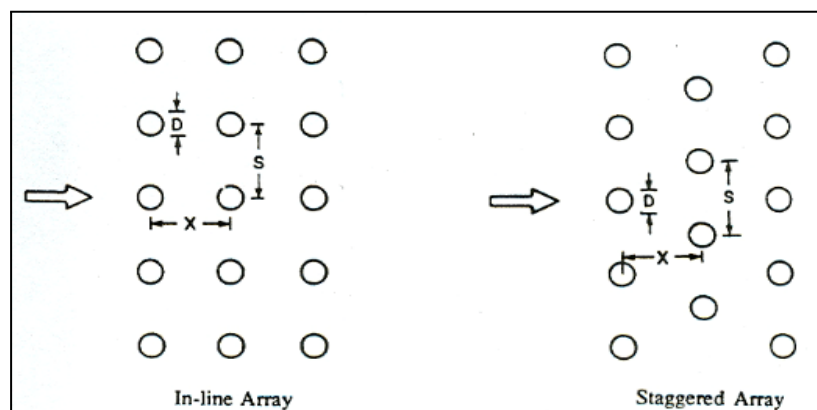


Figure 2.2: Fin array configurations

The configuration of fins in a heat sink directly affects the flow characteristics and consequently the heat transfer rate. Two common arrangements used in pin-fin or plate-fin heat sinks are the inline and staggered configurations.^[15]

2.6.1.1 INLINE FIN ARRAY

In an inline array (As shown in Figure 2.2), fins are arranged in aligned rows and columns. Air flows through uniform channels with minimal obstruction, resulting in lower flow resistance and a relatively uniform pressure distribution. However, this configuration often results in laminar flow development and boundary layer growth along the fins, which can reduce the local convective heat transfer coefficient further downstream.^[15]

2.6.1.2 STAGGERED FIN ARRAY

In a staggered array (As shown in Figure 2.3), fins are offset such that each downstream fin is placed between the gaps of the upstream row. This arrangement causes flow disruption, enhances turbulence, and breaks thermal boundary layers, resulting in increased heat transfer coefficients.^[15]

2.6.2: EFFECT OF CHANGE IN VELOCITY ON PERFORMANCE METRICS

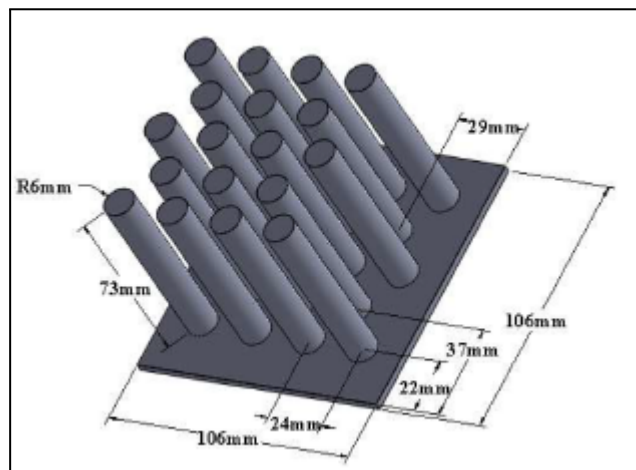


Figure 2.3: Geometry of the Staggered Pin Fin Heatsink used by the Author.^[16]

The study by *Ismail et al. (2024)* provides comprehensive insights into the thermal and hydraulic behavior of pin-fin (Figure 2.3) heat sinks under varying operating conditions. The research employs both numerical modeling using ANSYS Fluent and experimental validation to evaluate the influence of geometry and flow characteristics on the performance of pin-fin configurations.

Key findings indicate that the heat transfer performance of pin-fin heat sinks improves with increasing Reynolds number. As the airflow becomes more turbulent, the Nusselt number rises, leading to enhanced convective heat transfer. However, this benefit is accompanied by a corresponding increase in the pressure difference, resulting from elevated flow resistance through the densely packed pin arrays.

The study demonstrates that while conventional pin-fin designs provide a balance between manufacturability and thermal performance, their performance is limited by flow bypass and wake formation, which reduce local heat transfer in certain zones.

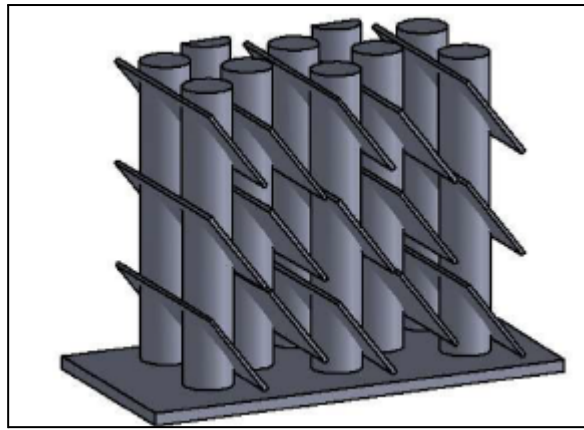


Figure 2.4: Geometry of the Staggered plate Fin Heat Sink used by the Author.^[16]

Additionally, pin-fins were used as a reference geometry for comparison against plate-fin configurations (As shown in Figure 2.4). It was observed that plate-fins inclined at optimized angles could outperform traditional pin-fins in terms of hydrothermal efficiency. However, pin-fins still offer advantages such as structural simplicity and adaptability for compact heat exchanger designs and are widely used. ^[16]

2.6.3 MESHING INFLATION LAYERS ON HEAT TRANSFER

The paper by *Ismail et al. (2024)* also highlights the importance of inflation layers in mesh generation for accurately capturing near-wall flow behavior in heat sink simulations. In the numerical analysis, inflation layers (As shown in Figure 2.5) were introduced near the surfaces of pin-fins and plate-fins to resolve the thermal and hydraulic boundary layers with greater precision. These layers are essential for computing critical gradients in temperature and velocity close to the solid walls, where most of the convective heat transfer occurs. ^[16]

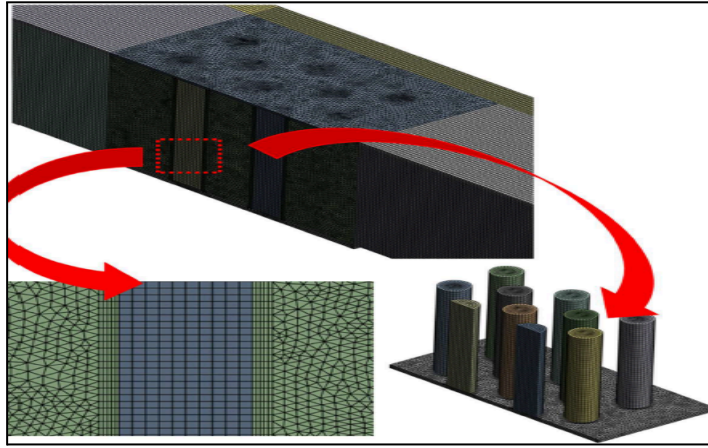


Figure 2.5: Meshing Inflation done by the Author.^[16]

CHAPTER 3: METHODOLOGY ON PIN FIN HEAT SINK

To ensure the accuracy and reliability of the CFD setup used in this project, an initial validation exercise was conducted using the published results^[17], which investigated the Designing and optimizing a novel heatsink for the enhancement of hydrothermal performances modelling and analysis using artificial neural networks. The study provided comprehensive numerical data—including temperature distribution, Nusselt number, and pressure drop—under controlled conditions using ANSYS Fluent. By replicating the same model geometry and applying the identical boundary conditions, the numerical results were successfully validated, establishing confidence in the CFD framework for further analysis.

Following the validation, the project proceeds to utilize the same baseline model as a reference configuration to carry out a detailed parametric optimization study. The aim was to evaluate the impact of key geometric and operational parameters on the performance of heat sinks. The parameters varied included fin diameter, fin spacing, fin height, and inlet velocity, focusing on two types of fin geometries: square pin-fins and cylindrical pin-fins, each analyzed under staggered and inline arrangements.

By systematically modifying these parameters, this project sought to identify optimized heat sink designs that offer maximum thermal efficiency (quantified by Nusselt number) with minimal pressure drop, under forced convection conditions. The outcome of this investigation is expected to provide valuable insights into the performance behavior of advanced fin structures and contribute to the development of more efficient thermal management solutions.^[17]

3.1: PROBLEM DESCRIPTION OF THE SELECTED PAPER.

The study aims to enhance heat transfer in heat sinks while keeping pressure drop minimal. A base plate of 100 mm × 100 mm × 3 mm was used, on which staggered pin fin arrays were mounted. The fins are 8 mm in diameter and 50 mm in height, with 14 pins placed in a staggered arrangement due to its known superior heat transfer performance over inline configurations.

The simulation domain includes an inlet and outlet zone (745 mm upstream, 195 mm downstream) to avoid reverse flow, ensuring accurate results. The study investigates four types of staggered cylindrical pin fin arrays:

1. Solid fins
2. Perforated fins (5 holes of 3 mm per fin)
3. Fins with hemispherical caps (8 per fin, 2.5 mm radius)
4. Hybrid fins (both perforations and hemispheres)

The flow is assumed to be 3D, incompressible, steady-state, and turbulent, with an inlet velocity range of 1.9–5.81 m/s. Fluid and solid properties are considered constant, and radiation is neglected.

It is important to note that since our project is based on widely used heatsinks we have only validated **pin fin heat sinks** from this research paper.^[17]

3.2: GEOMETRY AND CONFIGURATION OF THE HEAT SINK AND THE DOMAIN.

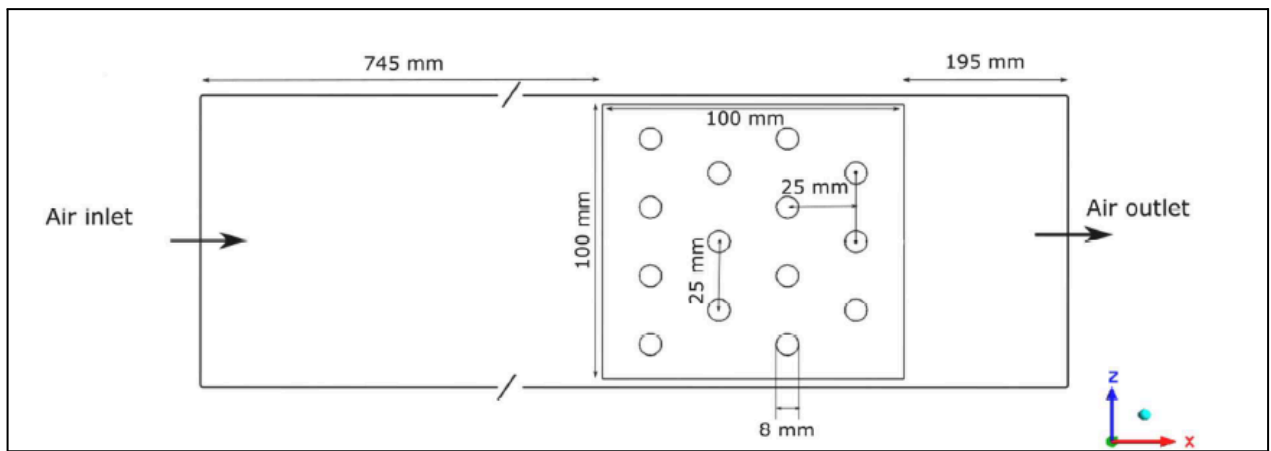


Figure 3.1: Dimensions of the Staggered Pin Fin Heat sink used in the paper.^[17]

The heat sink modeled in this study features cylindrical pin fins arranged in a staggered configuration over a rectangular baseplate. The geometry of the heat sink, including fin height, diameter, and spacing, is kept constant throughout the analysis. The staggered arrangement is chosen due to its potential to disrupt thermal boundary layers more effectively, thereby enhancing convective heat transfer.

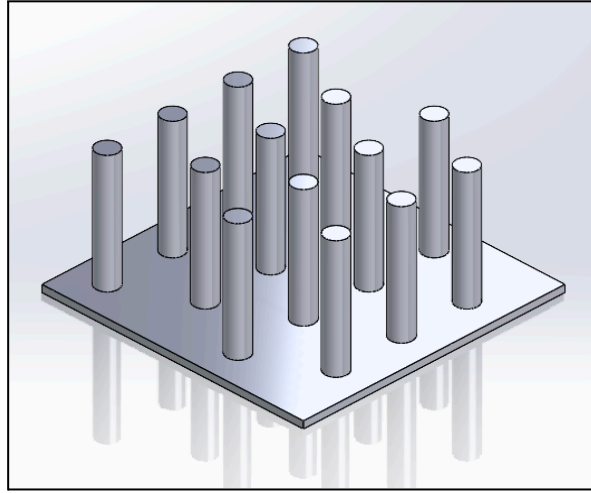


Figure 3.2: Geometry of the Staggered Pin Fin Heat sink used in the paper.^[17]

The base of the Heat sink is 100 mm x 100 mm and is 3 mm thick. The pins are 8 mm in diameter and 50 mm high, each fin is at a distance of 25 mm from each other. The heatsink is enclosed in a duct/domain which has 1040 mm, 100 mm, 50mm of length, breadth and height respectively. The heatsink is located at a distance of 745 mm from the inlet.^[17]

3.3: BOUNDARY CONDITIONS AND MATHEMATICAL FORMULATIONS

The CFD simulations in this study were performed using ANSYS Fluent, employing the SIMPLEC algorithm for pressure–velocity coupling. For spatial discretization, a second-order scheme was used for pressure, while the second-order upwind scheme was applied to the energy and momentum equations, as well as the transport equations for turbulence kinetic energy and turbulent dissipation rate. The Realizable k- ϵ turbulence model was adopted, suitable for steady, incompressible turbulent flow over complex geometries.

No.	Boundary Location	Mass Transport Condition	Heat Transfer Condition	Physical Motivation
1.	Heat sink inlet	$U_{in} = 1.9 - 5.81$ m/s	Fluid air = 27°C Solid $\frac{dT}{dy} = 0$	Model unidirectional inflow of air; sets initial air temperature; assumes no temperature gradient On solid at inlet.
2.	Heat sink	$P_{air} = 1$ atm		Ensures standard

	outlet			atmospheric pressure at outlet;
3.	Heat sink right wall	symmetry	Solid $\frac{dT}{dy} = 0$	Assumes no heat flux through the left wall due to symmetry
4.	Heat sink walls		$\frac{\partial T}{\partial x} = \frac{\partial T}{\partial y} = \frac{\partial T}{\partial z} = 0$	Adiabatic and no slip condition
5.	Heat sink base		$Q = 5903 \text{ W.m}^{-2}$	Applies constant heat flux through the base
6.	Interface between solid and fluid		$\lambda_s \frac{\partial T}{\partial n} = \lambda_f \frac{\partial T}{\partial n}$	Enforces no-slip condition at the interface; ensures thermal interface balance between solid and fluid

Table 3.1: Boundary Conditions

Several assumptions were made:

- The fluid (air) was considered incompressible and steady-state.
- Flow was assumed to be turbulent with no-slip conditions on walls.
- The thermophysical properties of the fluid and solid were assumed constant.
- Radiative heat transfer was neglected.
- Heat sink base and fins were made of aluminium, modeled as a solid with known thermal properties.
- Uniform and steady flux was imposed.
- Constant inlet velocity and temperature of the air.

A symmetric boundary condition was used along the mid-plane to reduce computational load and convergence time.

The governing equations used in this simulation are based on the principles of conservation of mass, momentum, and energy, which form the foundation of computational fluid dynamics. These equations are solved for a steady, incompressible flow regime with heat transfer.

The **continuity equation** ensures mass conservation and is expressed as:

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u_i)}{\partial x_i} = 0$$

The **momentum equation**, derived from Newton's second law, accounts for pressure forces, viscous forces, and body forces, and is given by:

$$\frac{\partial (\rho u_i)}{\partial t} + \frac{\partial (\rho u_i u_j)}{\partial x_j} = - \frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right] + \rho g_i$$

The **energy equation** accounts for convective and conductive heat transfer is given by:

$$\frac{\partial}{\partial t} (\rho h) + \frac{\partial}{\partial x_j} (\rho u_j h) = \frac{\partial}{\partial x_j} \left(k \frac{\partial T}{\partial x_j} \right)$$

To accurately capture turbulence effects, the **Realizable k-ε turbulence model** was employed. This model solves two additional transport equations: one for the turbulent kinetic energy (k) and another for the turbulent dissipation rate (ϵ). These are defined as follows:

$$\begin{aligned} \frac{\partial}{\partial x_j} (\rho k u_j) &= \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + G_k + G_b - \rho \epsilon + Y_M \\ \frac{\partial}{\partial x_j} (\rho \epsilon u_j) &= \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x_j} \right] - \rho C_2 \frac{\epsilon^2}{k + \sqrt{\nu \epsilon}} + C_1 \frac{\epsilon}{k} C_3 G_b \end{aligned}$$

In these equations, the constants are taken as:

$$\sigma k = 1.0, \quad \sigma \epsilon = 1.2, \quad C_1 = 1.44, \quad C_2 = 1.9$$

Following the solution of the governing equations, post-processing and data reduction were performed to evaluate the thermal and hydraulic performance of the heat sink designs. The analysis was carried out based on quantities such as pressure drop (ΔP), Nusselt number (Nu), Reynolds number (Re), and the convective heat transfer coefficient(h).

The **Reynolds number** was used to characterize the flow regime and is defined as:

$$Re = \frac{\rho u_{in} D_h}{\mu}$$

Where,

- ρ is the air density
- u_{in} is the inlet velocity
- D_h is the hydraulic diameter of the flow passage
- μ is the dynamic viscosity of air.

The hydraulic diameter was calculated from the channel dimensions as:

$$D_h = \frac{4A}{P} = \frac{2ab}{a+b}$$

where A is the cross-sectional area and P is the perimeter of the flow channel, with a and b representing its width and height respectively.

The **average heat transfer coefficient** was calculated from the imposed heat flux q and the average temperature difference between the fin surface and the air:

$$h_{avg} = \frac{q}{T_{avg} - T_m}$$

The mean air temperature T_m is computed as the arithmetic average of the inlet and outlet temperatures:

$$T_m = \frac{T_{in} + T_{out}}{2}$$

Using the above quantities, the **average Nusselt number** is calculated as:

$$Nu = \frac{h_{ave} D_h}{k}$$

Where k is the thermal conductivity of the air.

The physical properties of fluid and solid are mentioned below:

	k (W/m.k)	μ (kg/m.s)	ρ (kg/m³)	C_p (J/kg.k)
Air	0.0242	1.7894×10^{-5}	1.225	1006.43
Aluminium	202.4	-	2719	871

Table 3.2: Thermo-Physical properties

3.4: MESH INDEPENDENCE

Mesh Elements	Average Temperature
14 lakhs	344.494 K
12 lakhs	344.587 K
10 lakhs	344.635 K
8 lakhs	345.350 K
6 lakhs	345.280 K
4 lakhs	345.060 K
3.5 lakhs (Default)	345.983 K

Table 3.3: Mesh Convergence.

A mesh independence study was conducted by varying the total number of mesh elements from 3.5 lakh to 14 lakh and observing the resulting average base temperature. The temperature values initially showed slight fluctuations; however, beyond 6 lakh elements, the changes became minimal. The difference in temperature between the 10 lakh and 14 lakh element cases was only 0.141 K, indicating that the numerical solution had stabilized.

A high-quality unstructured mesh was generated using tetrahedral volume elements with triangular surface discretization. Local mesh refinement was applied in regions of high velocity and thermal gradients, particularly around the cylindrical pin surfaces.

Additionally, five inflation layers with smooth transition were employed along the pin-fin boundaries to accurately resolve the boundary layer.

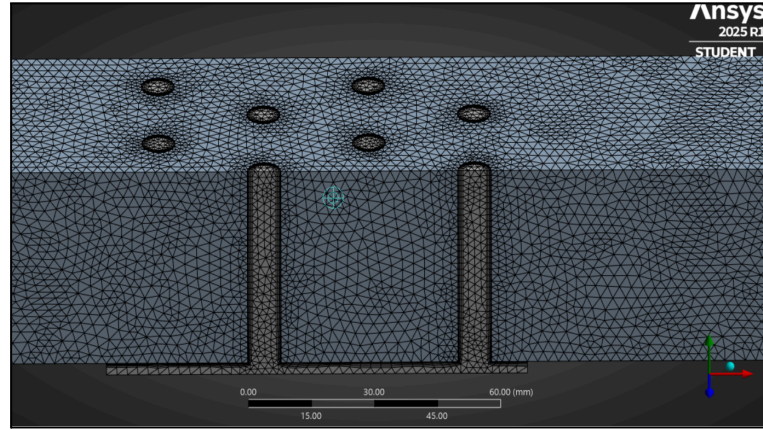


Figure 3.3: Figure of the mesh proposed by the Author.

The mesh maintains a good balance between accuracy and computational efficiency, with a total element count of approximately 9.9 lakh. Mesh independence was also validated, confirming that the chosen mesh provides reliable simulation results.

3.5: VALIDATION FOR THE CREDIBILITY OF CFD

To validate our numerical setup, we adopted the same geometric parameters, boundary conditions, and meshing strategies as mentioned in the original paper. Two key performance indicators—average temperature and Nusselt number—were chosen for comparison. The results from our simulations were plotted alongside those from the original authors for solid fins.

The Reynolds numbers are [8731, 13221, 17603, 22071, 26671] and the corresponding velocity calculated for simulation using the hydraulic diameter of 0.067 is [1.9, 2.88, 3.85, 4.81, 5.81] respectively.

Figure 3.4 shows the comparison of Average Temperature vs Reynolds Number, where it can be observed that our simulation closely follows the trend exhibited in the authors' data. Although a slight deviation is present at lower Reynolds numbers, the overall temperature drop pattern is consistent, indicating that the thermal behavior of the fins has been well captured by our model.

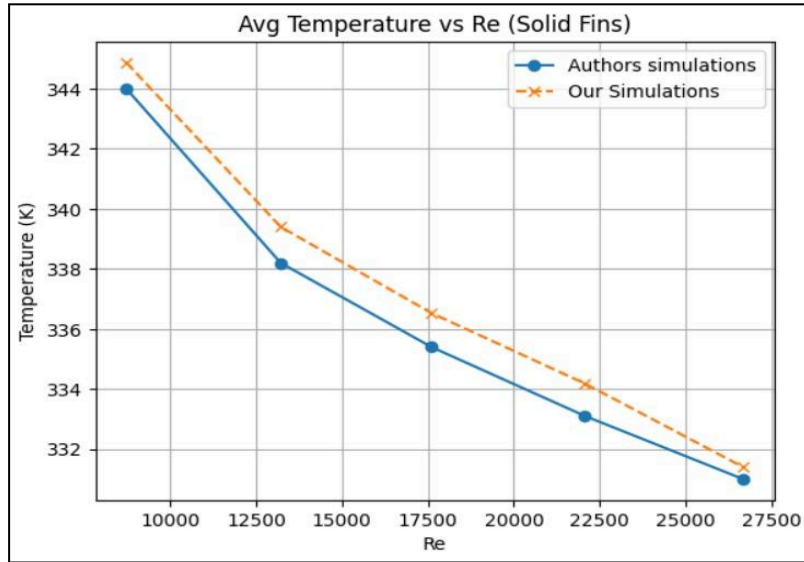


Figure 3.4: Comparison average temperature of authors' simulations with our simulations.

Figure 3.5 presents the Nusselt Number vs Reynolds Number comparison. Our simulation results align well with the author's, particularly at higher Reynolds numbers, demonstrating that the convective heat transfer prediction is reliable. The minor variance observed at intermediate Reynolds values is within acceptable error limits for CFD-based studies.

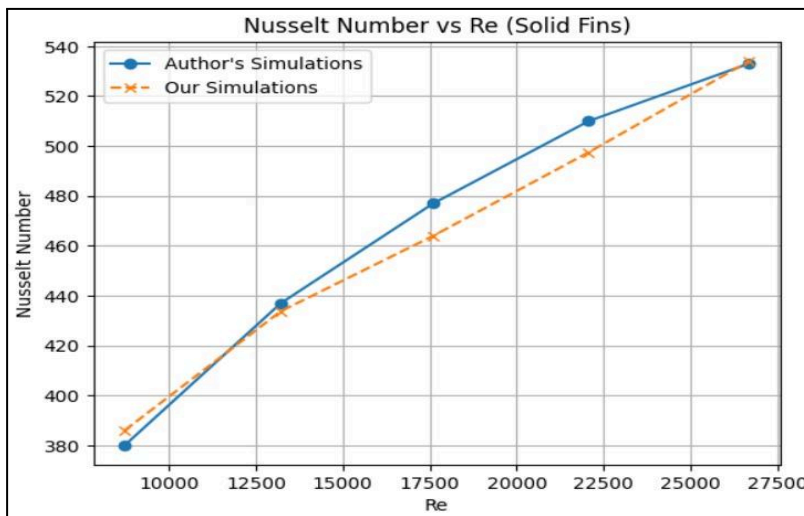


Figure 3.5: Comparison of Nusselt number with authors' simulations

This validation exercise confirms that our CFD model is sufficiently accurate and can be considered credible for further investigation. With this validation, we proceed to perform parametric analysis on various fin geometries (including square and cylindrical fins in staggered and inline configurations) by varying parameters such as fin diameter, height, spacing, and inlet velocity, using the same baseline setup.

3.6 EFFECT OF GEOMETRIC CHANGES FOR SOME PIN FIN HEAT SINKS

Pin fin heat sinks are widely employed in electronics cooling and high-heat-flux applications due to their enhanced surface area and ability to disrupt thermal boundary layers. Among the various configurations, cylindrical and square pin fin arrays are the most commonly used owing to their manufacturability, symmetry, and well-documented performance.

In this study, both cylindrical and square pin fin geometries were considered as the base designs. To develop a comprehensive understanding of how geometry influences thermal and fluid performance, a parametric study was conducted. The following key geometric parameters were systematically varied:

- Fin diameter / square edge length
- Fin height
- Fin spacing (both longitudinal and transverse)
- Array orientation – *inline* and *staggered*

The base simulation setup, boundary conditions, fluid properties along with the meshing were adopted directly from the validated reference paper discussed earlier. This was done deliberately to maintain consistency and enable a comparative analysis between our geometrically varied models and the established baseline from literature.

By analyzing the resulting trends in Nusselt number, pressure drop, average base temperature, and heat transfer coefficient, the study aims to extract insights on how individual geometric modifications impact the overall thermal performance. This comparative framework ensures that the influence of each parameter can be isolated, quantified, and used to guide optimal design strategies for next-generation heat sinks.

3.6.1 CYLINDRICAL PIN FIN HEAT SINKS

3.6.1.1 BASE GEOMETRIES OF THE STAGGERED AND INLINE ARRAYS

To establish a baseline for performance evaluation, cylindrical pin fin heat sinks were designed in two distinct array configurations—inline and staggered—as part of the initial simulation models.

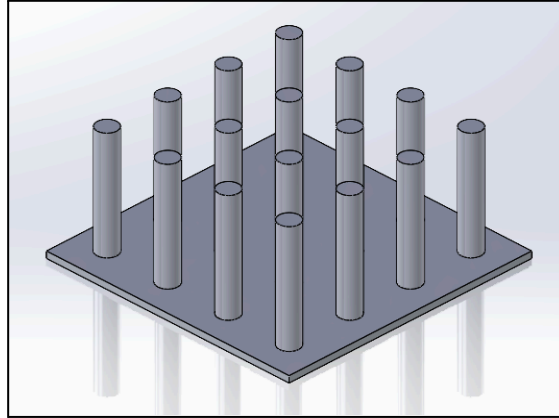


Figure 3.6: Inline Pin Fin Geometry.

The inline configuration consists of a uniform 4×4 grid, comprising a total of 16 cylindrical pins. The fins are aligned along both the longitudinal and transverse directions.

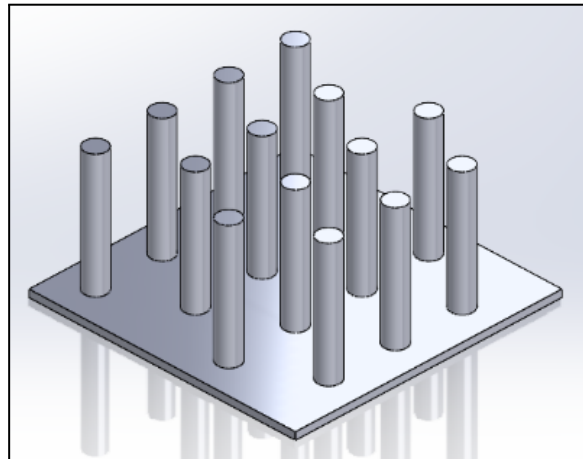


Figure 3.7: Staggered Pin Fin Geometry.

The staggered configuration, on the other hand, features 14 cylindrical pins arranged with alternating offsets in each row.

Parameters	Inline Heatsinks	Staggered Heatsink
Fin Diameter	8mm	8mm
Fin height	50mm	50mm
Fin spacing (Longitudinal and Transverse)	25mm	25mm
Base length	100mm	100mm
Base width	100mm	100mm
Base thickness	3mm	3mm
Number of pins	16	14

Table 3.4 Dimensions of heatsinks.

Dimensions	Values
Length	1040 mm
Breadth	100 mm
Height	100 mm
Distance of the heatsink from the inlet	745 mm

Table 3.5 Dimensions of Domain and heatsink placement (Both Inline and Staggered).

3.6.1.2 MESHING

A high-quality unstructured mesh was generated for both the staggered and inline cylindrical pin fin heat sink configurations to ensure numerical accuracy and convergence stability in the CFD simulations.

As shown in the figure below, a triangular surface mesh with tetrahedral volume elements was employed. The global element size was set to 2.5 mm, resulting in a total element count of approximately 9.9 lakh elements. A triangular surface mesh with tetrahedral volume elements was used due to its ability to accurately capture the complex geometry of cylindrical pin fins. This mesh type conforms well to curved surfaces and allows automatic generation without manual blocking. It also supports localized refinement near high-gradient zones, such as fin bases. Smooth transitions between fine and coarse regions help maintain mesh quality.

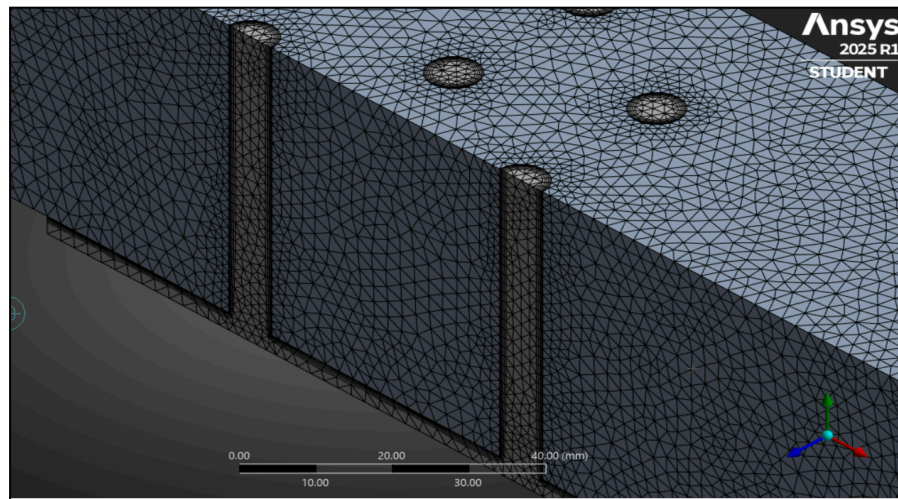


Figure 3.8: Type of meshing done on both staggered and inline pin fin.

To accurately capture near-wall effects and steep thermal gradients around the fin surfaces, inflation layers were applied. A total of 5 inflation layers with smooth transition and automatic growth rate control were incorporated. This approach ensures the development of well-resolved boundary layers without introducing abrupt skewness in the mesh.

3.6.1.3 MATHEMATICAL FORMULATION

Same as that mentioned in section 3.3.

3.6.1.4 RESULTS AND DISCUSSIONS

3.6.1.4.1 VARYING VELOCITY / REYNOLDS NUMBER.

3.6.1.4.1.1 AVERAGE TEMPERATURE

Figure 3.9 shows the variation of average base temperature with increasing inlet velocity for both staggered and inline cylindrical pin fin configurations. As expected, increasing the airflow velocity enhances convective heat transfer, resulting in a consistent decrease in average temperature for both cases.

Across the entire velocity range, the inline configuration exhibits lower average temperatures than the staggered arrangement. At the maximum velocity of 5.81 m/s, the average temperature for the inline case reaches 327.88 K, compared to 331.40 K for the staggered design. This implies that the inline array is more effective thermally, likely due to its higher pin density (16 fins vs 14), which provides greater total surface area for heat dissipation.

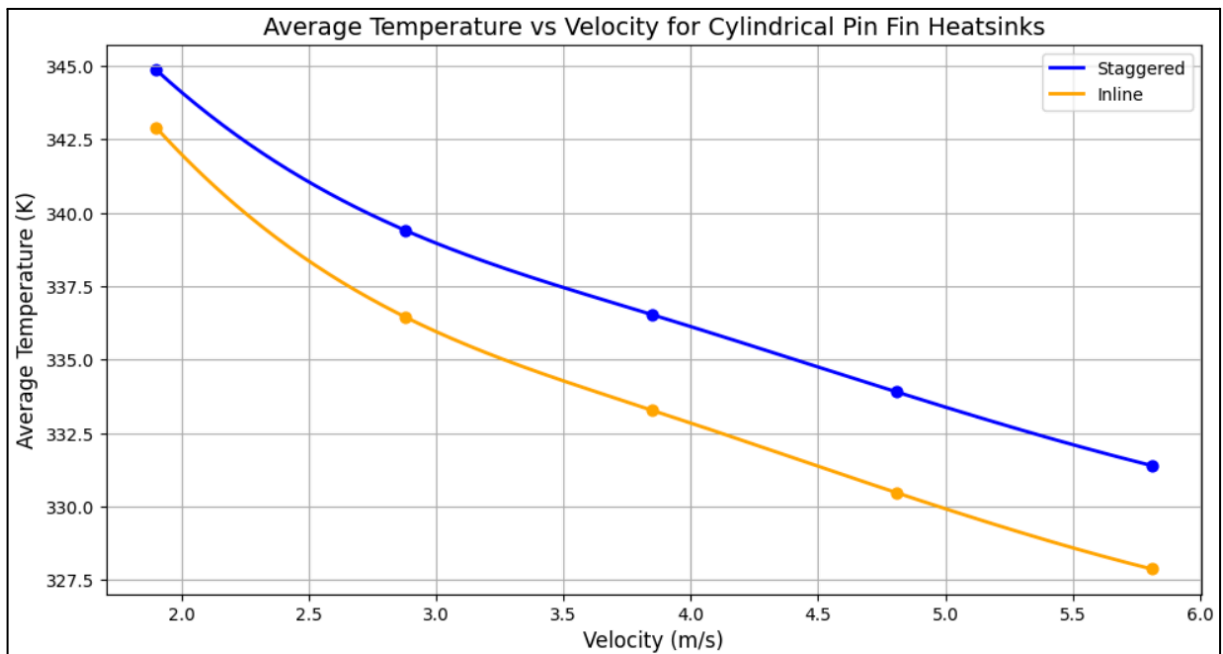


Figure 3.9: Average temperature vs Velocity.

However, this improved cooling comes at the cost of increased pressure drop, as shown in the previous section. Hence, while inline fins offer better temperature reduction, the staggered configuration remains more favorable in systems constrained by pressure limits.

3.6.1.4.1.2 PRESSURE DIFFERENCE

The graph (Figure 3.10) illustrates the variation of pressure difference (ΔP) with inlet velocity for both staggered and inline cylindrical pin fin configurations. As expected, pressure drop increases non-linearly with velocity due to the quadratic dependence of dynamic pressure on flow speed.

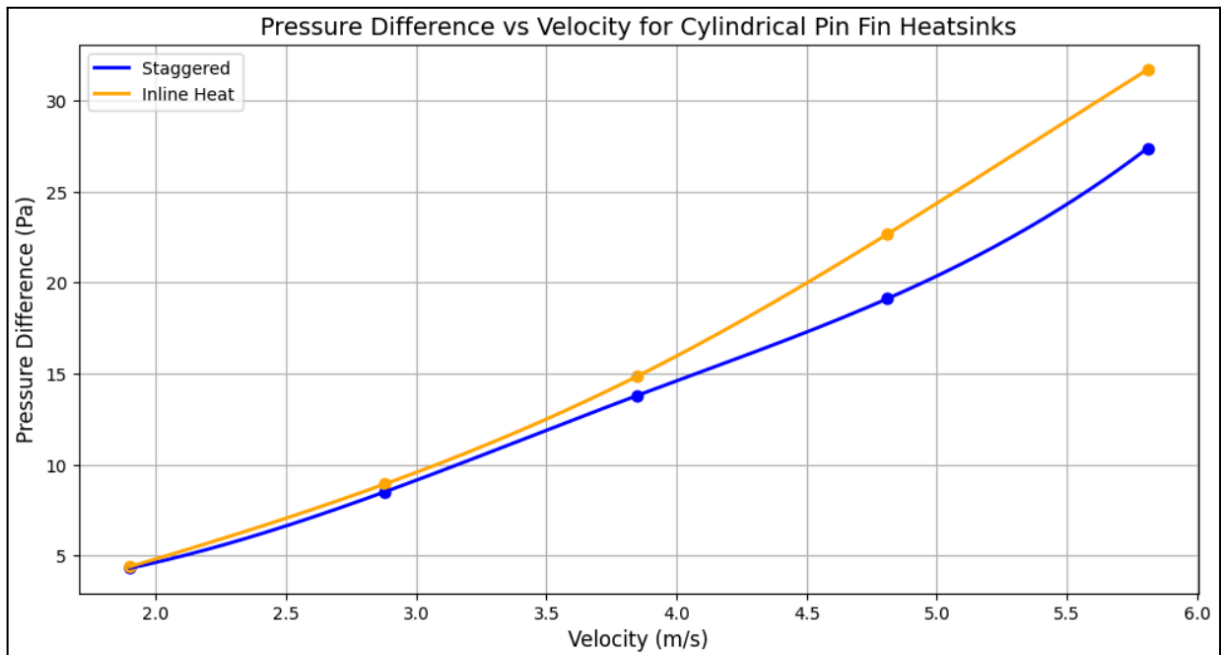


Figure 3.10: Pressure Difference vs Velocity.

However, it is observed that the inline configuration consistently exhibits higher pressure drop compared to the staggered one at all velocity levels. This behavior can be attributed to the higher number of pins (16 in inline vs 14 in staggered) and their aligned positioning, which causes more pronounced flow obstruction and stagnation zones.

At the maximum velocity of 5.81 m/s, the inline array records a pressure drop of 31.72 Pa, while the staggered array records 27.40 Pa, showing a ~16% reduction. This indicates that the staggered design offers lower flow resistance, making it more favorable in systems where minimizing pumping power is critical.

3.6.1.4.1.3 NUSSELT NUMBER

The figure 3.11 represents the variation of the Nusselt number (Nu) with inlet velocity for both staggered and inline cylindrical pin fin heat sinks. As the velocity increases, the Nusselt number shows a clear upward trend, reflecting improved convective heat transfer due to higher turbulence and increased Reynolds number.

The inline configuration consistently achieves higher Nusselt numbers than the staggered one across all velocity levels. At the maximum velocity of 5.81 m/s, the inline array reaches a Nu of 604.16, while the staggered configuration attains 534.96. This superior thermal performance can be attributed to the higher fin density and surface area in the inline layout, allowing more effective heat removal.

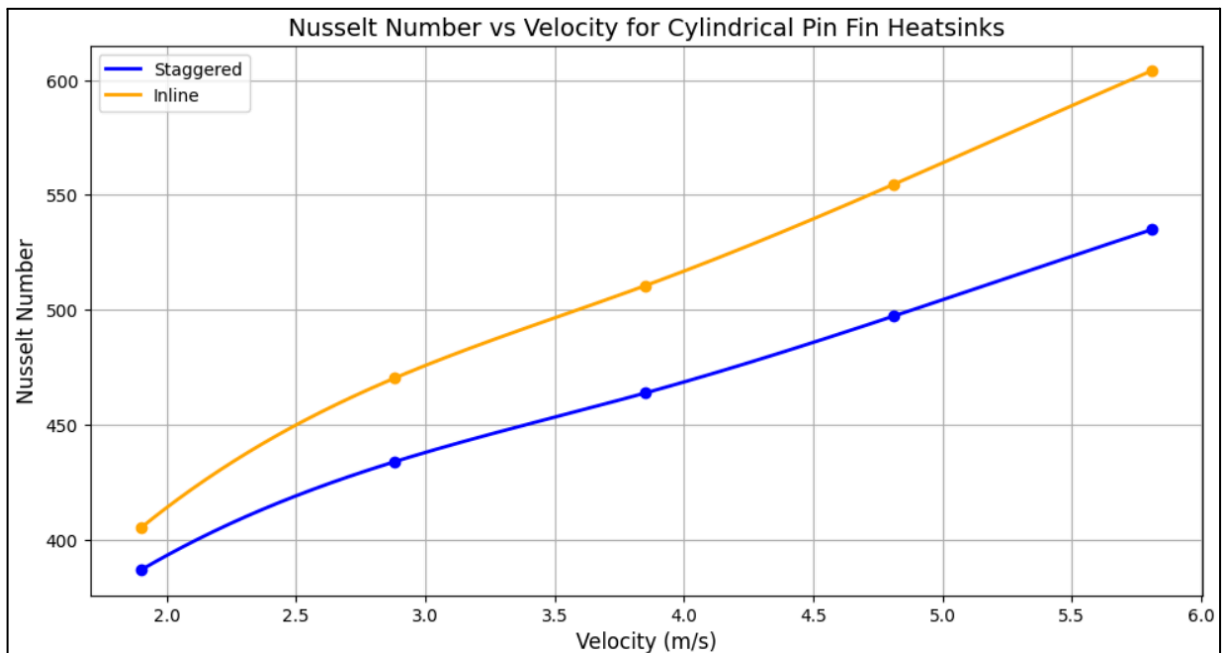


Figure 3.11: Nusselt Number vs Velocity.

Despite slightly reduced flow mixing compared to staggered, the inline array benefits from increased conduction area, making it thermally more efficient in terms of convective transfer. However, this enhancement must be weighed against the higher pressure drop seen in earlier results.

3.6.1.4.1.4 HTC / PRESSURE DIFFERENCE

The graph (In Figure 3.12) illustrates the ratio of heat transfer coefficient to pressure drop ($h/\Delta P$) as a function of inlet velocity for both staggered and inline cylindrical pin fin heat sinks. This parameter serves as a key indicator of thermal-hydraulic performance, representing the trade-off between thermal enhancement and flow resistance.

As velocity increases, $h/\Delta P$ drops significantly for both configurations due to the nonlinear rise in pressure drop outpacing the improvement in heat transfer. At lower velocities (1.90–2.88 m/s), the inline configuration shows marginally better thermal-hydraulic efficiency, with a peak value of 33.40 W/m²K·Pa, compared to 32.48 W/m²K·Pa for staggered.

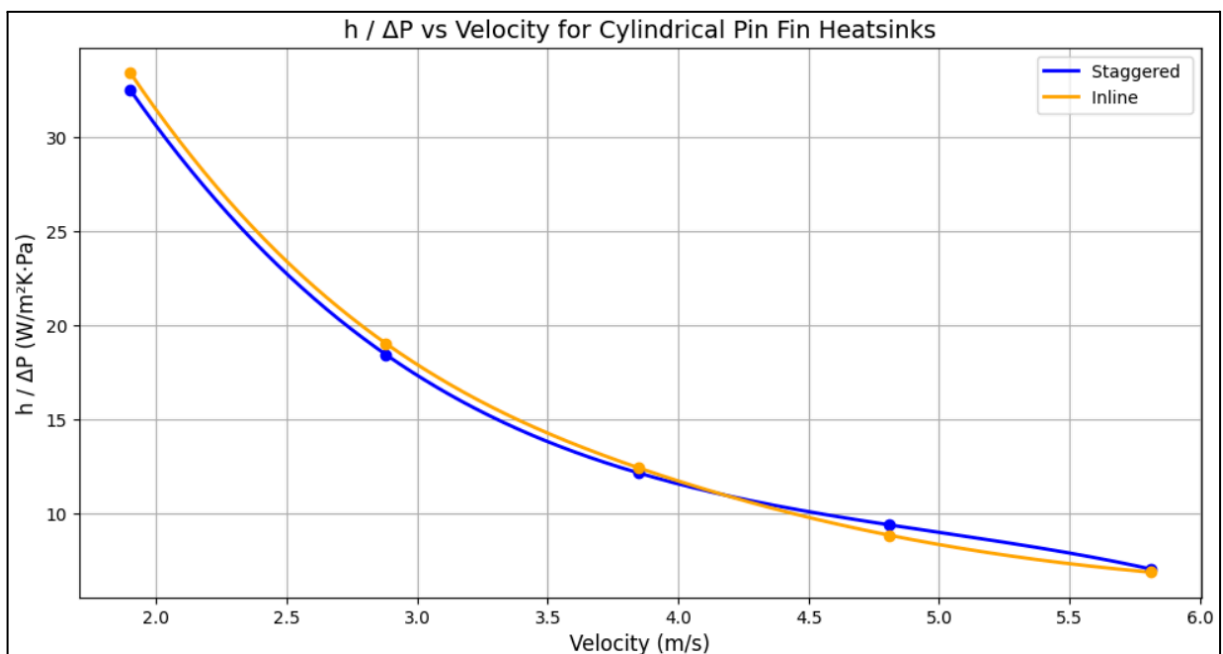


Figure 3.12: Heat Transfer Coefficient / Pressure Difference vs Velocity.

However, at higher velocities, the staggered configuration outperforms in terms of $h/\Delta P$, particularly from 4.81 m/s onward. At 5.81 m/s, the staggered array achieves 7.05 W/m²K·Pa, slightly higher than 6.88 for inline. This indicates that staggered fins offer a more favorable balance between heat transfer and pressure loss, especially in high-velocity or high-flow-rate applications.

3.6.1.4.2 VARYING HEIGHT.

3.6.1.4.2.1 AVERAGE TEMPERATURE

Figure 3.13 illustrates the impact of increasing fin height on the average temperature of cylindrical pin fin heat sinks for both staggered and inline configurations. A clear downward trend is observed, where longer fins lead to lower average temperatures due to increased heat transfer surface area and improved thermal conduction pathways.

At the lowest fin height of 10 mm, the average temperature for the staggered array is 439.57 K, which decreases to 344.87 K at 50 mm. Similarly, the inline configuration drops from 433.12 K to 342.91 K over the same height range. This validates that taller fins significantly enhance cooling performance by providing more area for convective heat exchange.

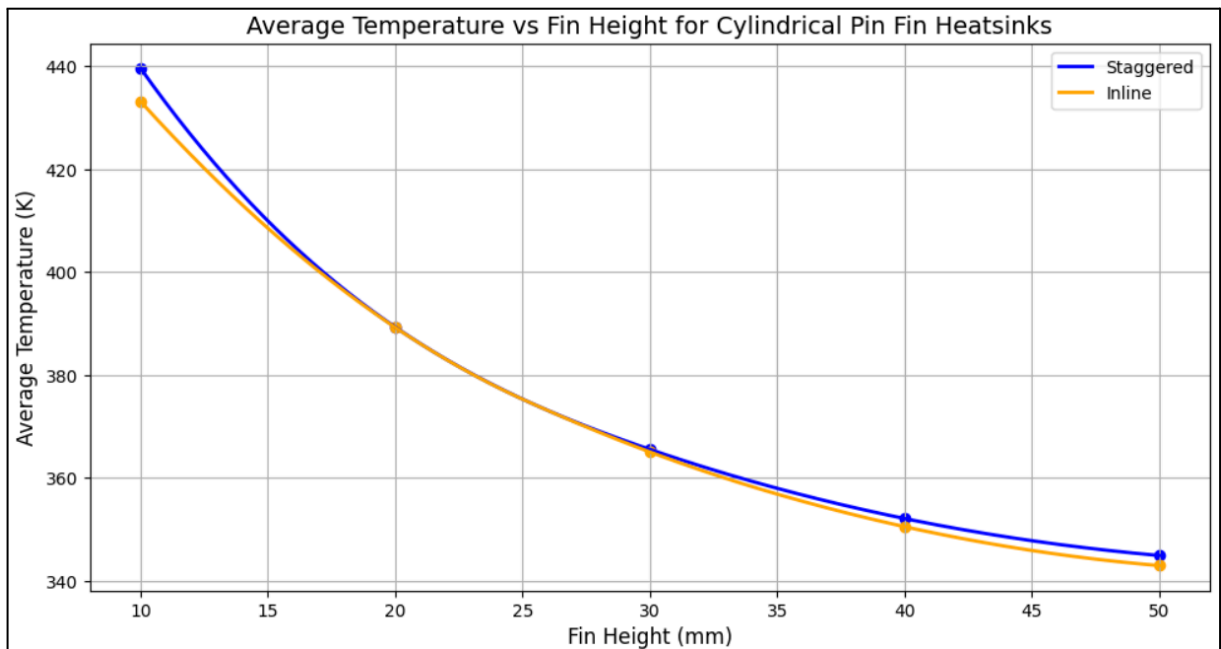


Figure 3.13: Average temperature vs Fin Height.

Although both configurations exhibit nearly identical trends beyond 30 mm, the inline array shows slightly better thermal performance at all heights, which aligns with previous observations on its superior Nusselt number and HTC. This improvement is largely due to the greater fin count and more uniform heat spreading in the inline setup.

3.6.1.4.2.2 PRESSURE DIFFERENCE

The graph (Figure 3.14) illustrates how pressure difference (ΔP) varies with increasing fin height for both staggered and inline cylindrical pin fin configurations. As expected, an increase in fin height leads to a rise in pressure drop due to greater flow resistance and extended surface obstruction in the channel.

The trend is nonlinear, with ΔP increasing sharply beyond 30 mm height for both configurations. At the maximum height of 50 mm, the inline configuration shows a slightly higher pressure drop (4.38 Pa) compared to the staggered case (4.30 Pa), indicating marginally more flow blockage due to higher pin count and alignment.

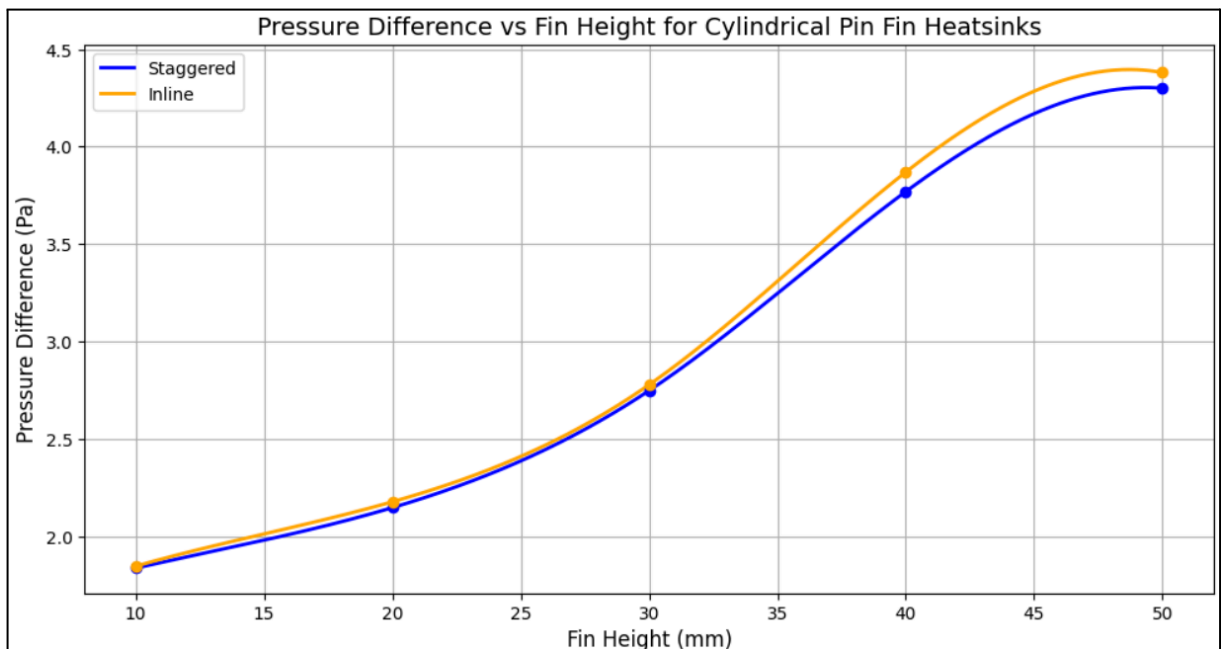


Figure 3.14: Pressure Difference vs Fin Height.

Overall, both configurations show similar ΔP trends, but the staggered array consistently maintains slightly lower pressure drops, reinforcing its suitability in scenarios where pressure losses need to be minimized while still benefiting from increased heat transfer surface area.

3.6.1.4.2.3 NUSSELT NUMBER

The graph (In Figure 3.15) shows the variation of the Nusselt number (Nu) with increasing fin height for both staggered and inline cylindrical pin fin configurations. As expected, the Nusselt number increases steadily with height, reflecting improved convective heat transfer due to larger surface area and enhanced thermal gradients.

At a height of 10 mm, the inline configuration records a Nu of 125.77, slightly higher than the staggered value of 120.02. As fin height increases to 50 mm, the Nusselt number rises to 404.98 for inline and 386.63 for staggered. This consistent improvement highlights the positive influence of fin height on thermal performance, as taller fins provide more surface for heat dissipation and increase residence time of fluid around the fins.

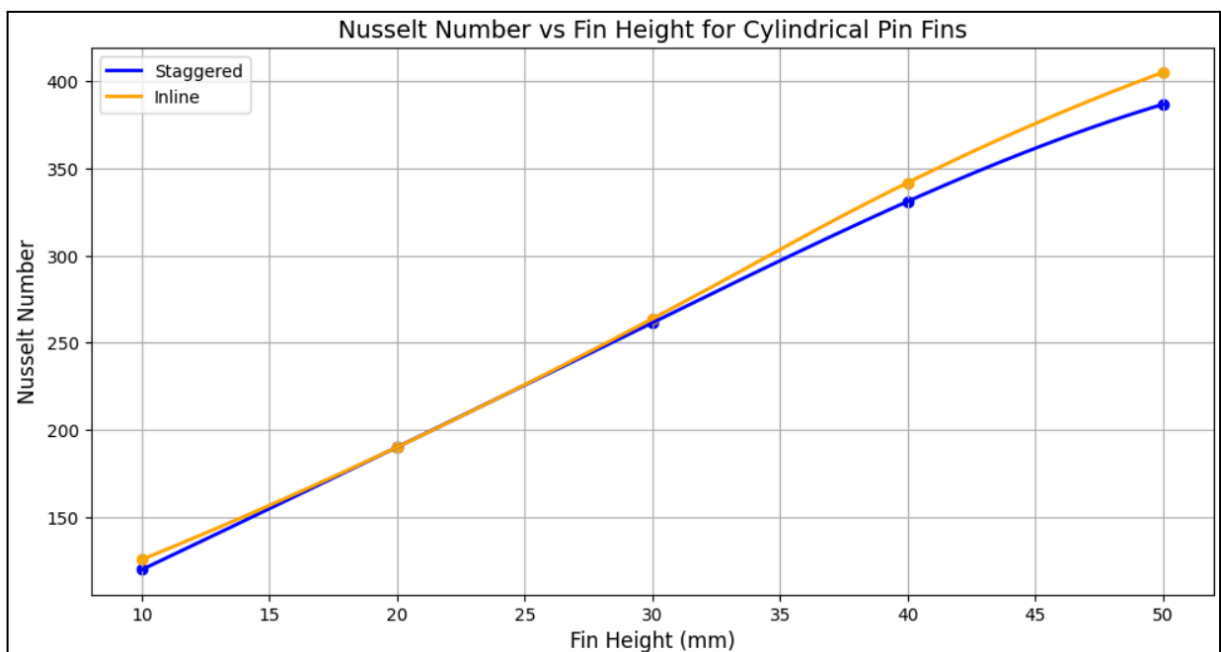


Figure 3.15: Nusselt Number vs Fin Height.

The inline configuration outperforms the staggered array at all heights, albeit by a small margin. This trend aligns with prior results, where inline fins exhibited better base temperature reduction and higher HTC due to their higher pin density and better surface coverage.

3.6.1.4.2.4 HTC / PRESSURE DIFFERENCE

The graph (In Figure 3.16) shows the variation of thermal-hydraulic performance, represented by the ratio $HTC/\Delta P$, with increasing fin height for both staggered and inline cylindrical pin fin heat sinks. This parameter captures the balance between heat transfer efficiency and the corresponding pressure penalty.

Both configurations show a peak performance at 30 mm height, where $HTC/\Delta P$ is maximized — 34.34 $W/m^2K \cdot Pa$ for staggered and 34.27 $W/m^2K \cdot Pa$ for inline. This indicates that 30 mm is the optimum fin height for achieving the best trade-off between thermal performance and flow resistance.

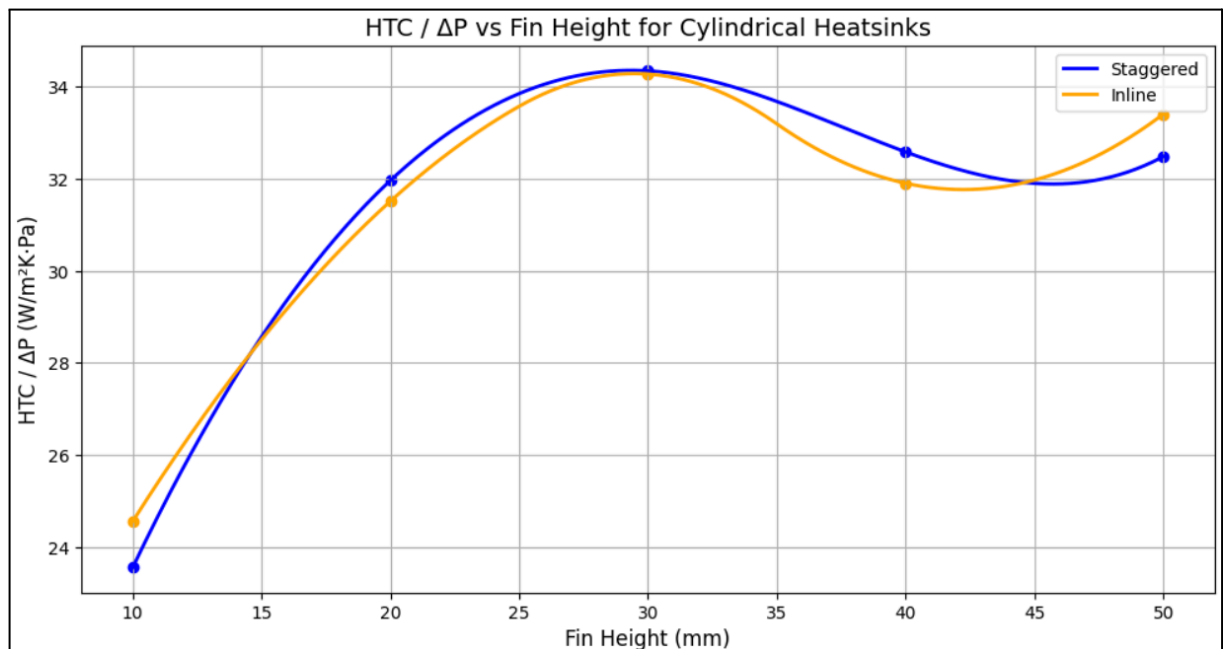


Figure 3.16: $HTC / \text{Pressure Difference}$ vs Fin Height.

Beyond 30 mm, the ratio slightly declines or flattens, even though HTC continues to increase. This is because the rate of pressure drop rises faster than the heat transfer gain, making the system less efficient per unit pressure loss. Interestingly, at 50 mm, inline regains a slight edge, recording 33.40 $W/m^2K \cdot Pa$, compared to 32.48 for staggered. Overall, while both geometries follow similar trends, the staggered configuration performs marginally better at the optimal point, reinforcing its suitability in applications where pressure drop is a design constraint.

3.6.1.4.3 VARYING PIN DIAMETER TO PIN SPACING RATIO.

3.6.1.4.3.1 AVERAGE TEMPERATURE

The graph (In Figure 3.17) depicts the variation in average temperature with increasing pin diameter-to-spacing ratio (D/S) for both staggered and inline cylindrical pin fin configurations. As the D/S ratio increases, the average temperature consistently decreases, indicating improved cooling performance due to higher fin density and reduced bypass flow.

At a D/S ratio of 0.32, the average temperature for the staggered and inline configurations is 344.87 K and 342.91 K respectively. As the D/S increases to 0.92, these values drop to 315.26 K and 312.72 K, reflecting better heat dissipation as the flow is forced through narrow gaps and contacts more surface area.

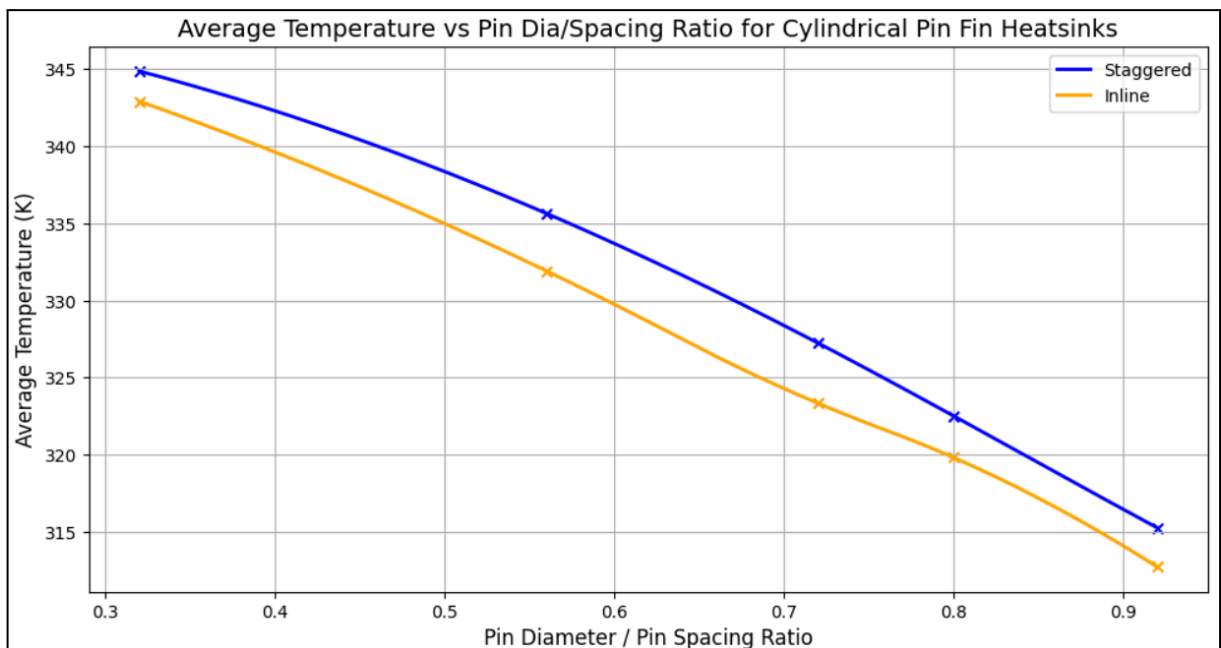


Figure 3.17: Average Temperature vs D/S Ratio.

The inline configuration outperforms staggered at all ratios, maintaining a lower average temperature throughout. This improvement is primarily due to higher pin count and more uniform distribution, which enhances the overall surface area for heat transfer. However, this comes with the trade-off of a much higher pressure drop, as seen in the previous graph.

3.6.1.4.3.1 PRESSURE DIFFERENCE

The graph (Figure 3.18) illustrates the variation of pressure difference (ΔP) with changing pin diameter-to-spacing ratio (D/S) for both staggered and inline cylindrical pin fin configurations. As expected, the pressure drop increases sharply with increasing D/S ratio due to greater blockage of flow area and higher form drag.

At lower D/S ratios (0.32–0.56), both configurations exhibit relatively low ΔP values, with the staggered and inline arrays showing similar performance. However, as the ratio exceeds 0.7, a dramatic rise in pressure drop is observed—especially in the inline configuration. At $D/S = 0.92$, the inline configuration reaches a peak ΔP of 376.19 Pa, significantly higher than 222.54 Pa for staggered.

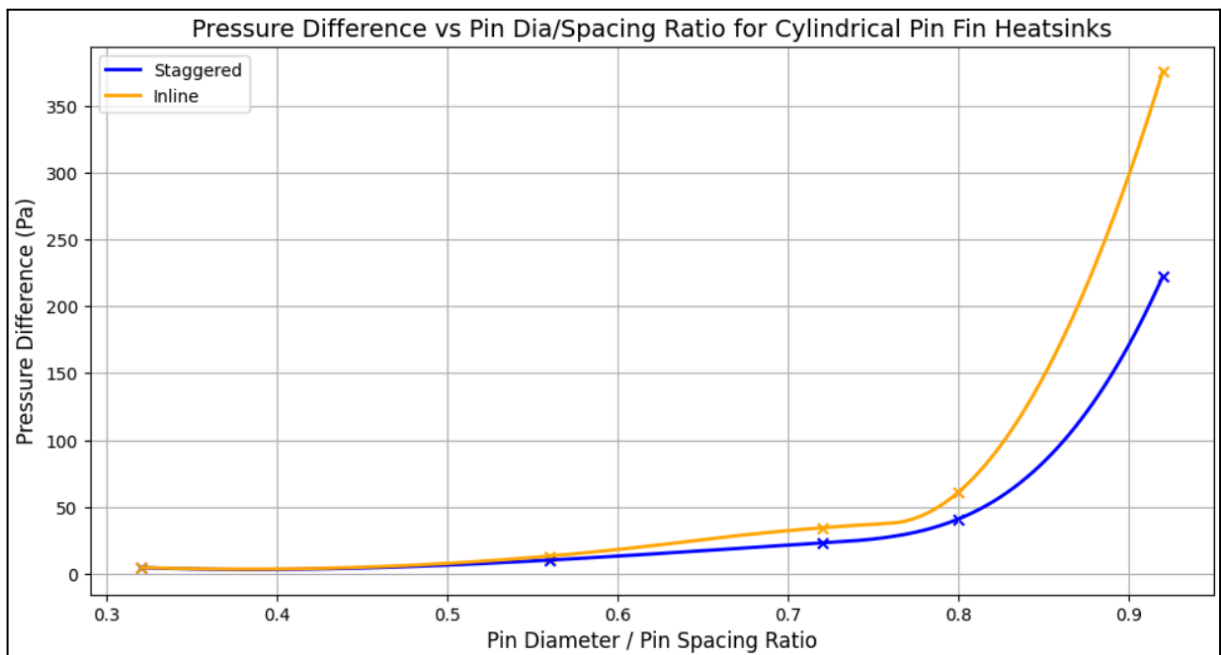


Figure 3.18: Pressure Difference vs D/S Ratio.

This behavior is attributed to the denser pin arrangement at higher D/S ratios, which severely restricts airflow and leads to turbulence intensification. The staggered array, due to its offset pin alignment, allows better flow redistribution and mixing, resulting in lower resistance to flow compared to the inline counterpart.

3.6.1.4.3.3 NUSSELT NUMBER

The graph (In Figure 3.19) displays the relationship between the Nusselt number (Nu) and the pin diameter-to-spacing ratio (D/S) for both staggered and inline cylindrical pin fin configurations. A sharp, nonlinear increase in Nu is observed with increasing D/S, reflecting improved convective heat transfer due to tighter pin spacing and larger fin surface area.

At a D/S of 0.32, the Nusselt number is relatively low—386.63 for staggered and 404.98 for inline. As the D/S ratio rises to 0.92, Nu surges to 1282.81 for staggered and an impressive 1604.62 for inline, indicating a significant enhancement in thermal performance.

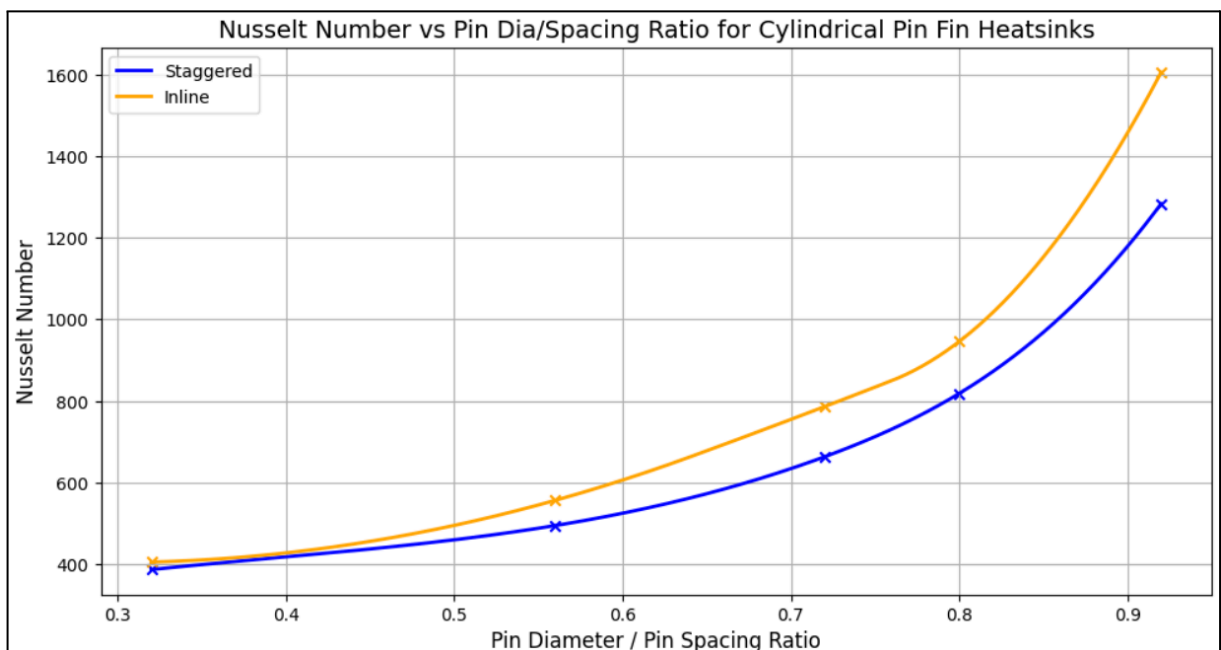


Figure 3.19 : Nusselt Number vs D/S Ratio.

The inline configuration consistently outperforms the staggered one across all D/S values, primarily due to its higher fin density, which increases thermal contact area and reduces bypass regions. This leads to more effective heat transfer, albeit at the cost of significantly higher pressure drops, as shown earlier.

3.6.1.4.3.4 HTC / PRESSURE DIFFERENCE

The graph (In Figure 3.20) illustrates the variation of thermal-hydraulic performance, expressed as $HTC/\Delta P$, with increasing pin diameter-to-spacing ratio (D/S) for both staggered and inline cylindrical pin fin configurations. This ratio provides a measure of heat transfer effectiveness relative to the pressure drop penalty.

At the lowest D/S ratio of 0.32, both configurations achieve their peak thermal efficiency — 33.40 $W/m^2K \cdot Pa$ for inline and 32.48 $W/m^2K \cdot Pa$ for staggered. As the D/S ratio increases, the $HTC/\Delta P$ ratio declines exponentially due to a disproportionately high rise in pressure drop compared to the gain in heat transfer.

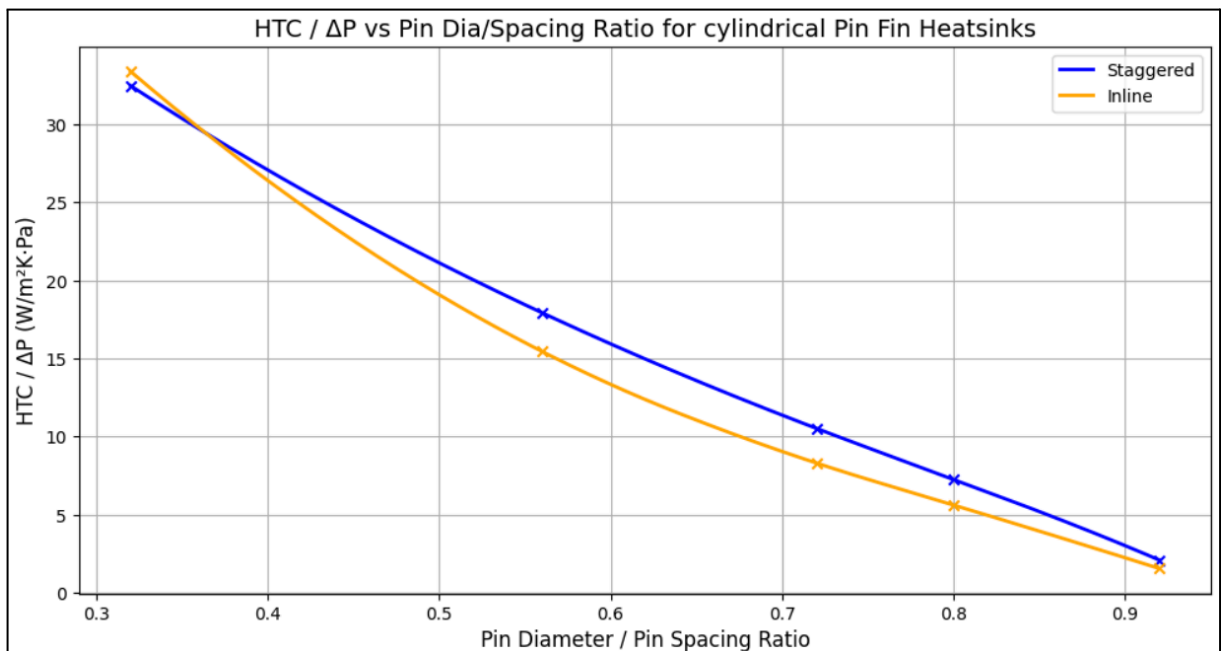


Figure 3.20: $HTC /$ Pressure Difference vs D/S Ratio.

At the highest D/S ratio of 0.92, the staggered design shows a final value of 2.08, whereas inline drops further to 1.54, indicating diminishing returns in thermal performance at extreme pin densities. Notably, the staggered configuration consistently maintains higher $HTC/\Delta P$ values beyond $D/S = 0.56$, proving to be more hydraulically efficient at denser fin arrangements.

3.6.1.4.4 Interpretation of Velocity Streamlines: Inline vs Staggered Pin Fin Arrays

The two velocity streamline plots visually compare the flow development in inline and staggered cylindrical pin-fin heat sinks. These flow visualizations offer crucial insight into how pin arrangement affects thermal and hydraulic behavior.

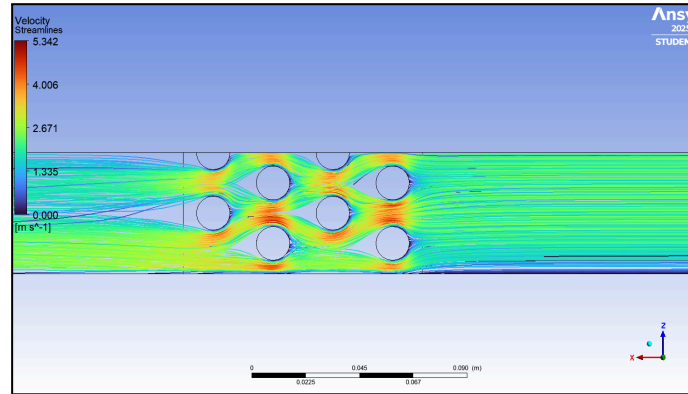


Figure 3.21: Streamline for staggered fin for D/S , 0.56.

In the staggered configuration (Figure 3.21), the flow path is visibly more disturbed and chaotic, with frequent directional changes and swirling near the pins. This promotes strong fluid mixing, leading to thinner boundary layers and enhanced convective heat transfer. The presence of such turbulence typically correlates with increased pressure loss. However, in this case, despite the apparent turbulence, the pressure drop is observed to be lower compared to the inline array.

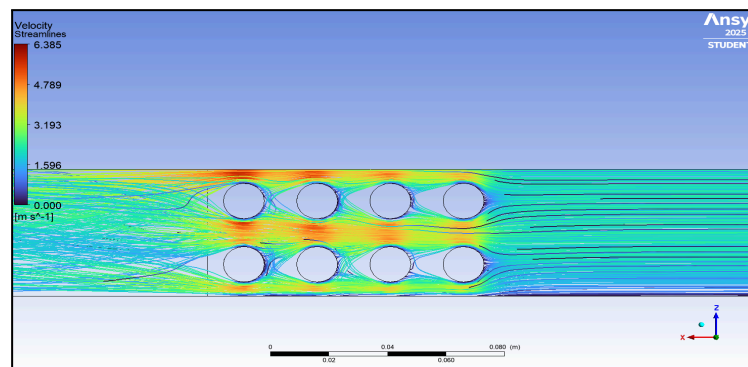


Figure 3.22 : Streamline for Inline fin for D/S , 0.56.

In contrast, the inline configuration (figure 3.22) demonstrates more streamlined flow with minimal cross-flow disturbance. While this leads to slightly reduced mixing, the pin density is higher, and the flow is forced through narrow passages. This results in significant flow constriction, causing higher velocity gradients between pins. Consequently, the inline setup experiences a higher pressure drop, despite having less turbulent mixing overall. This elevated

pressure drop is primarily due to the increased number of pins and surface area, which adds resistance to the fluid path.

From a thermal standpoint, the inline configuration yields a higher Heat Transfer Coefficient (HTC) due to the increased contact surface and localized velocity acceleration between fins. However, this comes at the cost of greater pumping power. The staggered configuration, while slightly lower in HTC, offers a more balanced performance with lower pressure drop and still effective convective cooling due to enhanced mixing.

3.6.2 SQUARE PIN HEAT SINKS

3.6.2.1 BASE GEOMETRIES OF THE STAGGERED AND INLINE ARRAYS

After validating the cylindrical pin-fin heat sink model from the reference study, the geometry was modified to incorporate square pin fins for further analysis.

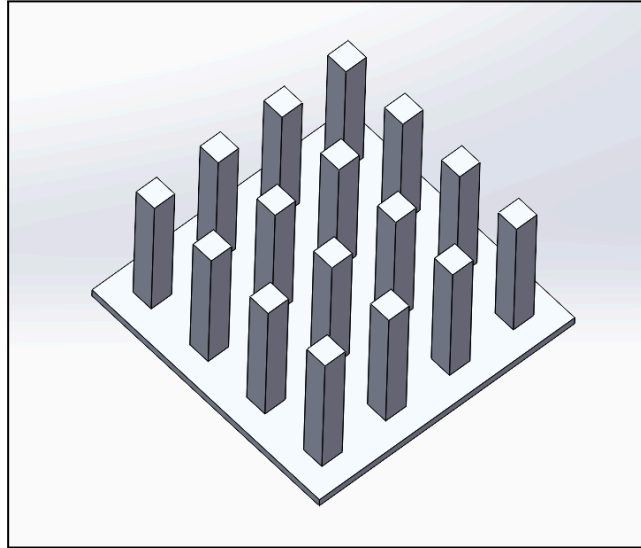


Figure 3.23: Inline Square Pin Fin Geometry.

The inline configuration consists of a uniform 4×4 grid, comprising a total of 16 square pins. The fins are aligned along both the longitudinal and transverse directions.

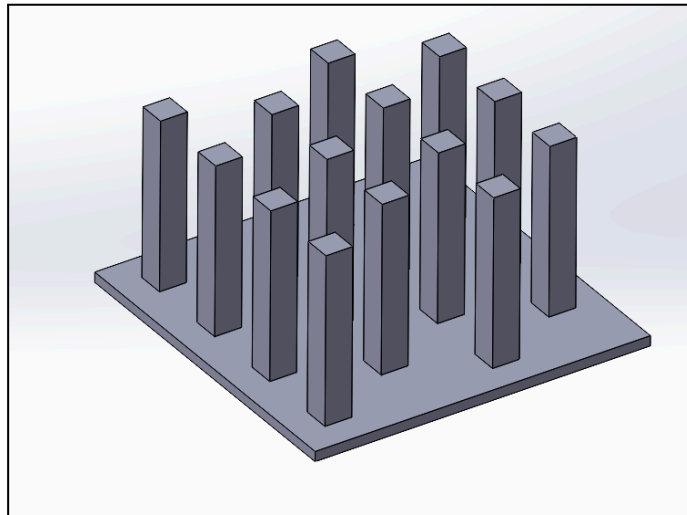


Figure 3.24: Staggered Square Pin Fin Geometry.

The staggered configuration, on the other hand, features 14 square pins arranged with alternating offsets in each row.

Parameters	Inline Heatsinks	Staggered Heatsink
Fin Edge Length	8mm	8mm
Fin height	50mm	50mm
Fin spacing (Longitudinal and Transverse)	25mm	25mm
Base length	100mm	100mm
Base width	100mm	100mm
Base thickness	3mm	3mm
Number of pins	16	14

Table 3.6: Dimensions of Base heatsinks.

Dimensions	Values
Length	1040 mm
Breadth	100 mm
Height	100 mm
Distance of the heatsink from the inlet	745 mm

Table 3.7: Dimensions of Domain and heatsink placement.

3.6.2.2 MESHING

A high-quality unstructured mesh was generated for the square pin-fin heat sink geometry, as shown in Figure 3.25. Similar to the cylindrical configuration, a triangular surface mesh with tetrahedral volume elements was employed to accurately capture the complex three-dimensional geometry. To ensure adequate resolution near the walls and high-gradient thermal regions, five inflation layers were incorporated around the fin surfaces using a smooth transition setting. This layering helped capture boundary layer effects critical to accurate heat transfer prediction. The mesh was generated using automatic transition control to maintain gradual refinement from dense regions near the fins to coarser regions in the far field, thus ensuring numerical stability.

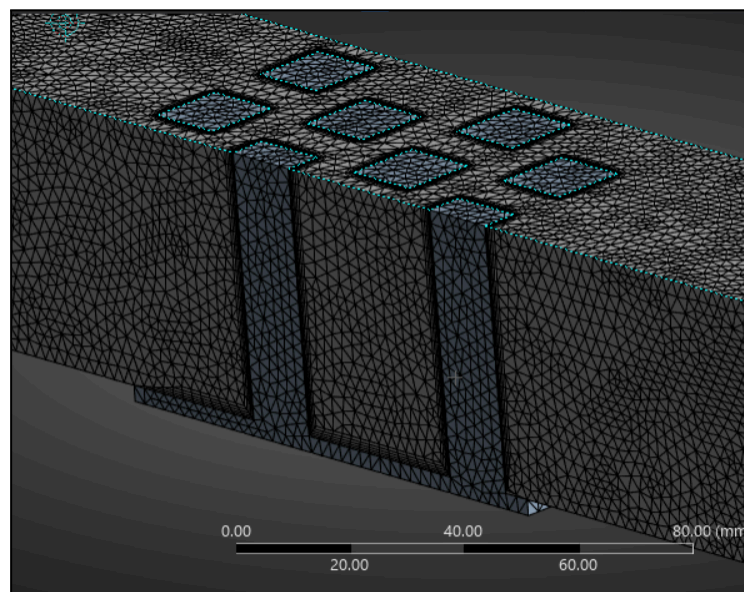


Figure 3.25: Meshing done on Square heatsink

3.6.2.3 RESULTS AND DISCUSSIONS

3.6.2.3.1 VARYING VELOCITY / REYNOLDS NUMBER.

3.6.2.3.1.1 AVERAGE TEMPERATURE

Figure 3.26 presents the variation in average temperature with increasing velocity for square pin fin heat sinks in inline and staggered configurations. As expected, average temperature consistently decreases with increasing velocity, due to enhanced convective heat transfer at higher airflow rates.

Between both configurations, the staggered arrangement shows superior thermal performance, maintaining a lower average temperature across all velocities. At 5.81 m/s, the staggered case drops to 325.38 K, while the inline remains slightly higher at 326.09 K. This performance gap is more prominent at lower velocities, where staggered pins generate stronger fluid mixing due to frequent wake interference, promoting better heat dissipation.

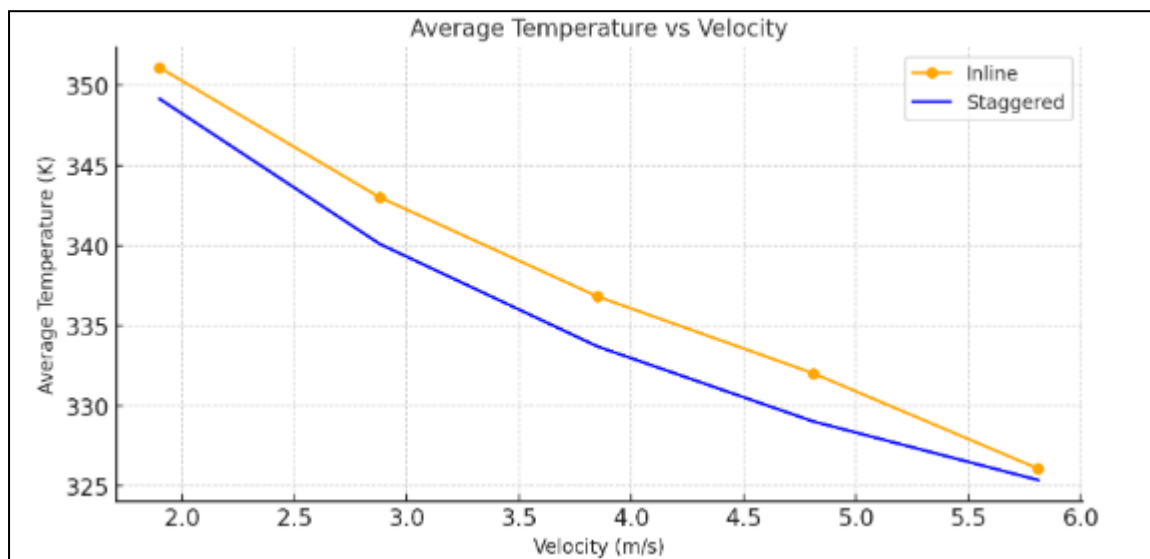


Figure 3.26: Average temperature vs change in velocity

The inline array, while offering less pressure drop, results in less turbulence and flow disruption, which limits its cooling efficiency. On the other hand, the staggered layout forces fluid to weave through alternating channels, increasing local velocity gradients and improving thermal boundary layer thinning — a key reason behind the lower temperature values observed.

3.6.2.3.1.2 PRESSURE DIFFERENCE

Figure 3.27 presents the variation of pressure difference (ΔP) with increasing inlet velocity for square pin fin heat sinks arranged in inline and staggered configurations. A consistent increase in pressure drop is observed with rising velocity due to elevated flow momentum and associated drag forces acting on the fin surfaces.

The staggered configuration shows significantly higher pressure drop across all velocities, reaching 103.69 Pa at 5.81 m/s compared to just 45.77 Pa in the inline case. This is attributed to the flow obstruction and chaotic wake regions created by the staggered layout, which enhances turbulence but restricts smooth passage of air through the fin channels.

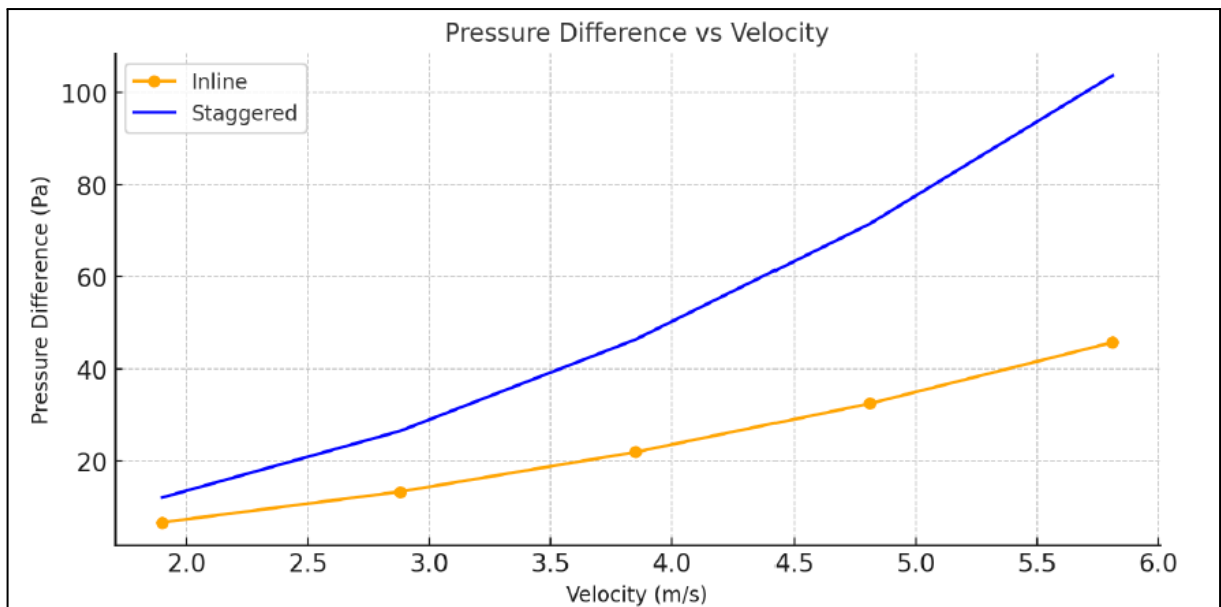


Figure 3.27: Pressure difference vs change in velocity

In contrast, the inline configuration offers more streamlined flow paths, resulting in lower form drag and reduced static pressure buildup. This makes inline square fin arrangements more suitable where lower hydraulic resistance is prioritized.

3.6.2.3.1.3 NUSSELT NUMBER

Figure 3.28 illustrates the variation of Nusselt number with respect to inlet velocity for square pin fin heat sinks arranged in both inline and staggered configurations. A consistent rise in Nu is observed for both layouts as velocity increases, signifying enhanced convective heat transfer driven by thinner thermal boundary layers and increased turbulence at higher flow rates.

Across the full velocity range, the staggered configuration outperforms the inline layout, achieving a maximum Nusselt number of 666.33 at 5.81 m/s, compared to 599.47 for the inline counterpart. This superior performance is due to flow disruption and frequent vortex formation caused by the offset pin arrangement, which promotes stronger mixing and greater surface exposure.

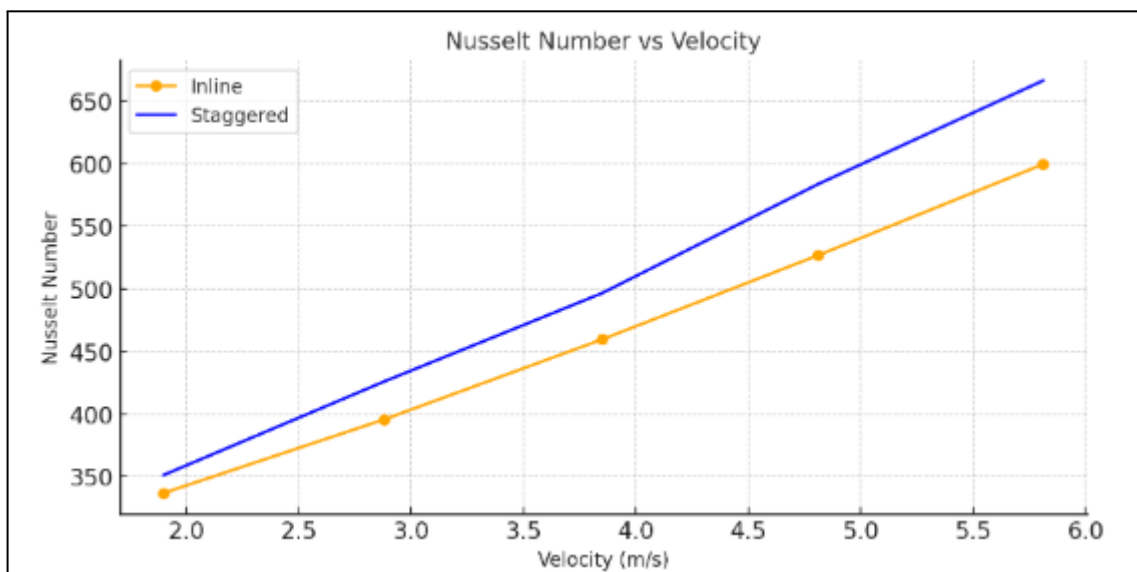


Figure 3.28: Nusselt Number vs change in velocity

On the other hand, the inline configuration yields more streamlined flow, resulting in relatively lower convective enhancement. While it may benefit from reduced drag, it does so at the cost of heat transfer efficiency.

3.6.2.3.1.4 HTC/PRESSURE DIFFERENCE

Figure 3.29 illustrates the variation of the $h/\Delta P$ ratio with increasing velocity for both inline and staggered square pin fin arrangements. The $h/\Delta P$ metric serves as an indicator of thermal-hydraulic performance, balancing heat transfer enhancement against the cost of pressure loss.

It is clearly observed that the inline configuration significantly outperforms the staggered one across the entire velocity range, with values dropping from 18.13 to 4.73 $\text{W/m}^2\text{K}\cdot\text{Pa}$ as velocity increases. In contrast, the staggered layout starts from 10.49 and drops sharply to 2.32 $\text{W/m}^2\text{K}\cdot\text{Pa}$, indicating poorer efficiency under hydraulic constraints.

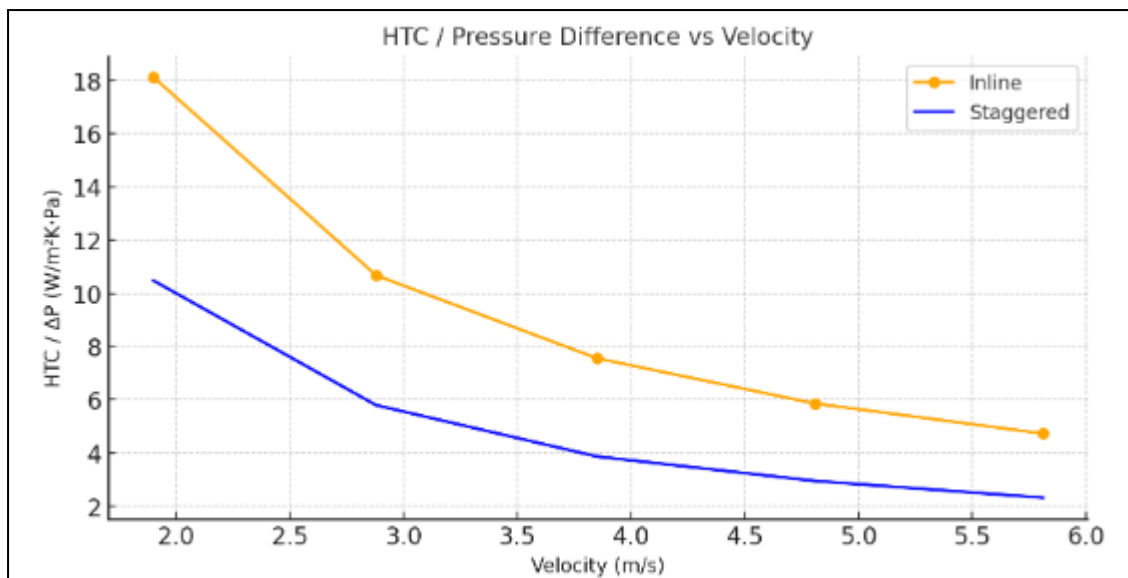


Figure 3.29: HTC/pressure difference vs change in velocity

The higher obstruction and more chaotic flow paths in the staggered array, while beneficial for heat transfer, result in substantially higher pressure drops. As a result, despite slightly better thermal metrics, its thermal-hydraulic efficiency suffers.

3.6.2.3.2 VARYING HEIGHT

3.6.2.3.2.1 AVERAGE TEMPERATURE

Figure 3.30 illustrates the relationship between fin height and average temperature for square pin fin heat sinks under forced convection. As fin height increases from 10 mm to 50 mm, average temperature consistently decreases for both inline and staggered configurations, indicating improved heat dissipation with taller fins due to greater surface area.

For the staggered configuration, the average temperature reduces from 455.58 K at 10 mm to 349.14 K at 50 mm. The staggered arrangement offers improved fluid mixing and more turbulence, especially in the wake regions behind the fins, which contributes to higher convective heat transfer. This effective disruption of thermal boundary layers allows staggered fins to maintain a slightly better thermal performance across all fin heights compared to inline fins.

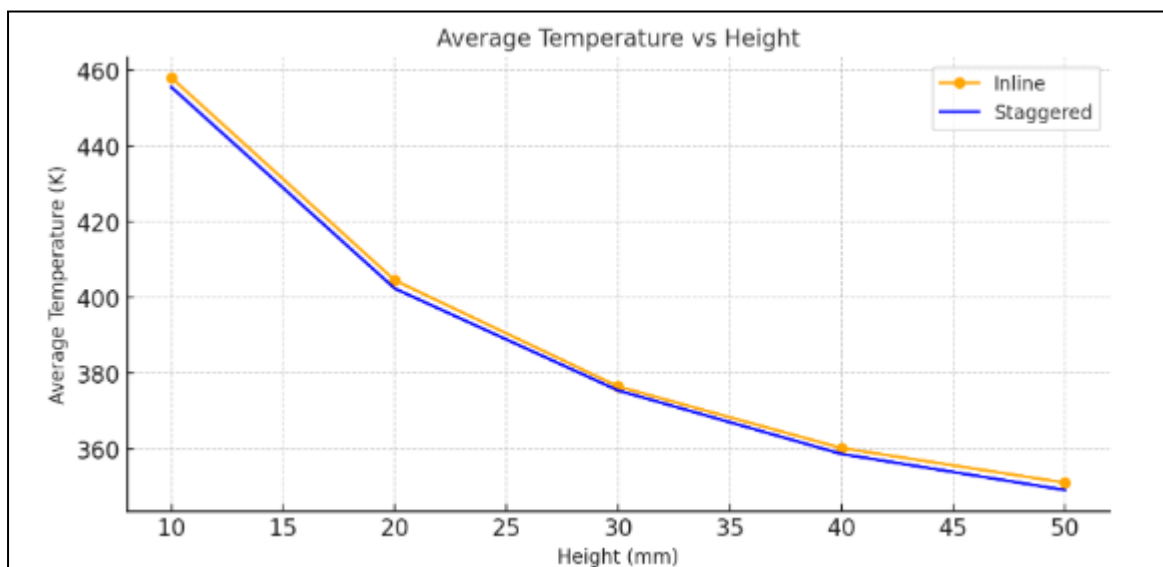


Figure 3.30: Average temperature vs change in height

For the inline configuration, the temperature decreases from 458.09 K to 351.09 K. Although the trend is similar, temperatures remain slightly higher than the staggered case due to less effective mixing and shadowing of downstream fins.

3.6.2.3.2.2 PRESSURE DIFFERENCE

Figure 3.31 illustrates the variation of pressure difference with respect to fin height for square pin fin heat sinks in both staggered and inline arrangements.

As the height of the pin fins increases, the pressure difference shows a rising trend for both configurations. This is expected, as taller fins create a larger obstruction to airflow, thereby increasing the flow resistance and resulting in a higher pressure drop.

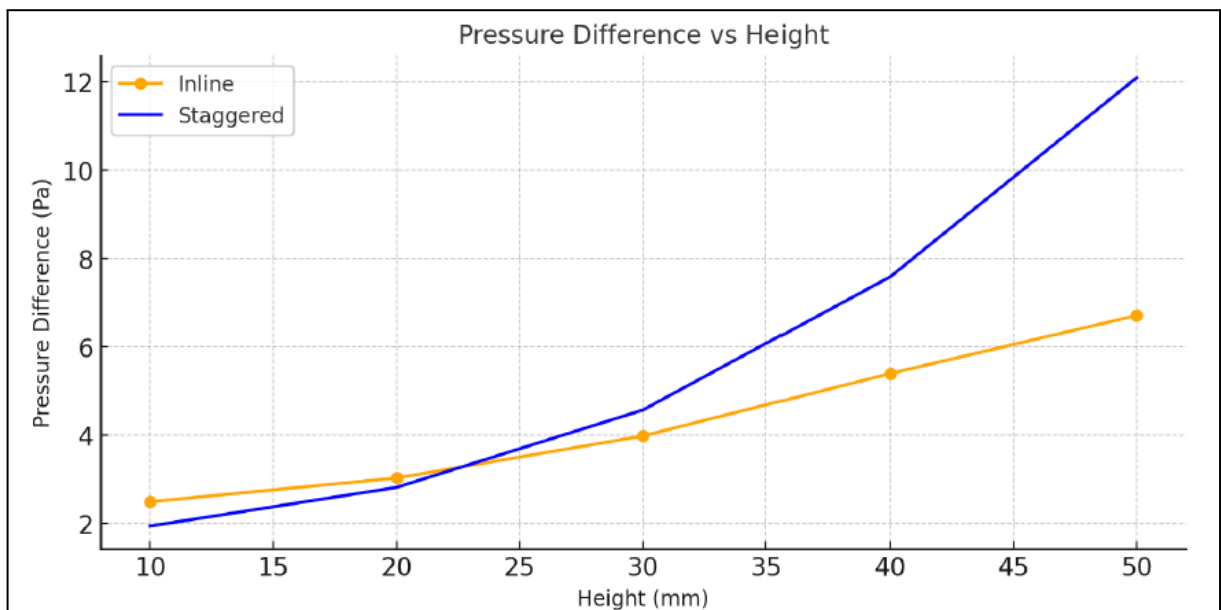


Figure 3.31: Pressure difference vs change in height

The staggered arrangement consistently exhibits a significantly higher pressure difference than the inline one. This is attributed to the increased turbulence and more frequent flow separations caused by the misaligned path of air in the staggered configuration. In contrast, the inline arrangement allows for a more streamlined flow, hence lower pressure losses.

3.6.2.3.2.3 NUSSELT NUMBER

Figure 3.32 illustrates the variation of Nusselt number with fin height for square pin fin heat sinks. As the fin height increases from 10 mm to 50 mm, the Nusselt number shows a clear linear rise, indicating improved convective heat transfer performance with taller fins.

In the staggered configuration, the Nusselt number increases from 107.32 to 351.15. This consistent growth reflects the enhanced turbulence and surface area contribution from staggered arrangements, which disrupt the thermal boundary layer more effectively.

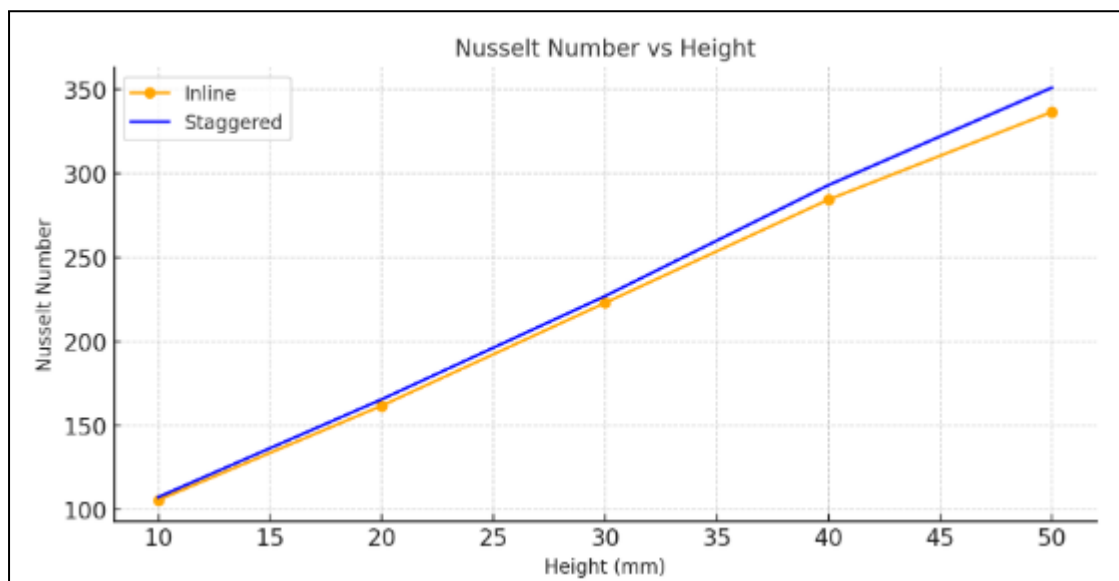


Figure 3.32: Nusselt number vs change in height

For the inline configuration, a similar trend is observed, with values increasing from 105.54 to 336.73. While the overall enhancement is good, the staggered layout still marginally outperforms the inline design due to better mixing and reduced thermal resistance.

3.6.2.3.2.4 HTC/PRESSURE DIFFERENCE

Figure 3.33 illustrates the variation of heat transfer coefficient to pressure drop ratio ($HTC/\Delta P$) with increasing fin height for square pin fin heat sinks. This performance metric is critical in evaluating the thermal effectiveness relative to the hydraulic penalty. An optimal fin height ideally provides a balance between high heat transfer and low pressure loss.

For the staggered configuration, the $HTC/\Delta P$ initially increases, peaking at 20 mm height with a value of 21.29 $W/m^2K \cdot Pa$, and then steadily declines to 10.49 at 50 mm. This trend suggests that after a certain point, increased fin height disproportionately increases pressure drop without corresponding thermal benefits, reducing the system's thermohydraulic efficiency.

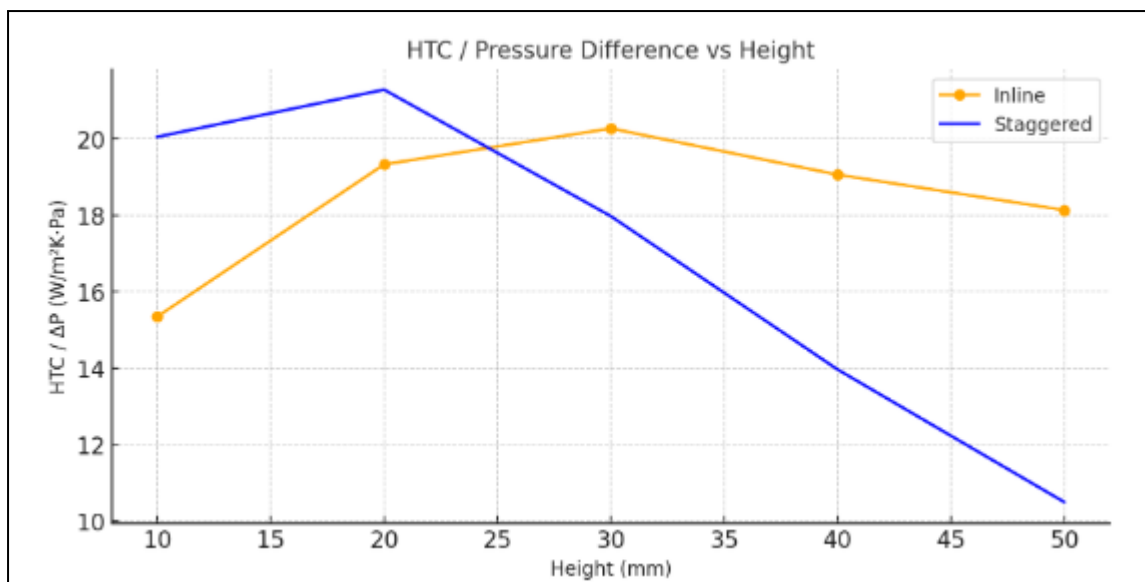


Figure 3.33: $HTC/\text{pressure difference}$ vs change in height

In the inline configuration, the $HTC/\Delta P$ shows a more stable and slightly parabolic rise and fall, peaking at 30 mm height (20.27), before tapering down to 18.13 at 50 mm. Inline fins maintain higher $HTC/\Delta P$ at greater heights compared to staggered ones, indicating better energy efficiency in terms of cooling per unit pressure penalty at taller fin configurations.

3.6.2.3.3 VARYING EDGE LENGTH

3.6.2.3.3.1 AVERAGE TEMPERATURE

Figure 3.34 illustrates the variation of average temperature with respect to edge length for both inline and staggered square pin fin configurations. As the edge length increases from 8 mm to 20 mm, there is a noticeable reduction in average temperature across both layouts, indicating enhanced thermal performance with increasing fin size.

In the staggered configuration, the average temperature consistently drops, reaching 314.26 K at 20 mm edge length from 349.14 K at 8 mm. This is due to improved fluid mixing and turbulence between offset pins, which enhances convective heat transfer and improves the fin's ability to pull heat away from the surface.

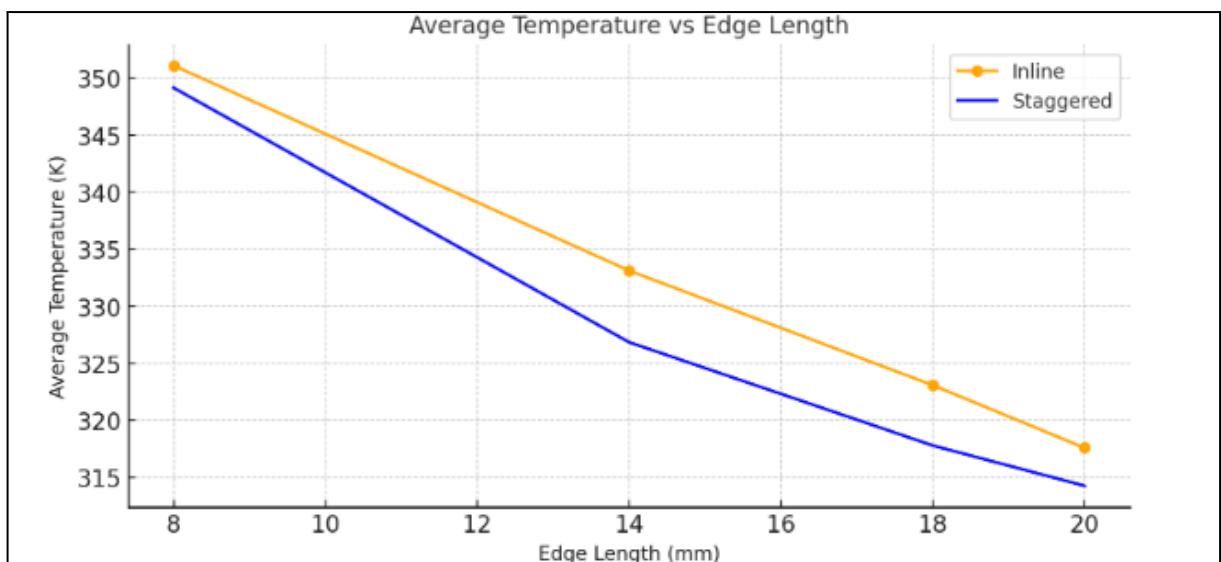


Figure 3.34: Average temperature vs change in edge length

In the inline configuration, a similar downward trend is observed, though slightly less effective than staggered. The temperature drops from 351.09 K to 317.58 K across the same range. The reduced performance compared to staggered is due to more streamlined flow paths and lower turbulence, resulting in slightly less convective interaction.

3.6.2.3.3.2 PRESSURE DIFFERENCE

Figure 3.35 illustrates the variation of pressure difference with edge length for square pin fin heat sinks in both staggered and inline configurations.

In the staggered configuration, pressure drop increases rapidly with increasing edge length, showing a strong nonlinear trend. This can be attributed to increased blockage and turbulence due to the tighter pin arrangement, which significantly restricts airflow and leads to higher form drag and recirculation.

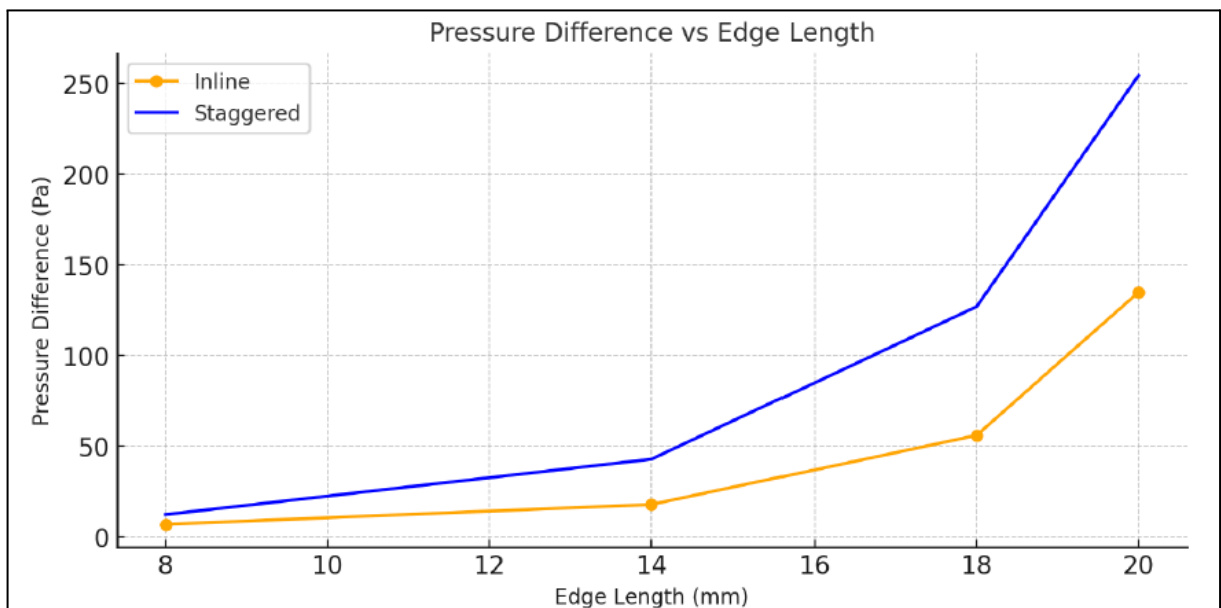


Figure 3.35: Pressure difference vs change in edge length

In the inline configuration, although the pressure drop also increases with edge length, the rise is relatively more gradual than staggered. The inline layout offers clearer flow paths at lower fin densities, reducing the aerodynamic resistance compared to the staggered setup at the same fin dimensions.

3.6.2.3.3.3 NUSSELT NUMBER

Figure 3.36 illustrates how the Nusselt number varies with increasing edge length for both staggered and inline fin arrangements. A consistent rise in Nusselt number is observed as the edge length increases from 8 mm to 20 mm, signifying enhanced convective heat transfer due to a larger surface area and increased flow obstruction.

In the staggered configuration, the Nusselt number increases sharply from 351.15 at 8 mm to 1285.97 at 20 mm. This substantial growth indicates superior heat transfer performance, primarily due to enhanced flow turbulence and better fluid mixing around the pins caused by the staggered arrangement.

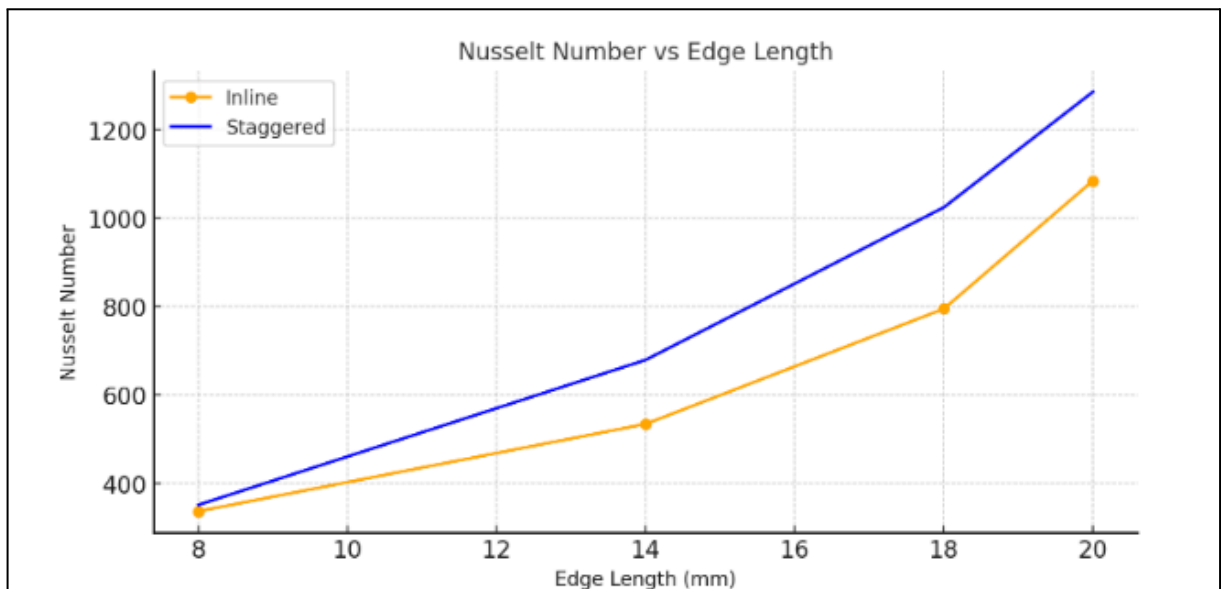


Figure 3.36: Nusselt number vs change in edge length

In the inline configuration, the Nusselt number also increases from 336.73 to 1084.85 over the same edge length range. Although the trend is similar, the values are consistently lower than those of the staggered layout, highlighting the relatively reduced convective interaction in inline fins due to more streamlined airflow paths.

3.6.2.3.4 HTC/PRESSURE DIFFERENCE

Figure 3.37 illustrates the variation of the performance ratio ($HTC/\Delta P$) with edge length for both inline and staggered square pin fin configurations. This metric provides a comparative assessment of thermal gain per unit of pressure drop, hence indicating the thermal-hydraulic performance efficiency.

In the staggered configuration, the $HTC/\Delta P$ ratio decreases steadily from $10.49 \text{ W/m}^2\text{K}\cdot\text{Pa}$ at 8 mm edge length to $1.83 \text{ W/m}^2\text{K}\cdot\text{Pa}$ at 20 mm. This drop suggests that although HTC increases with edge length, the corresponding pressure drop increases more aggressively—thereby diminishing efficiency at higher edge lengths.

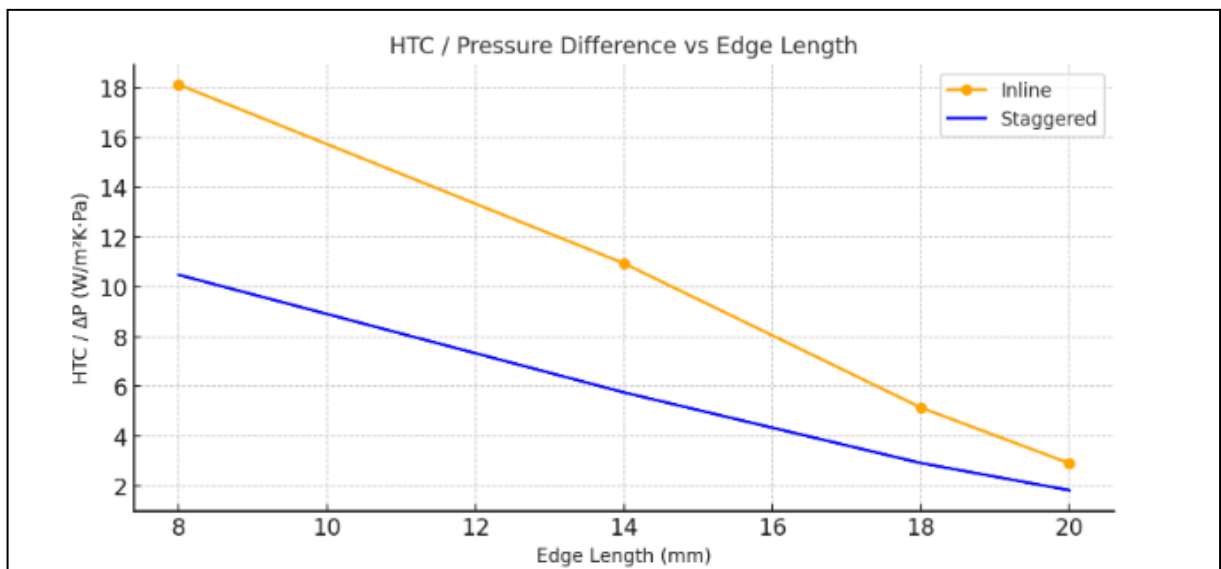


Figure 3.37: $HTC/\text{pressure difference}$ vs change in edge length

In the inline configuration, a similar declining trend is observed, with values falling from 18.13 to $2.91 \text{ W/m}^2\text{K}\cdot\text{Pa}$. The higher initial values compared to staggered indicate better thermal performance per unit pressure drop at smaller edge lengths. However, the steep decline implies that at larger edge lengths, inline configurations too suffer efficiency loss due to sharp increase in flow resistance.

3.6.2.4. Interpretation of Velocity Streamlines: Inline vs Staggered Pin Fin Arrays

The streamline plots (Figures 3.38 and 3.39) represent the velocity field within the square pin fin heat sink under identical inlet conditions but with two distinct fin arrangements: staggered and inline.

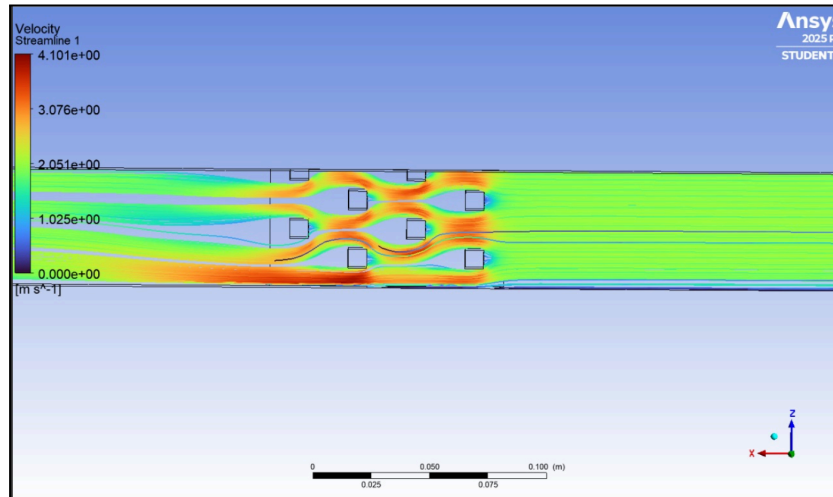


Figure 3.38 : Staggered orientation

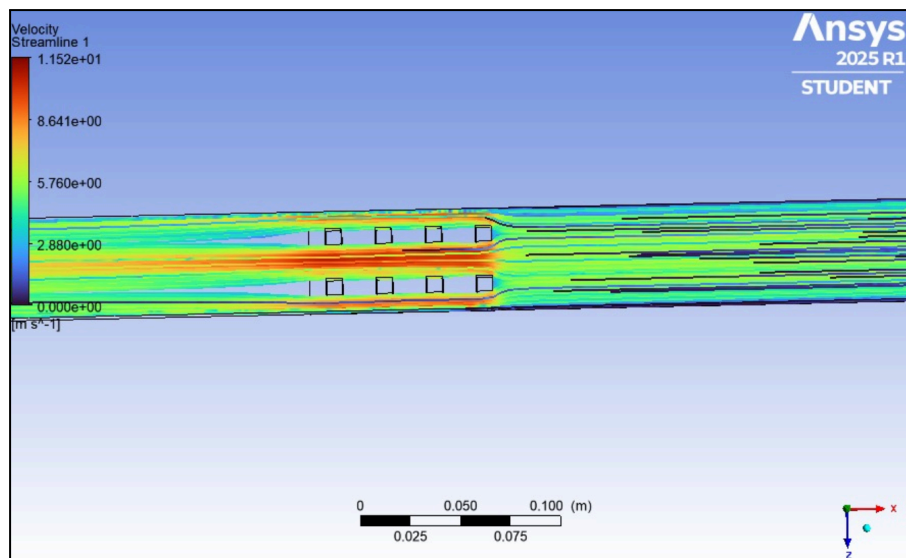


Figure 3.39 : Inline Orientation

In the staggered configuration (Figure 3.32), the flow is continuously disturbed as the square pins are offset row by row. Due to the sharp corners of the square pins, the air experiences abrupt flow separation at the edges, forming larger wake regions behind each pin. This strong

disruption generates greater turbulence and improves fluid mixing, which enhances convective heat transfer.

However, the frequent flow redirection and separation also cause a higher pressure drop across the heat sink. The sharp edges increase drag forces compared to circular pins, making staggered square arrays more hydraulically demanding.

In the inline configuration (Figure 3.33), the pins are aligned in straight rows, creating smoother channels for the airflow. The flow experiences fewer abrupt separations and maintains a more streamlined path through the gaps. This results in a lower pressure drop compared to the staggered case.

However, the more organized flow leads to less mixing, and although the square shape still causes some boundary layer thinning, the overall heat transfer performance is slightly reduced compared to the staggered arrangement.

The presence of sharp edges in square pins, compared to rounded pins, increases local turbulence but also raises flow resistance. Sharp corners act as natural separation points, leading to higher wake formation and stronger secondary flows, especially in the staggered configuration.

CHAPTER 4: METHODOLOGY ON RECTANGULAR HEAT SINKS

Adhikari^[19] analyzed the characteristics of a plate fin heat sink, under forced convection, turbulent conditions. The research paper aimed to describe the performance of a rectangular heat sink under turbulent flows. The numerical simulations validated against experimental values, giving them more credibility. Nusselt number was a parameter, which was compared with correlational equations. The simulation was shown to be in agreement with existing literature and the experimental setup and achieved its objective, thus making it a suitable validation case study^[19].

PROBLEM STATEMENT OF GIVEN PAPER

The heat sink in the given paper was analyzed for forced convection, with Reynold's number lying in the range of 2600 to 6800. Nusselt number was studied for the corresponding tested Reynold's numbers. The dimensions of the heat sink and enclosure used in the paper are -

Parameters	Values
1. Base Plate	75 * 42 mm ²
2. Fin Thickness	2 mm
3. Channel Spacing	6 mm
4. Base Thickness	Not mentioned in paper (assumed 1 mm)

Table 4.1 : Dimensions of Rectangular heat sink

Calculation for Hydraulic Diameter as per Adhikari's paper -

$$D_h = \frac{4 * A}{P}$$

$$A = s * H (m^2)$$

$$P = s + 2 * H (m)$$

4.1 MESH CONVERGENCE

A mesh convergence study was conducted to ensure independence of domain on the solution obtained for the heat sink. Meshing was done in ANSYS Workbench and a graph for maximum heat sink temperature versus number of elements was plotted.

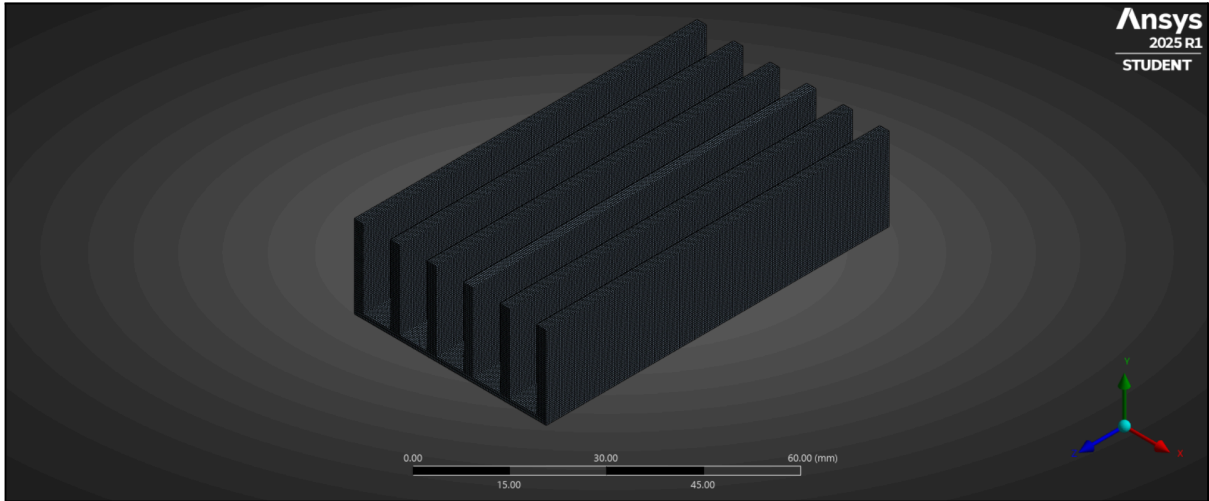


Figure 4.4 : Heat Sink Mesh

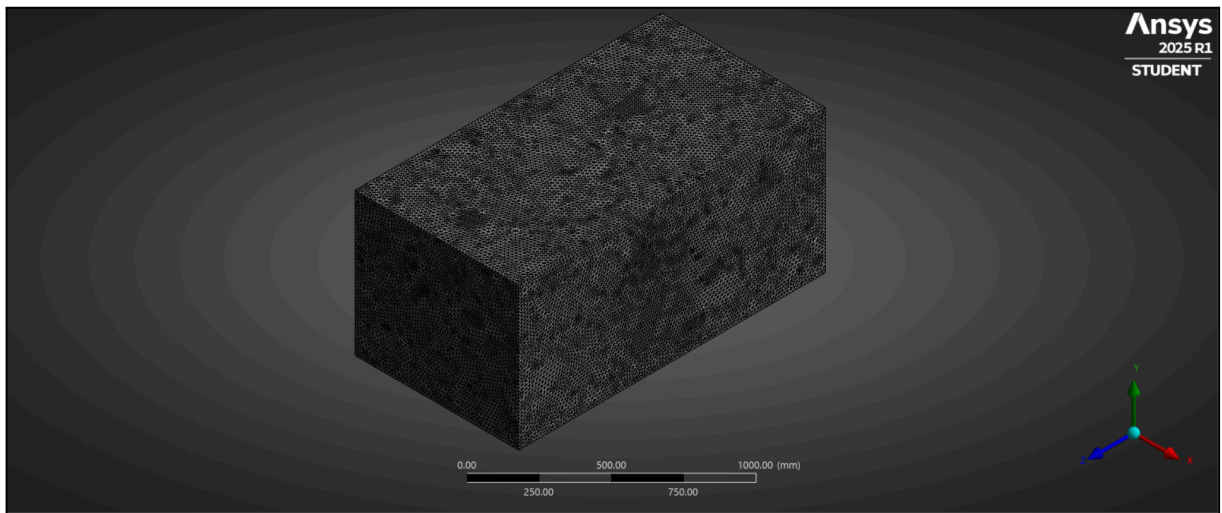


Figure 4.5 : Domain Mesh

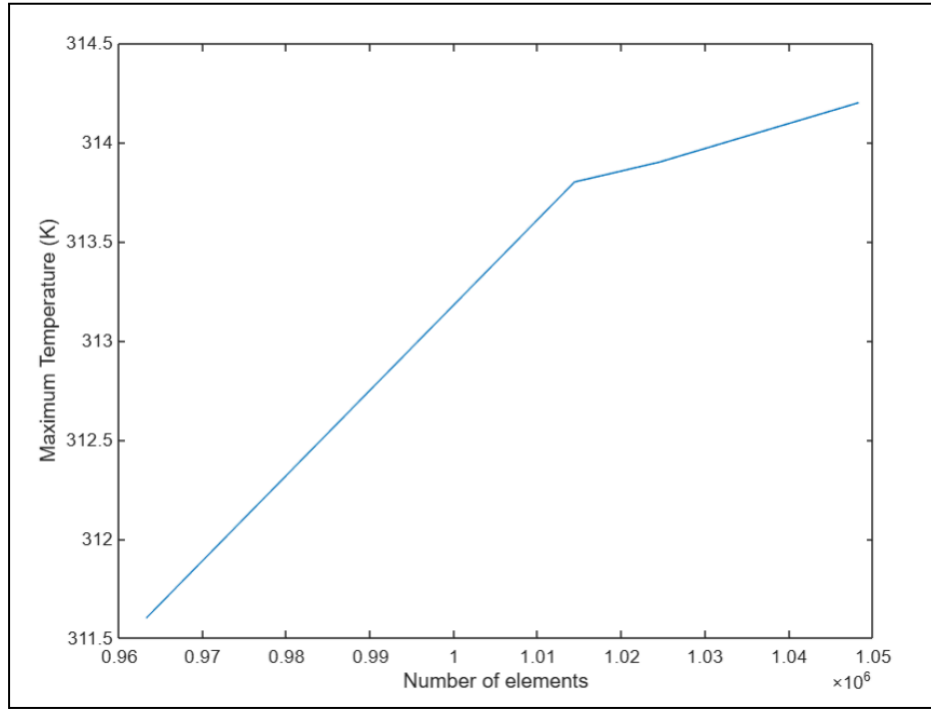


Figure 4.6 : Mesh convergence Plot

The convergence was stopped, as soon as consecutive solutions displayed a difference of 0.64%, thus ensuring domain independence.

4.2 SOLVER SETUP

ANSYS Fluent was used for solving the governing energy, momentum and viscous model equations. For implementation of viscosity, the SST k-omega (2 equation) model was used in conjunction with the governing equations (same as that for pin fin heat sinks).

For turbulence kinetic energy -

$$\frac{D}{Dt}(\rho k) = \nabla \cdot (\rho D_k \nabla k) + \rho G - \frac{2}{3} \rho k (\nabla \cdot \mathbf{u}) - \rho \beta^* \omega k + S_k$$

For turbulence specific dissipation rate -

$$\frac{D}{Dt}(\rho \omega) = \nabla \cdot (\rho D_k \nabla \omega) + \frac{\rho \gamma G}{\nu} - \frac{2}{3} \rho \gamma \omega (\nabla \cdot \mathbf{u}) - \rho \beta \omega^2 - \rho (F_1 - 1) C D_{k\omega} + S_\omega$$

Turbulence viscosity is obtained as -

$$\nu_t = a_1 \frac{k}{\max(a_1, \omega, b_1, F_{23})}$$

The constant used for analysis are -

Alpha_inf	0.52	TKE (Outer) Prandtl	1
Beta*_inf	0.09	SDR (Inner) Prandtl	2
a1	0.31	SDR (Outer) Prandtl	1.168
Beta_i (inner)	0.075	Energy Prandtl Number	0.85
Beta_i (outer)	0.0828	Wall Prandtl Number	0.85
TKE (Inner) Prandtl	1.176	Production Limiter Clip Factor	10

Table 4.2 : Value of Constants used

4.3 BOUNDARY CONDITIONS

- 1) Inlet velocity - 4 m/s
- 2) Outlet pressure - 0 Pa (Gauge)
- 3) Bottom wall for the enclosure was set to adiabatic. Side and top walls were set to free stream condition (No slip condition was applied to these walls)
- 4) Heat load of 9.67 W was applied to the heat sink base, which is equivalent to a heat flux of 3070 W/m² (for convergence analysis)
- 5) Inlet and outlet temperature was set to 20.2 degree celsius
- 6) Steady State condition is applied

The solution was run for 1000 iterations and the convergence was achieved after 180 iterations.

4.4 RESULTS

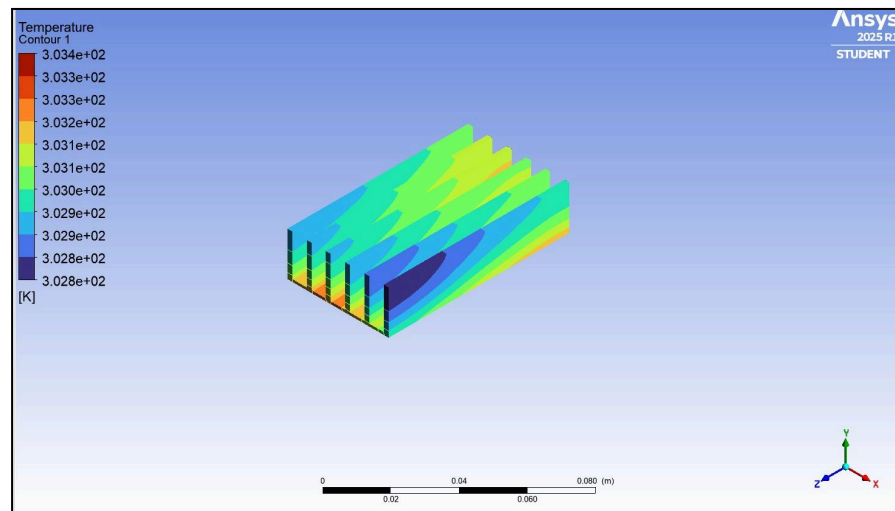


Figure 4.7 : Temperature contour plot for $Q = 4.71$ W and $v = 4$ m/s

The plots for velocity versus maximum temperature were plotted and compared with Adhikari's paper. The maximum error obtained was around 2.3%, thus verifying the numerical model from the paper.

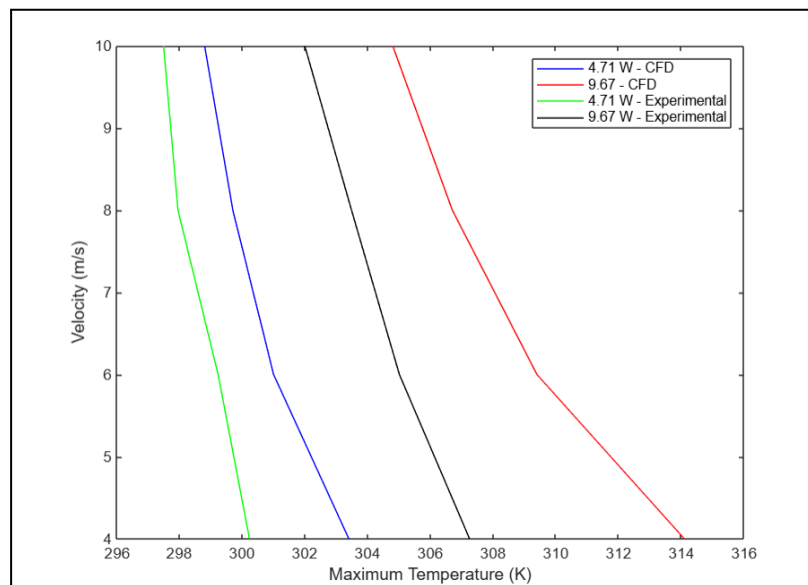


Figure 4.8: Velocity vs. Maximum Temperature

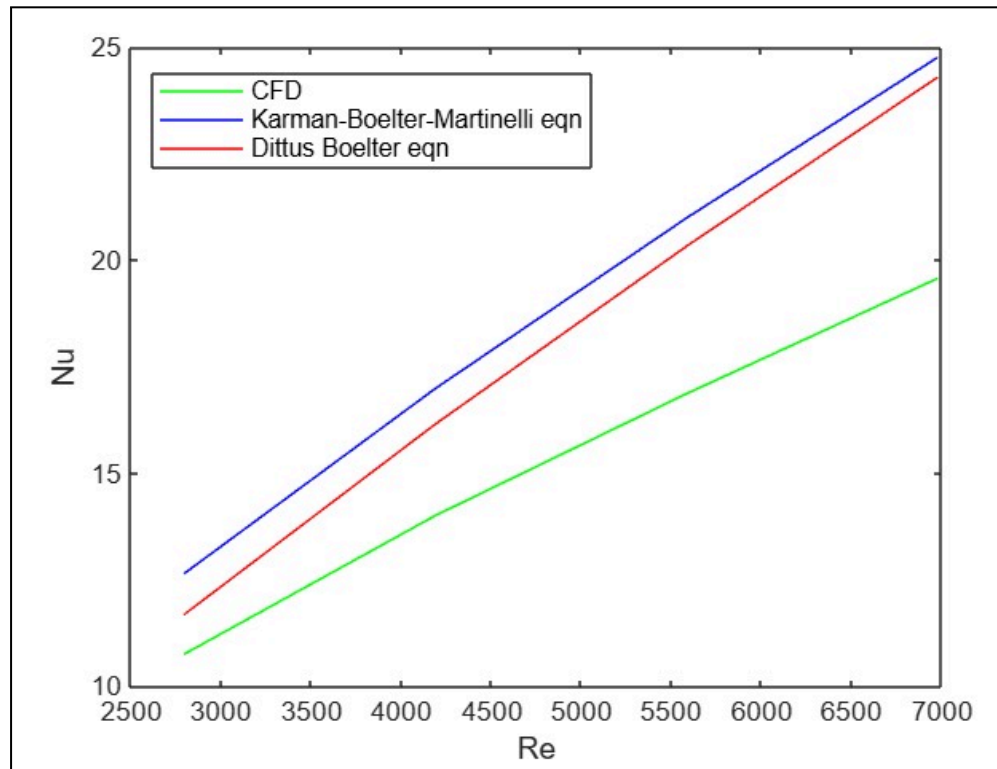


Figure 4.9: Nusselt Number vs. Re for given correlations (analysis for 4 m/s and 4.71 W)

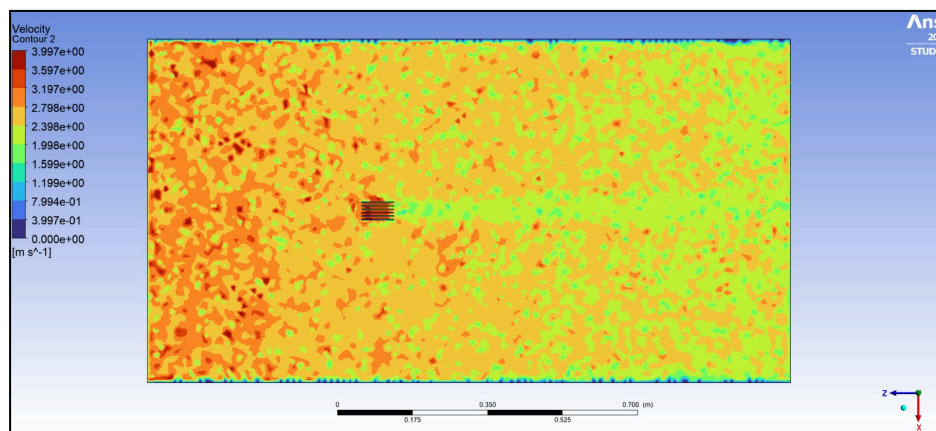


Figure 4.10: Bypass Flow

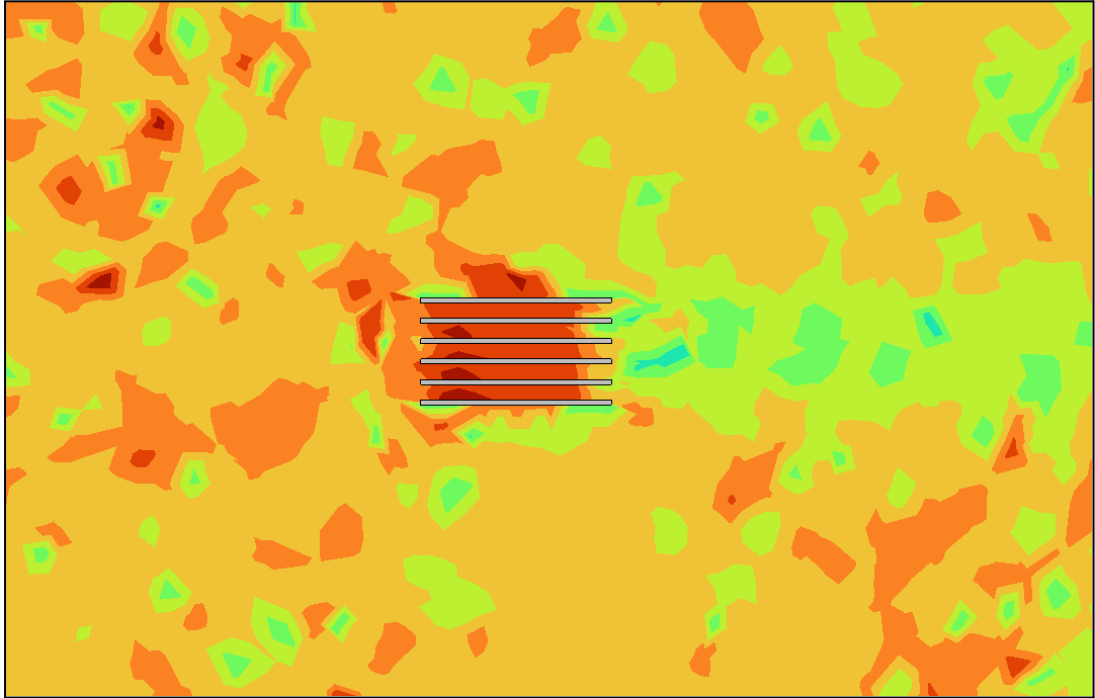


Figure 4.11 :Bypass Flow (Zoomed View)

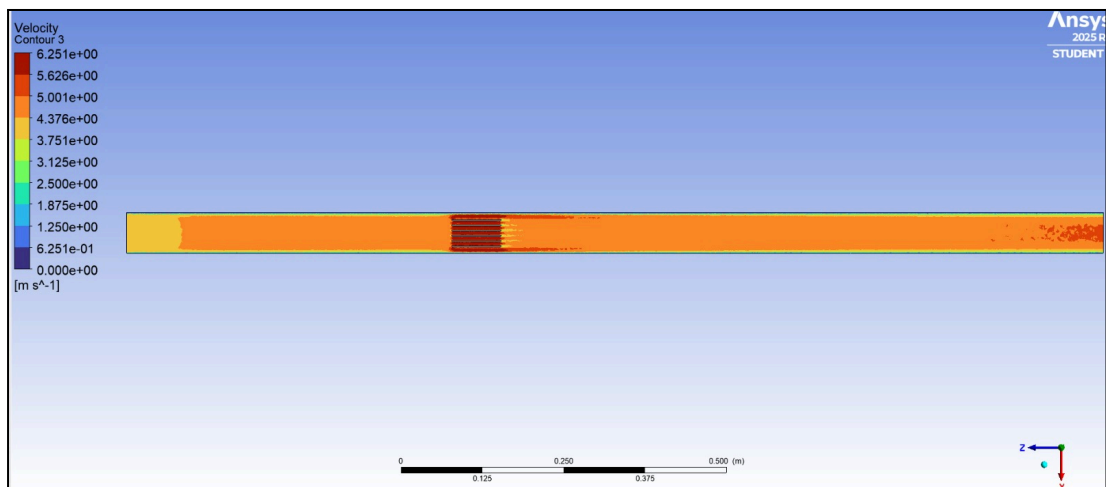


Figure 4.12 :Non - Bypass Flow

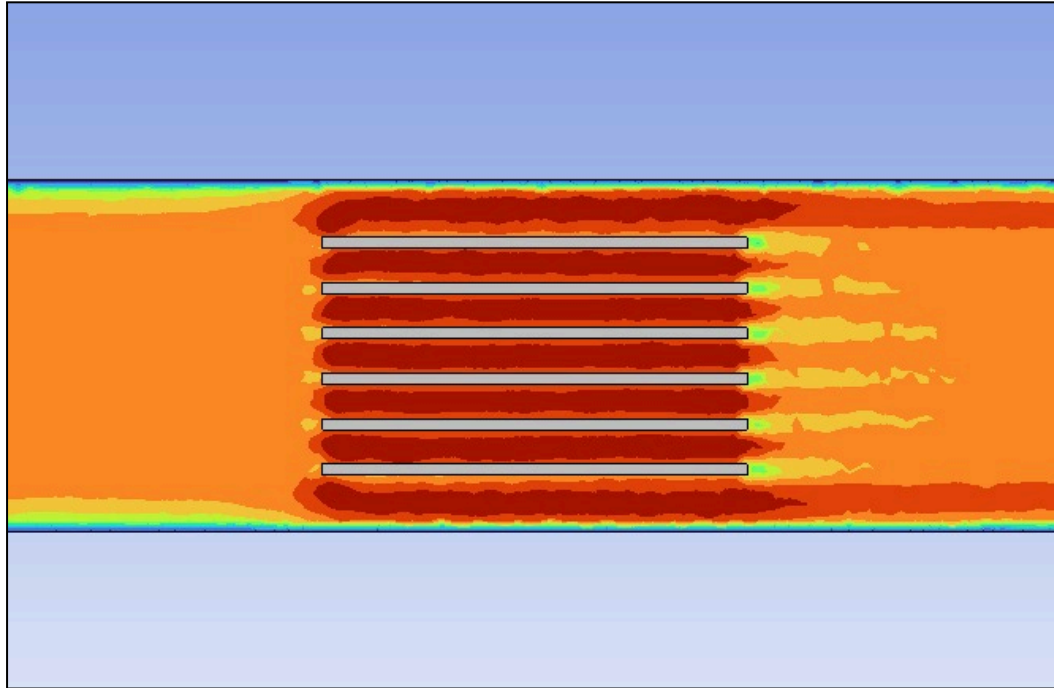


Figure 4.13: Non - Bypass Flow (Zoomed View)

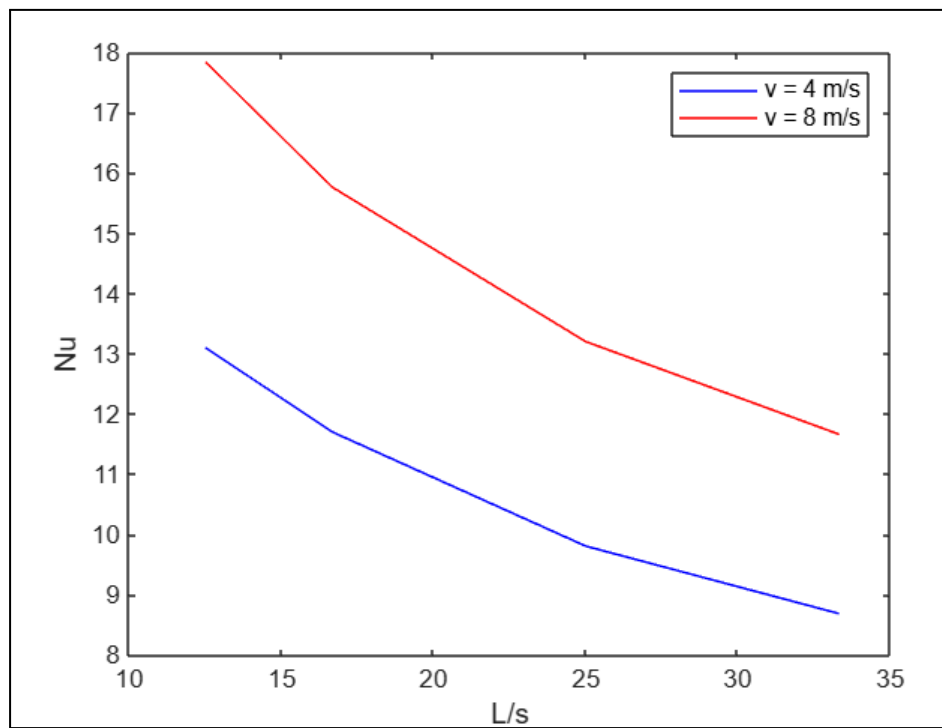


Figure 4.14 : Nusselt number was dimensionless length

Nusselt number decreases as length increases, due to fully developed flow. The fully developed flow results in development of the boundary layer between the plates and thus, obstructs the air flow between the plates, this reduces the effective heat transfer from the heat sink, a phenomenon known as choking.

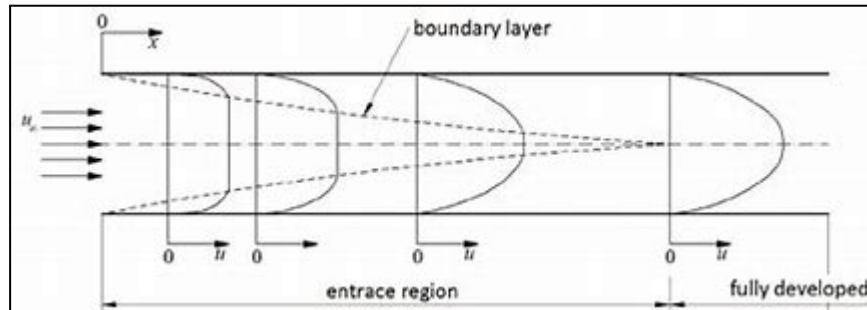


Figure 4.15 : Boundary layer development

The effect of bypass flow is to give the path of least resistance to the flow, and thus, pressure drop reduces, in the channel. A tradeoff is that the flow does not make contact with the fins effectively, thus reducing heat transfer rate. In non-bypass flow, the flow is forced through the channels by use of a constricted enclosure, thus enabling better heat transfer, while increasing pressure drop, due to sudden contraction and expansion during entry and exit of the air flow through the rectangular plates, whose effects will be considered in the correlations used in the subsequent part of the study.

The Karman-Boelter-Martinelli and Dittus Boelter equations were plotted against Reynold's numbers, calculated for the validation study.

Karman - Boelter - Martinelli equation (for circular pipes)

$$Nu = \frac{Re_d Pr \sqrt{f/8}}{0.833[5Pr + 5 \ln(5Pr + 1) + 2.5 \ln(R_d \sqrt{f/8}/60)]}$$

$$f = 0.3164/Re_d^{0.25} \text{ for } Re_d \leq 10^5$$

Dittus - Boelter equation

$$Nu = 0.023Re^{0.8}Pr^{0.4}$$

f - Friction factor

Re - Reynolds number

Pr - Prandtl Number

4.5 ANALYTICAL CORRELATIONS

Shaeri^[18], verified the use of analytical correlations for pressure drop and heat transfer coefficient for a heat sink channel using existing literature. The relations were validated using experiments and mathematical techniques. Shaeri employed an experimental setup, and validated the analytical correlations with the experimental data. The present study would verify the use of correlations for pressure drop and heat transfer coefficient, for a full heat sink. Assumptions for CFD based analysis are, for evaluating the correlations are -

- 1) Top bypass flow is obstructed
- 2) Small side bypass flow is permitted
- 3) Steady State analysis

The correlations given are as follows-

$$\Delta P = \frac{\rho U^2}{2} * [Kc + Ke + 4 * x^+ * f_{app} Re]$$

$$D_h = \frac{4 * Wch}{2 * (Wch + H)}$$

$$f_{app} Re = [(\frac{3.44}{\sqrt{x^+}})^2 + (fRe)^2]^{0.5}$$

$$Re = \frac{(\rho * U * Dh)}{\mu}$$

$$x^+ = \frac{L}{Dh * Re}$$

$$Kc = 0.42 * (1 - \epsilon)$$

$$Ke = (1 - \varepsilon)^2$$

$$\varepsilon = \frac{N*Wch*H}{W*(H+t_b)}$$

$$fRe = 24 * (1 - 1.3553 * (\frac{Wch}{H}) + 1.9467 * (\frac{Wch}{H})^2 - 1.7012 * (\frac{Wch}{H})^3 + 0.9564 * (\frac{Wch}{H})^4 - 0.2537 * (\frac{Wch}{H})^5)$$

$$h = \frac{kf}{Wch} * \frac{\tanh \sqrt{2*Nui * \frac{kf*H^2}{ks*Wch*t_f} * ((\frac{t_f}{L} + 1))}}{\sqrt{2*Nui * \frac{kf*H^2}{ks*Wch*t_f} * ((\frac{t_f}{L} + 1))}} * Nui$$

$$Nui = [(1/(Reb * Pr * 0.5)^3) + (1/(0.664 * \sqrt{Reb} * Pr^{1/3} * \sqrt{1 + (3.65/\sqrt{Reb})})^3)^{-\frac{1}{3}}]$$

$$Reb = \frac{\rho*U*Wch}{\mu} * \frac{Wch}{L}$$

$$U = V_{in} (1 + \frac{tf}{Wch})$$

$$h = \frac{Q}{(A)*(T_{avg}-T_i)}$$

A - Total area of heat sink, from which heat transfer is taking place (m²)

T_{avg} - Average base temperature (K)

T_i - Inlet air temperature (K)

U - Channel velocity (m/s)

V_{in} - Inlet velocity (m/s)

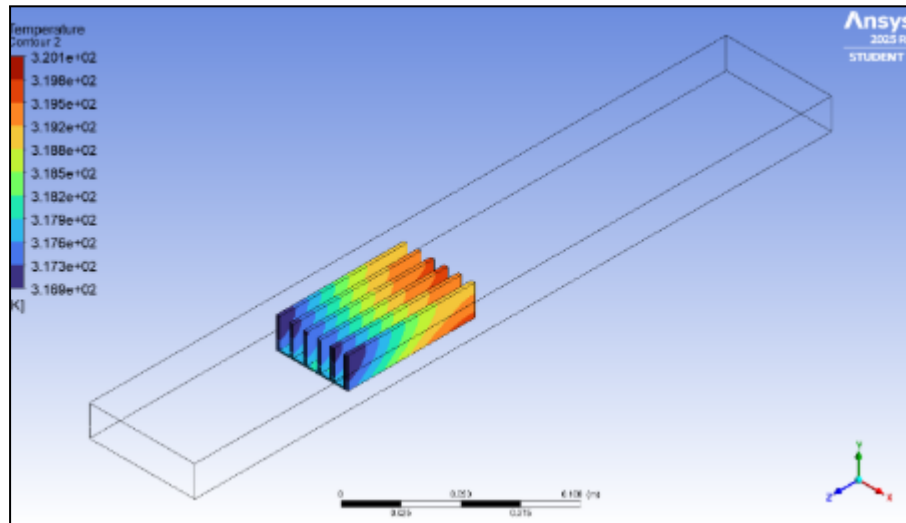


Figure 4.16 : Temperature contour plot for $Q = 9.67 \text{ W}$ and $v = 1 \text{ m/s}$

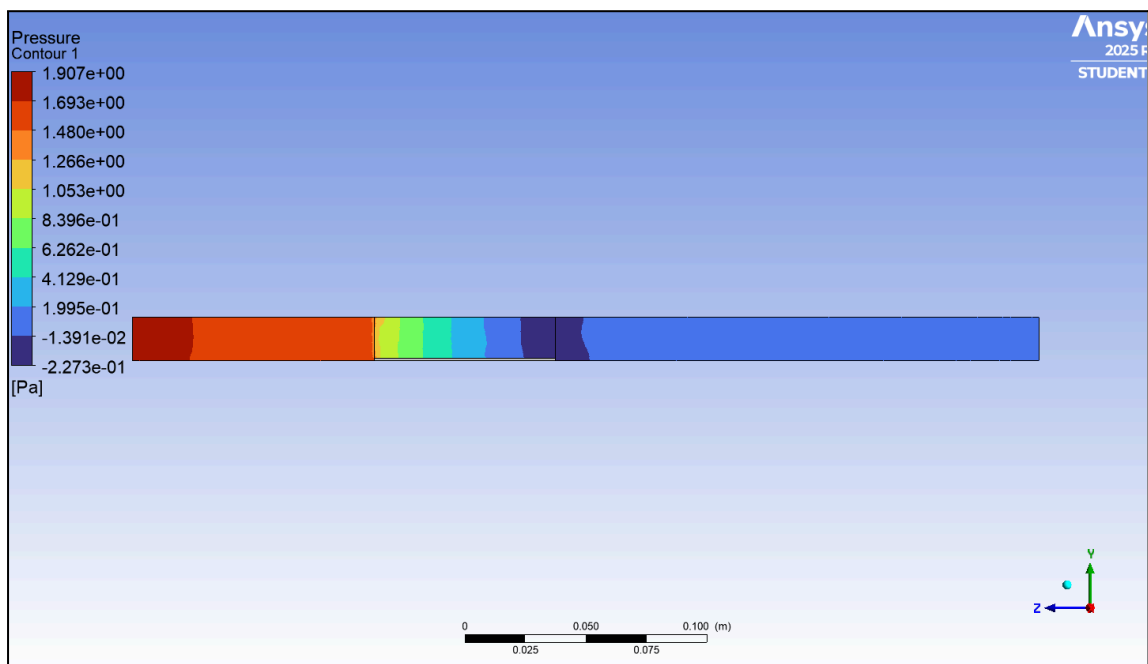


Figure 4.17 : Pressure contour plot for $Q = 9.67 \text{ W}$ and $v = 1 \text{ m/s}$

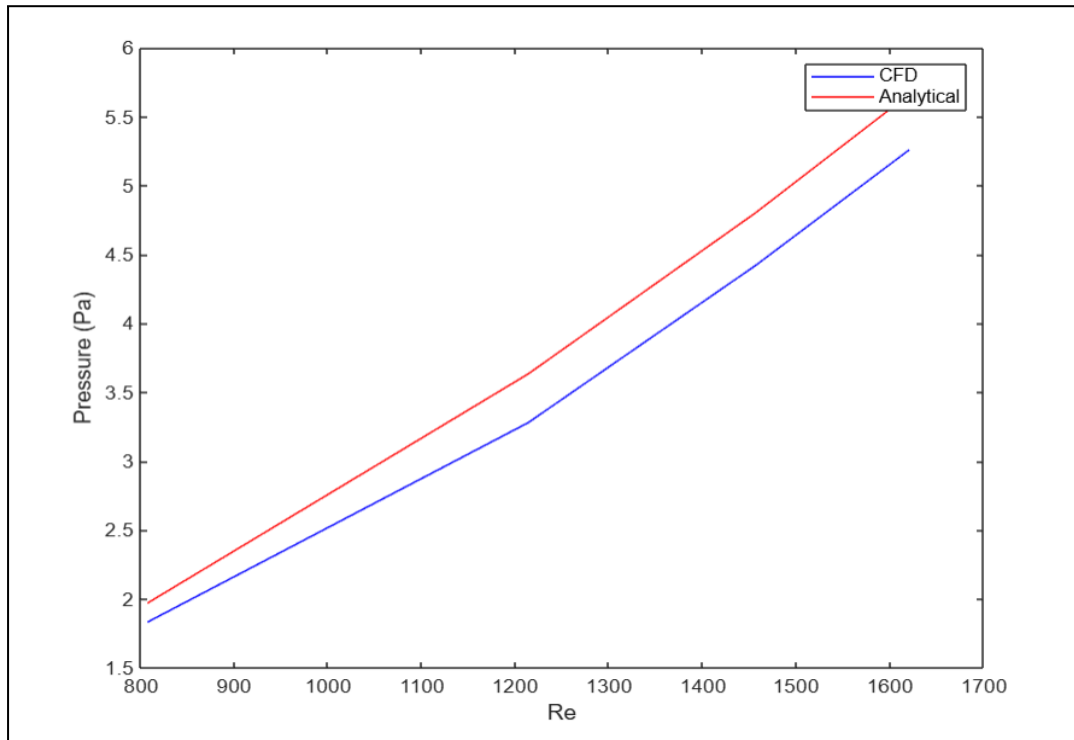


Figure 4.18 : Pressure vs. Reynolds Number

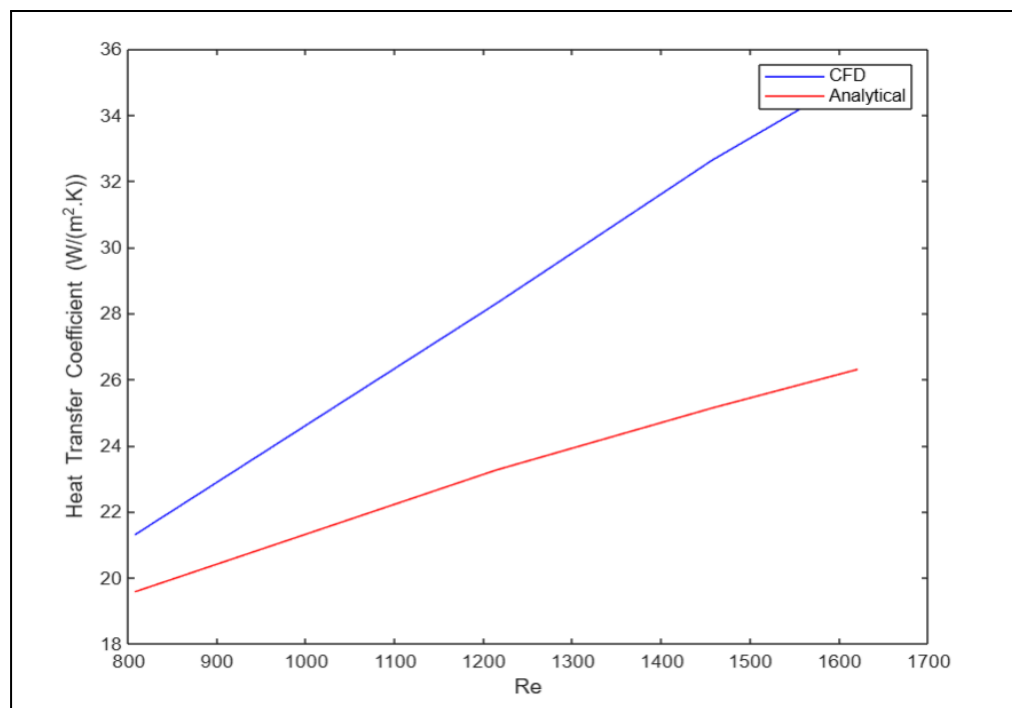


Figure 4.19 : HTC vs. Reynolds Number

Pressure plot displays a maximum error of 7.6%, while heat transfer coefficient plot displays a maximum error for 25.3%. Thus, the effective use of correlations is only at lower velocities, even in the laminar regime. Moreover, since the correlation is applicable for a single channel, it predicts the overall performance of the heat sink, relatively better, and the relationship between the heat transfer coefficient of an entire heat sink and a channel can be found out effectively using statistical or machine learning models.

CHAPTER 5: ANN AND OPTIMIZATION TECHNIQUES

5.1 PREDICTION OF PARAMETERS AND OPTIMIZATION

The parameters to be optimized for this study are the basic dimensional parameters based on objective functions obtained through analysis. An objective function is then optimized based on minimization or maximization of its value and the corresponding parameters are obtained. Assuming that we have the parameter values and the corresponding relations for the objective functions, the optimization process can be carried out feasibly.

In case they are not available, the relation between the input and output must be studied and obtained through statistical or machine learning techniques. The section discusses the use of Artificial Neural Networks and Extreme Learning Machine as a tool for predicting the relationship between input and output parameters. The objective functions for pressure drop and heat transfer coefficient are available in literature, so data acquisition would be through these equations as opposed to time consuming CFD and experimental analysis.

The main objective is to showcase the reliability of these techniques with respect to applications in numerical analysis. Neural Networks have shown great capabilities in predicting output patterns based on inputs with a non-linear relationship. The various types of neural networks are -

- 1) Artificial Neural Network - Used for regression and classification problems
- 2) Convolutional Neural Network - Image and text recognition
- 3) Recurrent Neural Network - Sentiment analysis

5.2 BASICS OF NEURAL NETWORKS

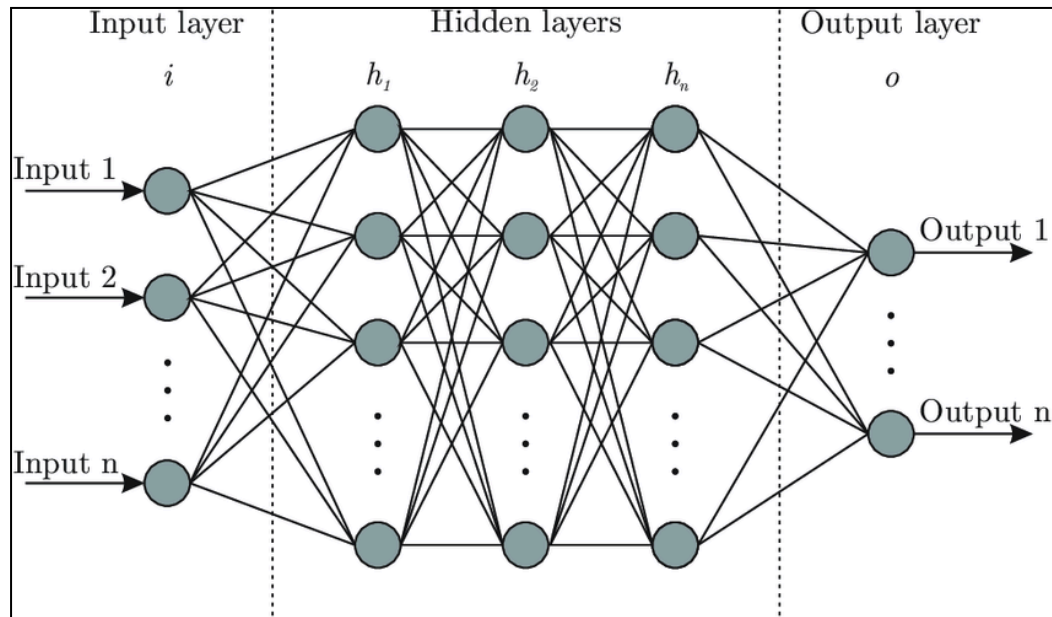


Figure 5.1 : ANN Topology

In machine learning, a neural network (also artificial neural network or neural net, abbreviated ANN or NN) is a computational model inspired by the structure and functions of biological neural networks.

A neural network consists of connected units or nodes called artificial neurons, which loosely model the neurons in the brain. Artificial neuron models that mimic biological neurons more closely have also been recently investigated and shown to significantly improve performance. These are connected by edges, which model the synapses in the brain. Each artificial neuron receives signals from connected neurons, then processes them and sends a signal to other connected neurons. The "signal" is a real number, and the output of each neuron is computed by some non-linear function of the sum of its inputs, called the activation function. The strength of the signal at each connection is determined by a weight, which adjusts during the learning process.

Typically, neurons are aggregated into layers. Different layers may perform different transformations on their inputs. Signals travel from the first layer (the input layer) to the last layer (the output layer), possibly passing through multiple intermediate layers (hidden layers). A network is typically called a deep neural network if it has at least two hidden layers.

Artificial neural networks are used for various tasks, including predictive modeling, adaptive control, and solving problems in artificial intelligence. They can learn from experience, and can derive conclusions from a complex and seemingly unrelated set of information. The use of artificial neural networks as a case study is inspired by Shaeri's^[18] paper on a similar topic.

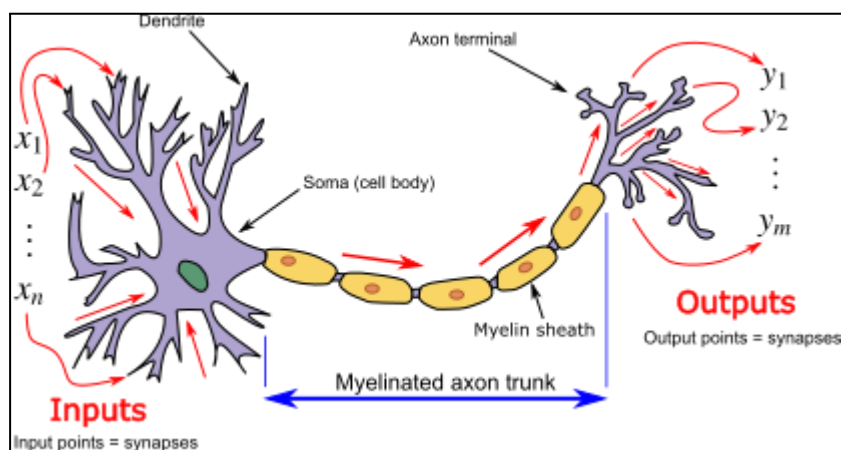


Figure 5.2 : Biological neuron

ANNs began as an attempt to exploit the architecture of the human brain to perform tasks that conventional algorithms had little success with. They soon reoriented towards improving empirical results, abandoning attempts to remain true to their biological precursors. ANNs have the ability to learn and model non-linearities and complex relationships. This is achieved by neurons being connected in various patterns, allowing the output of some neurons to become the input of others. The network forms a directed, weighted graph.

An artificial neural network consists of simulated neurons. Each neuron is connected to other nodes via links like a biological axon-synapse-dendrite connection. All the nodes connected by links take in some data and use it to perform specific operations and tasks on the data. Each link has a weight, determining the strength of one node's influence on another, allowing weights to choose the signal between neurons.

- **Artificial Neurons**

ANNs are composed of artificial neurons which are conceptually derived from biological neurons. Each artificial neuron has inputs and produces a single output which can be sent to multiple other neurons. The inputs can be the feature values of a sample of external data, such as images or documents, or they can be the outputs of other neurons. The outputs of the final output neurons of the neural net accomplish the task, such as recognizing an object in an image.

To find the output of the neuron we take the weighted sum of all the inputs, weighted by the weights of the connections from the inputs to the neuron. We add a bias term to this sum. This weighted sum is sometimes called the activation. This weighted sum is then passed through a (usually nonlinear) activation function to produce the output. The initial inputs are external data, such as images and documents. The ultimate outputs accomplish the task, such as recognizing an object in an image.

- **Organization**

The neurons are typically organized into multiple layers, especially in deep learning. Neurons of one layer connect only to neurons of the immediately preceding and immediately following layers. The layer that receives external data is the input layer. The layer that produces the ultimate result is the output layer. In between them are zero or more hidden layers. Single layer and unlayered networks are also used. Between two layers, multiple connection patterns are possible. They can be 'fully connected', with every neuron in one layer connecting to every neuron in the next layer. They can be pooling, where a group of neurons in one layer connects to a single neuron in the next layer, thereby reducing the number of neurons in that layer. Neurons with only such connections form a directed acyclic graph and are known as feedforward networks. Alternatively, networks that allow connections between neurons in the same or previous layers are known as recurrent networks.

- **Hyperparameter**

A hyperparameter is a constant parameter whose value is set before the learning process begins. The values of parameters are derived via learning. Examples of hyperparameters include learning rate, the number of hidden layers and batch size. The values of some hyperparameters can be dependent on those of other hyperparameters. For example, the size of some layers can depend on the overall number of layers.

- **Learning**

Learning is the adaptation of the network to better handle a task by considering sample observations. Learning involves adjusting the weights (and optional thresholds) of the network to improve the accuracy of the result. This is done by minimizing the observed errors. Learning is complete when examining additional observations does not usefully reduce the error rate. Even after learning, the error rate typically does not reach 0. If after learning, the error rate is too high, the network typically must be redesigned. Practically this is done by defining a cost function that is evaluated periodically during learning. As long as its output

continues to decline, learning continues. The cost is frequently defined as a statistic whose value can only be approximated. The outputs are actually numbers, so when the error is low, the difference between the output (almost certainly a cat) and the correct answer (cat) is small. Learning attempts to reduce the total of the differences across the observations. Most learning models can be viewed as a straightforward application of optimization theory and statistical estimation.

- **Learning Rate**

The learning rate defines the size of the corrective steps that the model takes to adjust for errors in each observation. A high learning rate shortens the training time, but with lower ultimate accuracy, while a lower learning rate takes longer, but with the potential for greater accuracy. Optimizations such as Quickprop are primarily aimed at speeding up error minimization, while other improvements mainly try to increase reliability. In order to avoid oscillation inside the network such as alternating connection weights, and to improve the rate of convergence, refinements use an adaptive learning rate that increases or decreases as appropriate. The concept of momentum allows the balance between the gradient and the previous change to be weighted such that the weight adjustment depends to some degree on the previous change. A momentum close to 0 emphasizes the gradient, while a value close to 1 emphasizes the last change.

- **Forward Propagation**

In Forward propagation input data moves through each layer of the neural network where each neuron applies weighted sum, adds bias, passes the result through an activation function and makes predictions. This process is crucial before backpropagation updates the weights. It determines the output of a neural network with a given set of inputs and current state of model parameters (weights and biases). Understanding this process helps in optimizing neural networks for various tasks like classification, regression and more. The basic flow of components in a forward propagation algorithm is -

1. Input Layer

- The input data is fed into the network through the input layer.
- Each feature in the input dataset represents a neuron in this layer.
- The input is usually normalized or standardized to improve model performance.

2. Hidden Layers

- The input moves through one or more hidden layers where transformations occur.
- Each neuron in the hidden layer computes a weighted sum of inputs and applies activation function to introduce non-linearity.
- Each neuron receives inputs, computes:

$$Z=WX+b$$

Where,

W is the weight matrix

X is the input vector

b is the bias term

The activation function such as ReLU or sigmoid is applied.

3. Output Layer

- The last layer in the network generates the final prediction.
- The activation function of this layer depends on the type of problem:
 - a. Softmax (for multi-class classification)
 - b. Sigmoid (for binary classification)
 - c. Linear (for regression tasks)

4. Prediction

The network produces an output based on current weights and biases.

The loss function evaluates the error by comparing predicted output with actual values.

● **Backward Propagation**

Backpropagation is a technique used in deep learning to train artificial neural networks particularly feed-forward networks. It works iteratively to adjust weights and bias to minimize the cost function. In each epoch the model adapts these parameters reducing loss by following the error gradient. Backpropagation often uses optimization algorithms like gradient descent or stochastic gradient descent. The algorithm computes the gradient using the chain rule from calculus allowing it to effectively navigate complex layers in the neural network to minimize the cost function.

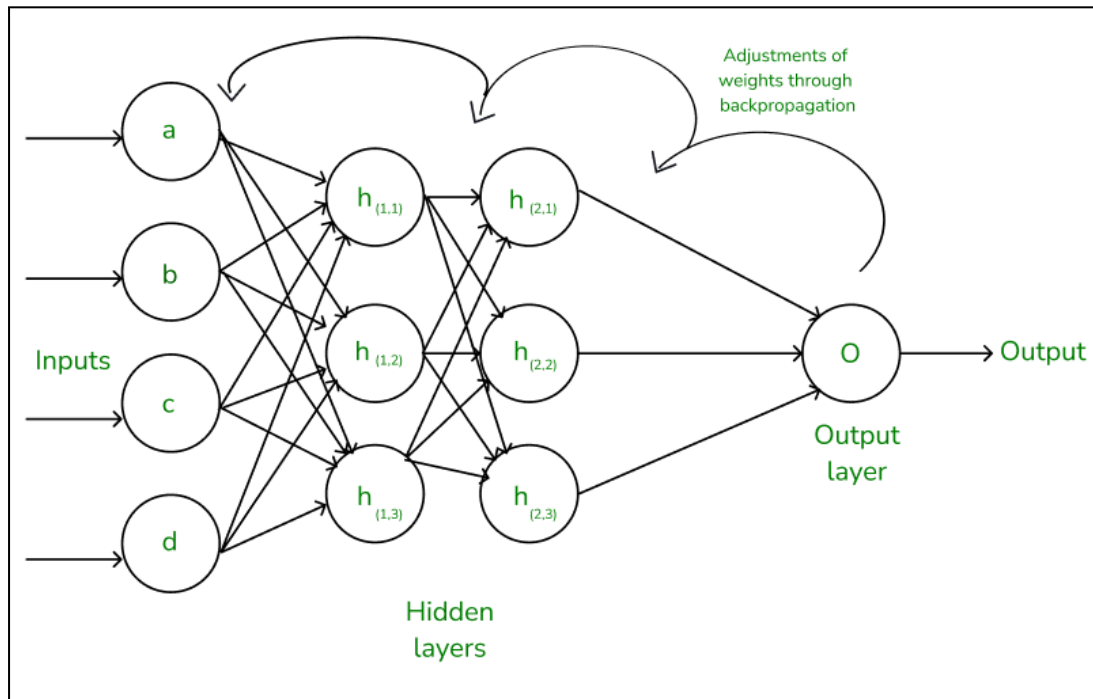


Figure 5.3 : Backpropagation algorithm

Backpropagation plays a critical role in how neural networks improve over time. Here's why:

Efficient Weight Update: It computes the gradient of the loss function with respect to each weight using the chain rule making it possible to update weights efficiently.

Scalability: The backpropagation algorithm scales well to networks with multiple layers and complex architectures making deep learning feasible.

Automated Learning: With backpropagation the learning process becomes automated and the model can adjust itself to optimize its performance.

In the backward pass the error (the difference between the predicted and actual output) is propagated back through the network to adjust the weights and biases. One common method for error calculation is the Mean Squared Error (MSE) given by:

$$\text{MSE} = (\text{Predicted Output} - \text{Actual Output})^2$$

Once the error is calculated the network adjusts weights using gradients which are computed with the chain rule. These gradients indicate how much each weight and bias should be adjusted to minimize the error in the next iteration. The backward pass continues layer by layer ensuring that the network learns and improves its performance. The activation function through its derivative plays a crucial role in computing these gradients during backpropagation.

5.3 TYPES OF LEARNING SCHEMES

- **Supervised Learning**

Supervised learning uses a set of paired inputs and desired outputs. The learning task is to produce the desired output for each input. In this case, the cost function is related to eliminating incorrect deductions. A commonly used cost is the mean-squared error, which tries to minimize the average squared error between the network's output and the desired output. Tasks suited for supervised learning are pattern recognition (also known as classification) and regression (also known as function approximation). Supervised learning is also applicable to sequential data (e.g., for handwriting, speech and gesture recognition). This can be thought of as learning with a "teacher", in the form of a function that provides continuous feedback on the quality of solutions obtained thus far.

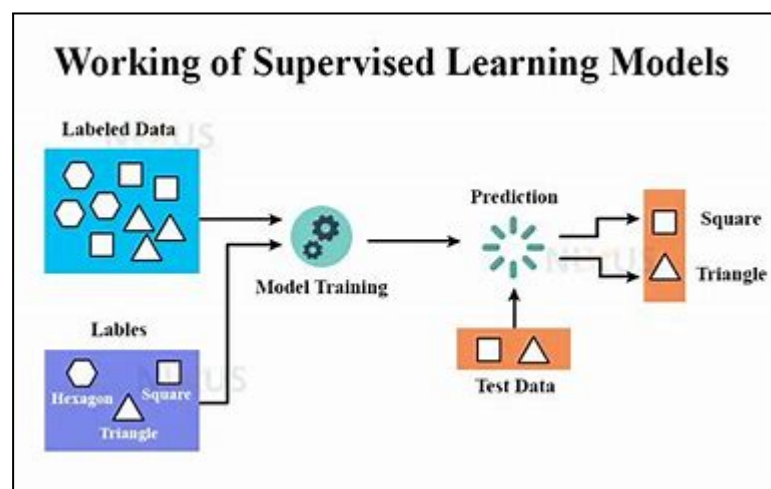


Figure 5.4 : Supervised Learning

- **Unsupervised Learning**

Unsupervised learning is a branch of machine learning that deals with unlabeled data. Unlike supervised learning, where the data is labeled with a specific category or outcome, unsupervised learning algorithms are tasked with finding patterns and relationships within the data without any prior knowledge of the data's meaning. These algorithms find hidden patterns and data without any human intervention, meaning the model only has input parameter values and discovers the groups or patterns on its own.

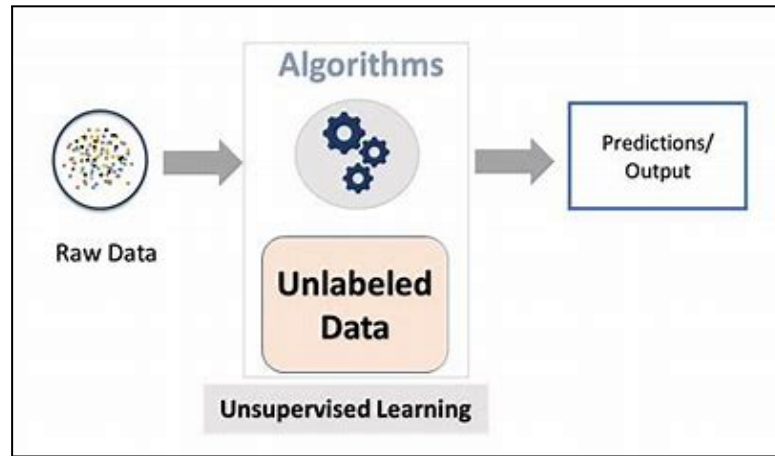


Figure 5.5 : Unsupervised Learning

- **Reinforcement Learning**

Reinforcement Learning (RL) is a branch of machine learning that focuses on how agents can learn to make decisions through trial and error to maximize cumulative rewards. RL allows machines to learn by interacting with an environment and receiving feedback based on their actions. This feedback comes in the form of rewards or penalties.

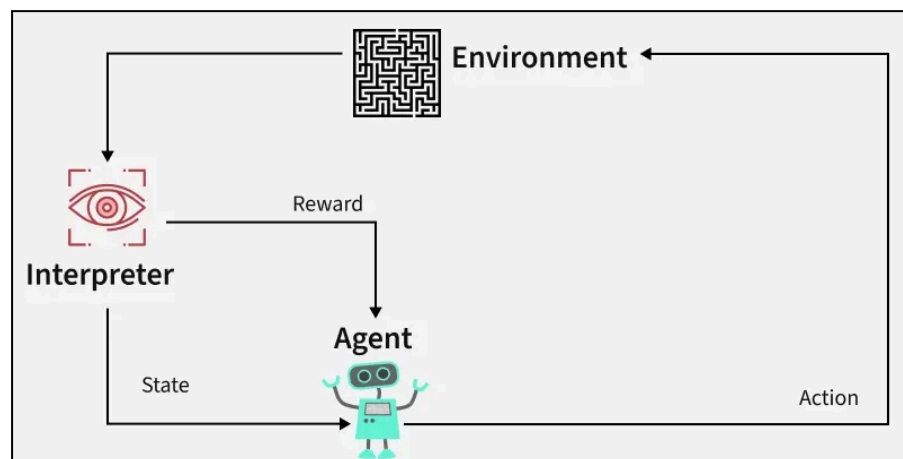


Figure 5.6 : Reinforcement Learning

Reinforcement Learning revolves around the idea that an agent (the learner or decision-maker) interacts with an environment to achieve a goal. The agent performs actions and receives feedback to optimize its decision-making over time.

Agent: The decision-maker that performs actions.

Environment: The world or system in which the agent operates.

State: The situation or condition the agent is currently in.

Action: The possible moves or decisions the agent can make.

Reward: The feedback or result from the environment based on the agent's action.

The present study uses supervised learning with known inputs and outputs and the data is divided into training, testing and validation data. Keras library is used to program an artificial neural network in Python, along with Pandas and NumPy as data processing libraries and Matplotlib as the graphical visualization library.

5.4 ACTIVATION FUNCTIONS

- **ReLU**

Rectified Linear Unit (ReLU) is a popular activation function used in neural networks, especially in deep learning models. It has become the default choice in many architectures due to its simplicity and efficiency. The ReLU function is a piecewise linear function that outputs the input directly if it is positive; otherwise, it outputs zero.

In simpler terms, ReLU allows positive values to pass through unchanged while setting all negative values to zero. This helps the neural network maintain the necessary complexity to learn patterns while avoiding some of the pitfalls associated with other activation functions, like the vanishing gradient problem.

The ReLU function can be described mathematically as follows:

$$f(x)=\max(0,x)$$

Where,

x is the input to the neuron.

The function returns x if x is greater than 0.

If x is less than or equal to 0, the function returns 0.

The formula can also be written as:

$$f(x)=\{x \text{ if } x>0$$

$$f(x)=\{0 \text{ if } x<0$$

This simplicity is what makes ReLU so effective in training deep neural networks, as it helps to maintain non-linearity without complicated transformations, allowing models to learn more efficiently.

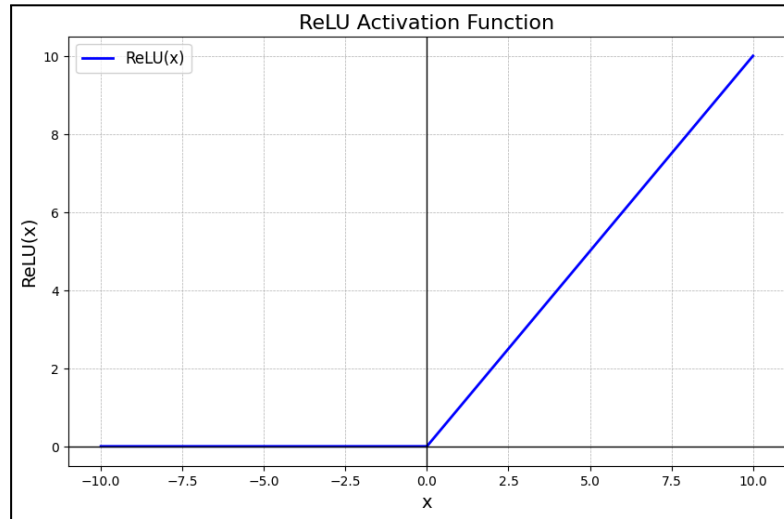


Figure 5.7 : ReLU function

- **Sigmoid**

Sigmoid is a mathematical function that maps any real-valued number into a value between 0 and 1. Its characteristic “S”-shaped curve makes it particularly useful in scenarios where we need to convert outputs into probabilities. This function is often called the logistic function.

Mathematically, sigmoid is represented as:

$$\sigma = \frac{1}{1+e^{-x}}$$

Where,

x is the input value,

e is Euler’s number (≈ 2.718)

Sigmoid function is used as an activation function in machine learning and neural networks for modeling binary classification problems, smoothing outputs, and introducing non-linearity into models.

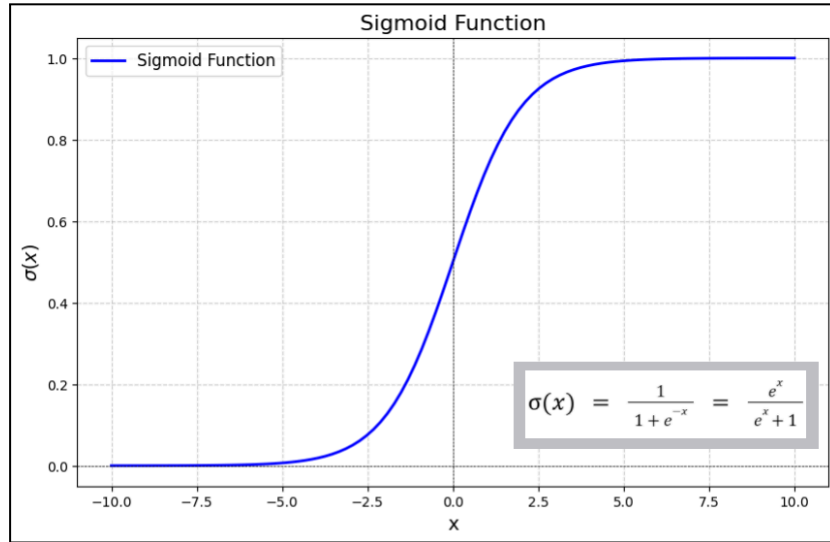


Figure 5.8 : Sigmoid function

The function used for the ANN case study is ReLU, since it is better suited against negative values and avoids the passage of negative weights and biases from one neuron layer to other, thus refining the training process of forward and backward propagation and is relatively better in eliminating the possibility of negative outputs, in case of prediction of very small values, although the negative prediction are a possibility in case of a noisy dataset.

5.5 DATA PREPARATION

As mentioned earlier, the data is pseudo-ad hoc, which means the study is only used for depicting the capabilities of a neural network in predicting output from a given set of training inputs. The data is generated using validated correlations and is sufficient in number to make a supportive argument for the study. The heat sink is assumed to be of 50*50 mm². The input parameters are channel width, length, height and velocity of flow. The data generated is such that the Reynold's number in the channel is lower than 2300, thus satisfying the applicability of the correlations. The analysis is only performed for a rectangular heat sink. The output parameters are pressure and heat transfer coefficient. The following parametric ranges are used for data generation -

Two types of neural networks, namely single layer and multi-layer neural networks are studied in terms of performance -

- Single Layer Neural Network
 - Input Layer - 5 nodes
 - Single Hidden Layer - 30 nodes
 - Output Layer - 1 node

- Multi Layer Neural Network
 - Input Layer - 5 nodes
 - First hidden layer - 30 nodes
 - Second hidden layer - 30 nodes
 - Output Layer - 1 node

The case study involves training with different learning rates. The batch size is kept as 4 and the study is run for 50 epochs. Batch size, basically, is the number of data points after which the predictions are averaged to be fed into the optimizer and calculate the loss between the expected and predicted values. The metric used for optimization is mean absolute error, since it is robust to outliers unlike the mean squared error (since it takes square of the errors into consideration). Adam optimizer is used for forward and backward propagation and optimization of weights after each epoch and batch size analysis. The dataset contains 800 data points, out of which 70% are used for training, 20% for testing and 10% for validation.

5.6 RESULTS

After each analysis the plots for MAE vs. epoch were noted and the coefficient of determination was calculated. The following tables consist of results from both types of neural networks for different learning rates. The number of neurons, batch size and epochs were selected after an iterative trial and error process and were kept constant for the entire study. The user can change the values, based on their dataset, or through advanced optimization techniques.

Single Layer Neural Network -

Learning rate	Coefficient of determination (R^2)
0.0001	-1.42
0.001	0.93
0.01	0.994

Table 5.1 : Learning rate vs R^2 (single layer)

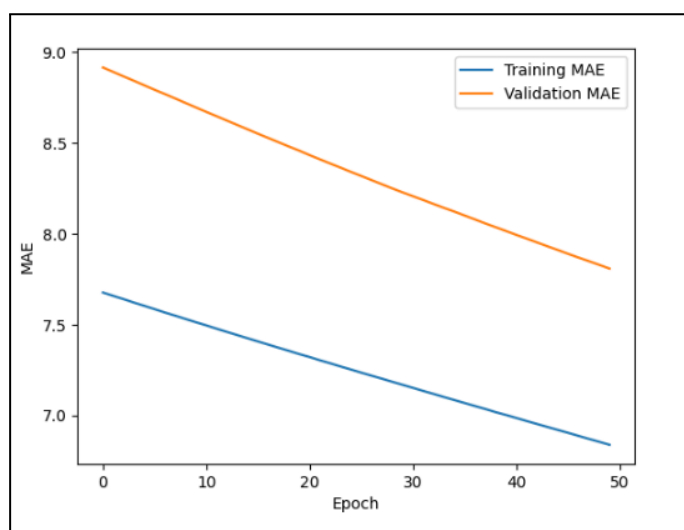


Figure 5.9 : MAE vs. Epoch plot for LR = 0.0001

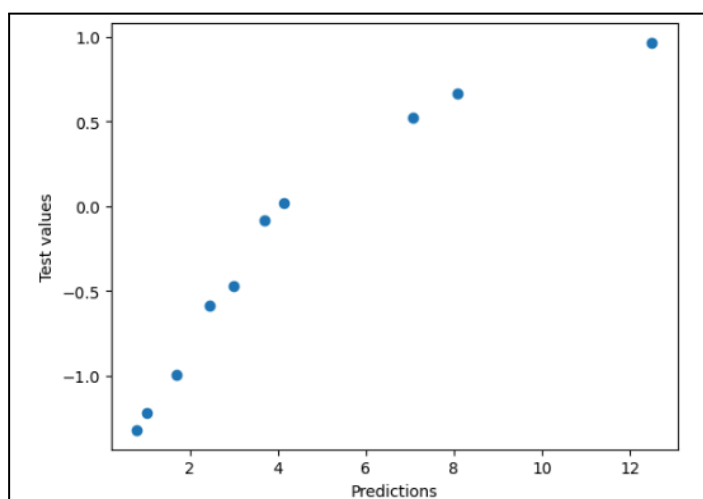


Figure 5.10 : Test values and predictions for single layer ANN for LR = 0.0001

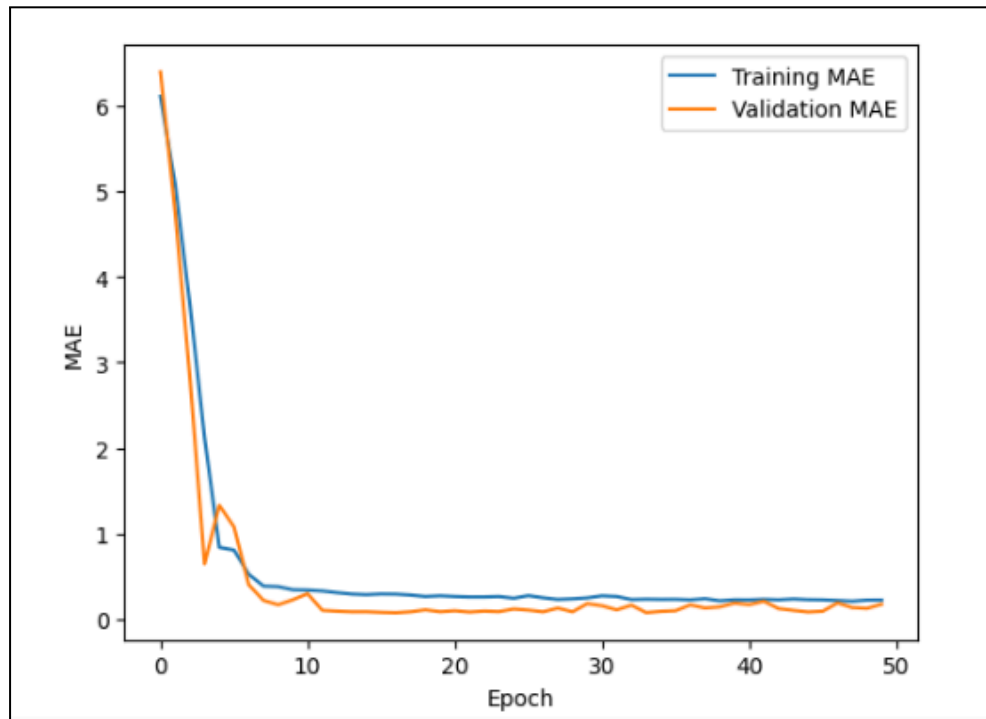


Figure 5.11 : MAE vs. Epoch plot for LR = 0.01

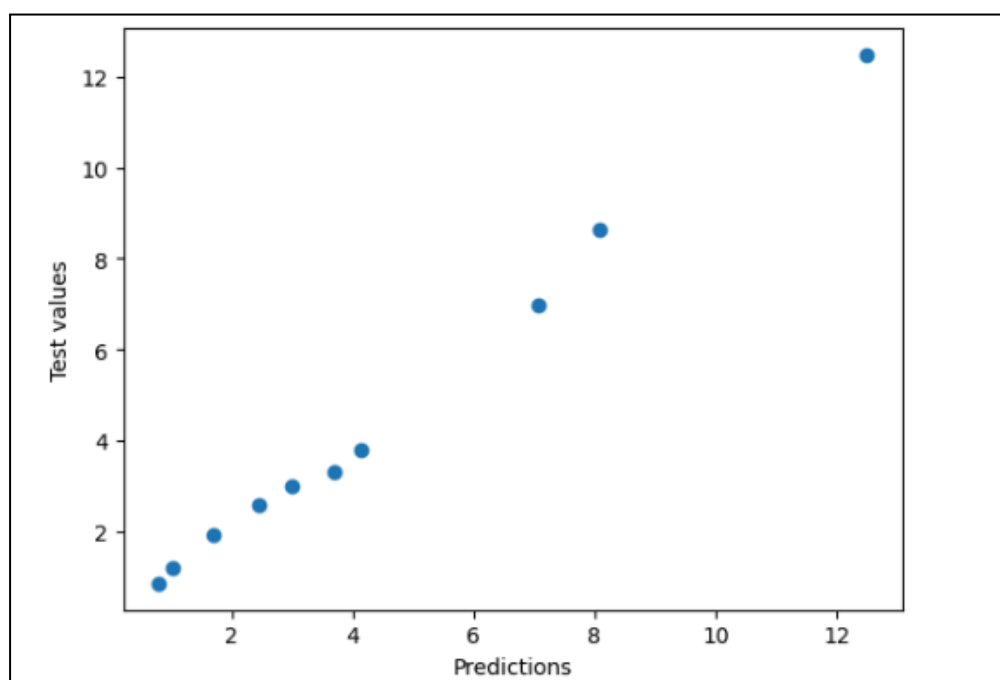


Figure 5.12 : Test values and predictions for single layer ANN for LR = 0.01

Multi-Layer Neural Network:

Learning rate	Coefficient of determination (R^2)
0.0001	-1.077
0.001	0.989
0.01	0.998

Table 5.2 : Learning rate vs R^2 (Multi layer)

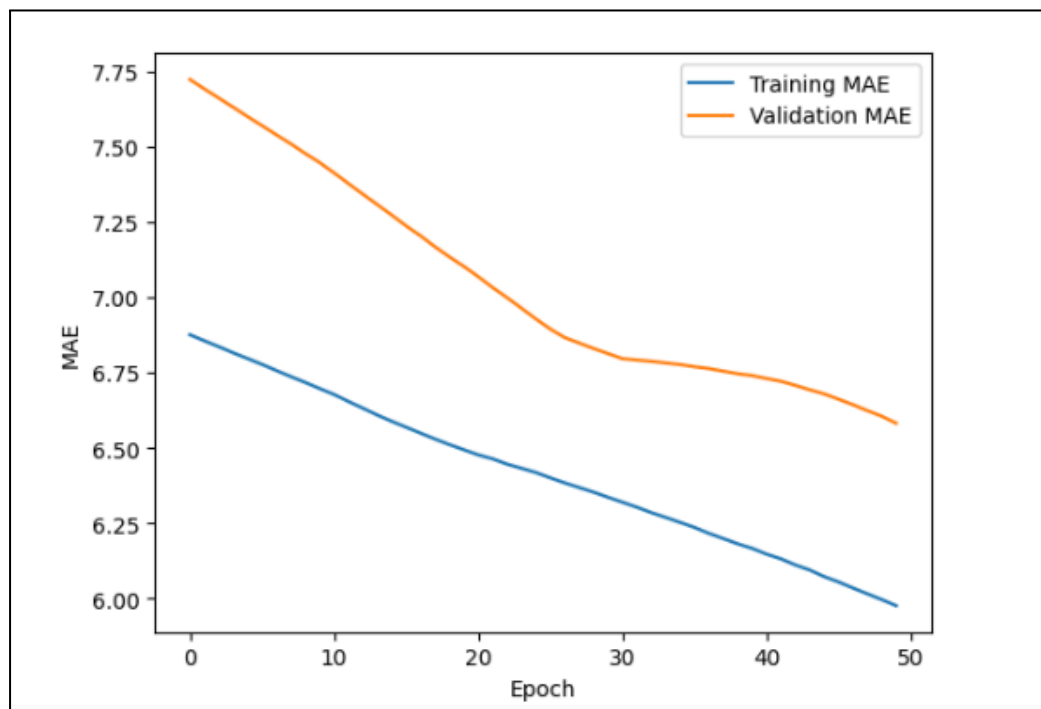


Figure 5.13 : MAE vs. Epoch plot for LR = 0.0001

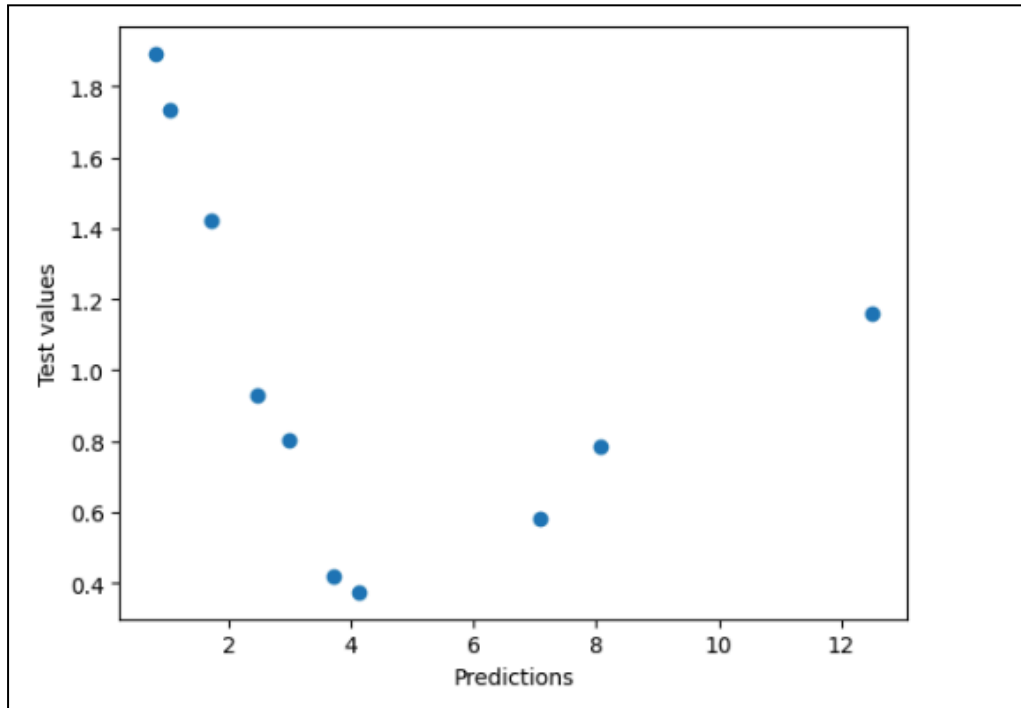


Figure 5.14 : Correlation between test values and predictions for MANN for LR = 0.0001

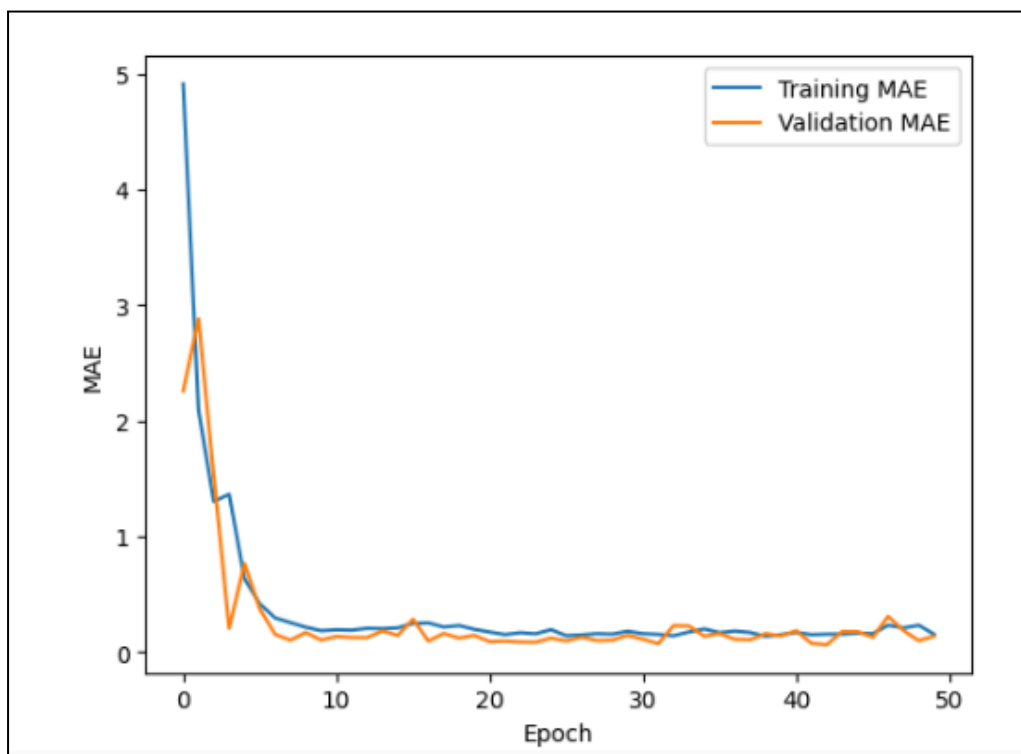


Figure 5.15 : MAE vs. Epoch plot for LR = 0.01

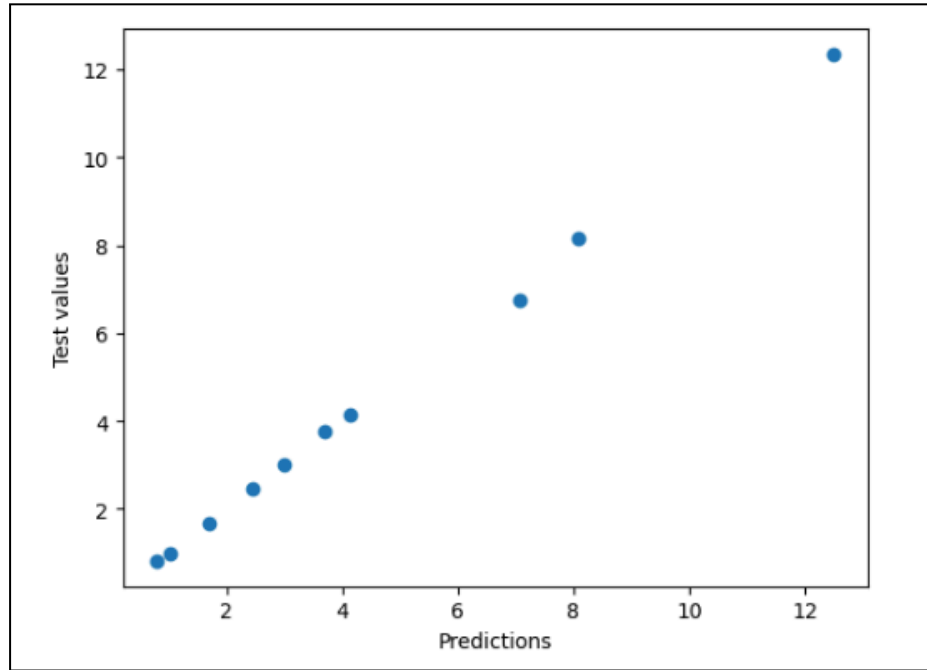


Figure 5.16 : Correlation between test values and predictions for MANN for LR = 0.01

Conclusion -

For both ANN's, the reliability of the results directly depends on the learning rate of the model. The tradeoff is that the computational time increases with increase in learning rate, but that depends on the volume of the dataset as well. Since, the analyzed dataset was relatively smaller in volume, the computational time was reduced. Moreover, the optimal number of epochs, and batch size also plays an important role.

For multilayer ANN's the performance is better as compared to single layer ANN's due to larger number of neurons, thus rendering higher processing power, similar to a biological neural structure. However, the number of neurons is not always directly proportional to performance. A large number of neurons might lead to duplication errors and possible misinterpretation of results, thus they should be calculated using standard empirical relations or through iterative testing as well. The model thus predicts the output values with great accuracy and can be used as a reliable tool for experimental and CFD based input data. The baseline model can be improved through K-Fold validation and regularization to avoid data fitting and understand data better at each stage.

5.7 EXTREME LEARNING MACHINE

In Deep learning, an Extreme Learning Machine (ELM) is a type of feedforward neural network utilized for tasks such as classifications and regression. ELM stands apart from traditional feedforward neural networks due to its unique training approach.

In ELM, the hidden layer's weights and biases are randomly initialized. However, these initial values are just starting points. The distinctive aspect of ELM lies in its ability to compute the output layer's weights using the Moore-Penrose generalized inverse of the hidden layer's output matrix. This approach enables ELM to learn from training data in a single step, setting it apart from traditional neural networks that often require iterative training procedures, such as backpropagation. It uses single-hidden layer feedforward neural networks (SLFN) instead of traditional feedforward neural networks. Thus it randomly selects hidden nodes and analytically finds their output weight. ELM's single-step training process makes it an efficient and versatile tool for a wide range of machine-learning applications.

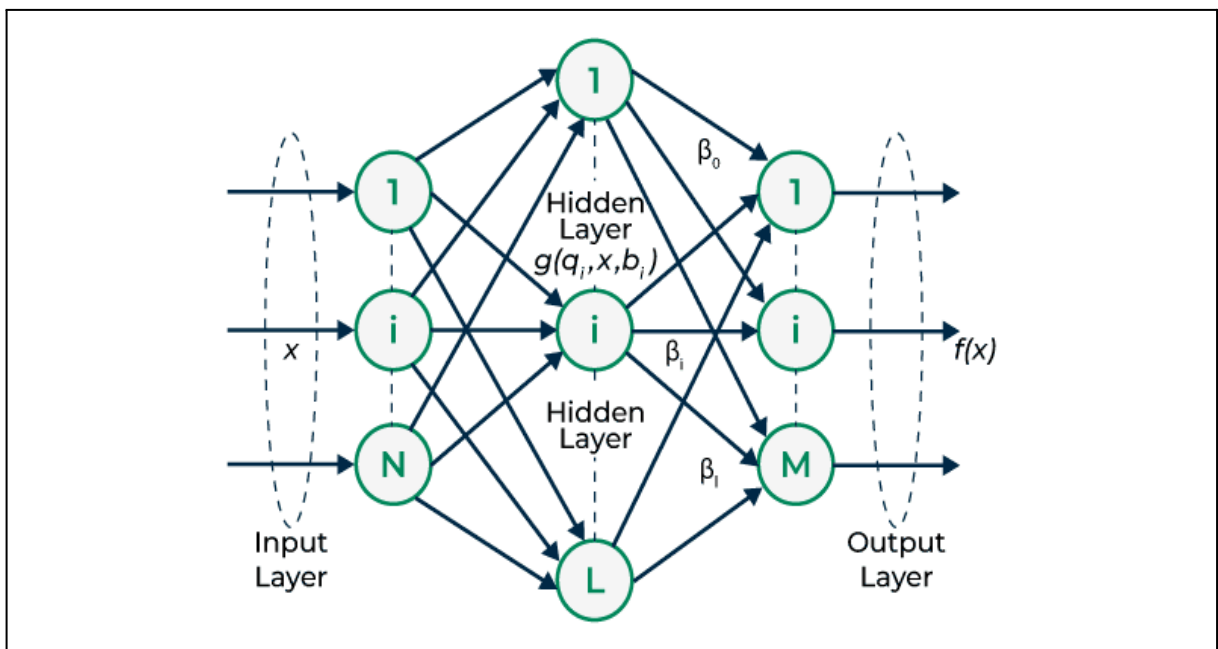


Figure 5.17 : ELM Topology

5.7.1 ARCHITECTURE OF ELM

The architecture of ELM is very simple and straightforward which involves three segments which are listed below,

- 1) Input layer
- 2) Single hidden layer
- 3) Output layer

- **ELM Training**

Input Training Data: The first step is to gather the training data which include input features and target variables to feed into ELM machine learning algorithm.

Random Initialization: Next the weights and bias are randomly initialized in hidden layers thus this process eliminates iterative weight adjustment.

Feature Mapping: The input data is transformed into a high-dimensional feature space using randomly assigned weights. This process is known as feature mapping and helps to capture complex relationships between the input features.

Hidden Layer Processing: Then the transformed input data is processed by the hidden layer, resulting in an output matrix. This output is a key component in ELM's unique single-step learning process.

Output Weight Calculation: In the fifth step ELM leverages the output matrix by applying Moore-Penrose generalized inverse to calculate the output weights. This mathematical technique ensures robustness, even in the presence of noise or missing data.

Model Evaluation: Once the output weights are determined by Moore-Penrose generalized inverse, the model's training output is generated. This output is evaluated by specific metrics based on the problem use cases.

Fine-Tuning and Hyperparameters Adjustment: Depending on the assessment results, fine-tuning or hyperparameter adjustments can be applied to improve model performance.

Deployment: Finally the training process is complete, the ELM model is ready for deployment in real-world applications, where it can make predictions and decisions based on the learned patterns.

- **Setup**

The ELM library from Python is used to build the model. Ridge Regression is used to avoid overfitting. Sigmoid function is used to obtain the weights based on the initialization of the Moore Penrose matrix based algorithm.

Results and Conclusions -

Neurons	Coefficient of Determination
100	0.9156
300	0.9967

Table 5.3 : Trained neurons and their corresponding R^2 values

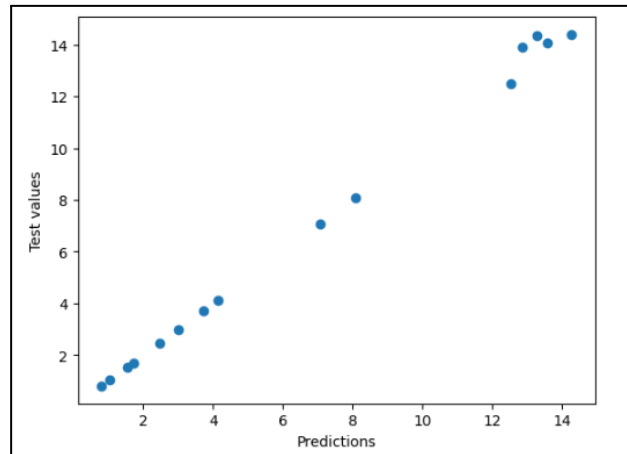


Figure 5.18 : Correlational plot for 300 neurons

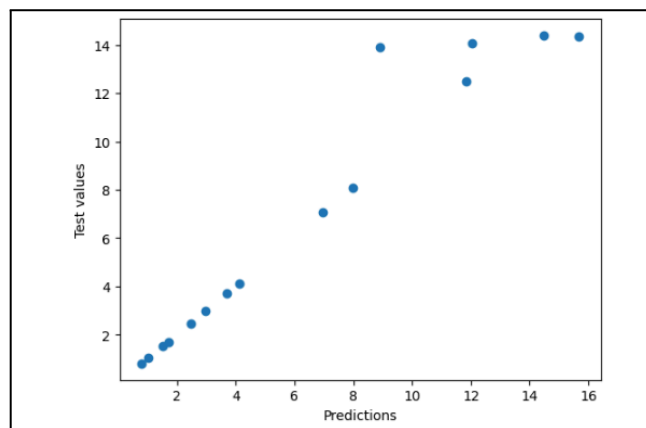


Figure 5.19 : Correlational plot for 100 neurons

The number of neurons affects the result as shown in the above images. However, each separate iteration leads to a different result, pivoted around a common base point (can be calculated using a convergence technique, but is more suitable for larger data). The model, however, accurately predicts the values based on the training data. Moreover, it does not have to undergo an iterative process, unlike neural networks and leads to accurate predictions with a single layer of neuron based on its topology, reducing computational time. However, the results can be improved after cross-validation and regularization and through setting of a convergence point.

5.8 GENETIC ALGORITHMS

Genetic Algorithms(GAs) are adaptive heuristic search algorithms that belong to the larger part of evolutionary algorithms. Genetic algorithms are based on the ideas of natural selection and genetics. These are intelligent exploitation of random searches provided with historical data to direct the search into the region of better performance in solution space. They are commonly used to generate high-quality solutions for optimization problems and search problems.

Genetic algorithms simulate the process of natural selection which means those species that can adapt to changes in their environment can survive and reproduce and go to the next generation. In simple words, they simulate “survival of the fittest” among individuals of consecutive generations to solve a problem. Each generation consists of a population of individuals and each individual represents a point in search space and possible solution. Each individual is represented as a string of character/integer/float/bits. This string is analogous to the chromosome.

Genetic algorithms are based on an analogy with the genetic structure and behavior of chromosomes of the population. Following is the foundation of GAs based on this analogy –

1. Individuals in the population compete for resources and mate
2. Those individuals who are successful (fittest) then mate to create more offspring than others
3. Genes from the “fittest” parent propagate throughout the generation, that is sometimes parents create offspring which are better than either parent.
4. Thus each successive generation is more suited for their environment

Search Space

The population of individuals is maintained within search space. Each individual represents a solution in search space for a given problem. Each individual is coded as a finite length vector (analogous to chromosome) of components. These variable components are analogous to Genes. Thus a chromosome (individual) is composed of several genes (variable components).



Figure 5.20 : Search Space

Fitness Score

A Fitness Score is given to each individual which shows the ability of an individual to “compete”. The individual having optimal fitness score (or near optimal) are sought.

The GAs maintains the population of n individuals (chromosome/solutions) along with their fitness scores. The individuals having better fitness scores are given more chance to reproduce than others. The individuals with better fitness scores are selected who mate and produce better offspring by combining chromosomes of parents. The population size is static so the room has to be created for new arrivals. So, some individuals die and get replaced by new arrivals, eventually creating a new generation when all the mating opportunities of the old population are exhausted. It is hoped that over successive generations better solutions will arrive while least fit die.

Each new generation has on average more “better genes” than the individual (solution) of previous generations. Thus each new generation has better “partial solutions” than previous generations. Once the offspring produced have no significant difference from offspring produced by previous populations, the population converges. The algorithm is said to be converged to a set of solutions for the problem.

Operation of Genetic Algorithms

Once the initial generation is created, the algorithm evolves the generation using following operators –

- 1) Selection Operator: The idea is to give preference to the individuals with good fitness scores and allow them to pass their genes to successive generations.
- 2) Crossover Operator: This represents mating between individuals. Two individuals are selected using a selection operator and crossover sites are chosen randomly. Then the genes at

these crossover sites are exchanged thus creating a completely new individual (offspring). For example –

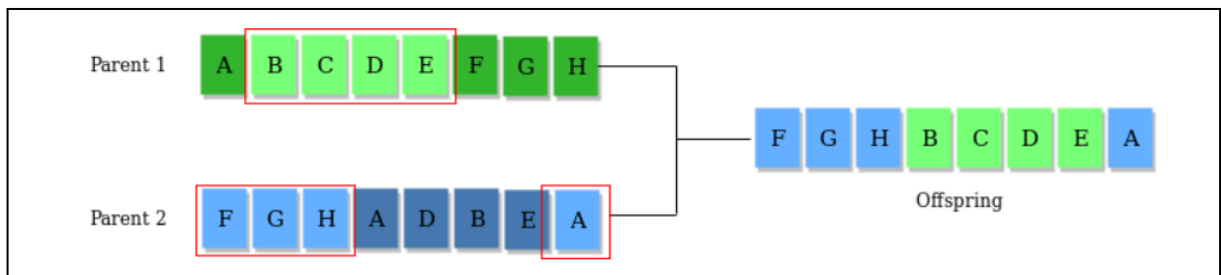


Figure 5.21 : Selection and Crossover

3) Mutation Operator: The key idea is to insert random genes in offspring to maintain the diversity in the population to avoid premature convergence. For example –

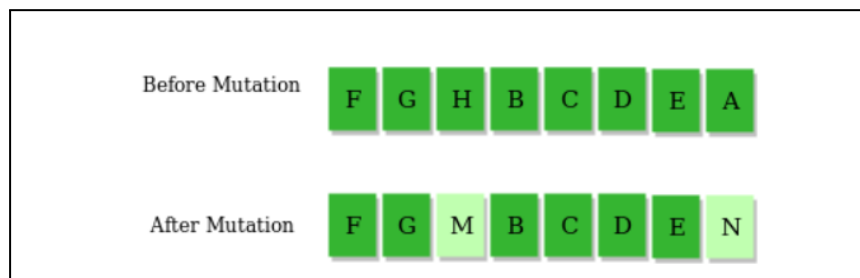


Figure 5.22 : Mutation

MATLAB's Genetic Algorithm Toolbox was used for creating an optimization model based on input variables to be optimized, such that objective functions of interest are minimized or maximized based on given requirement.

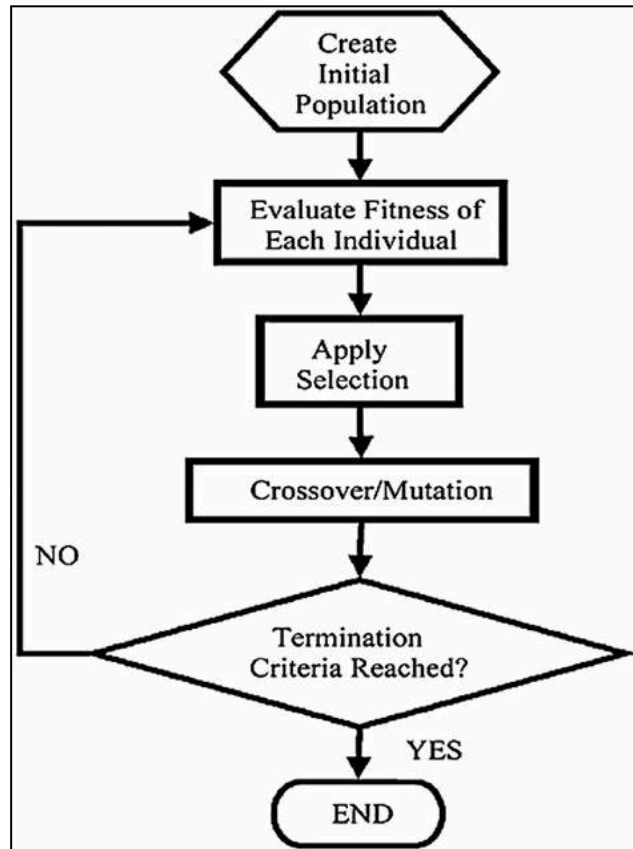


Figure 5.23 : Flowchart for Genetic Algorithm

PROBLEM STATEMENT

Ventola's^[20] paper employs the use of genetic algorithms as an optimization technique based on minimization of thermal resistance of the heat sink. The study employs the same approach for different objectives, which will be mentioned subsequently.

For the given optimization problem, Shaeri's^[18] paper is referred, in which the following heat sink is used-

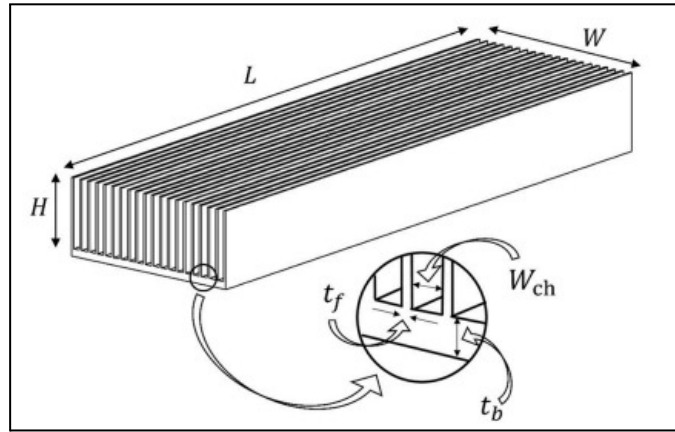


Figure 5.24 : Schematic of heat sink used in Shaeri's paper

Based on the above nomenclature -

$W_{ch} = 2.18 \text{ mm}$, $t_f = 0.96 \text{ mm}$, $t_b = 2.54 \text{ mm}$, $H = 22.86 \text{ mm}$, $L = 203 \text{ mm}$, $N = 19$
(number of channels)

The paper makes use of the same correlations mentioned in the chapter on rectangular plate heat sinks and demonstrates an agreeable validation between experimental and analytical values. The following plots are for the above heat sink with a heat input of 100 W.

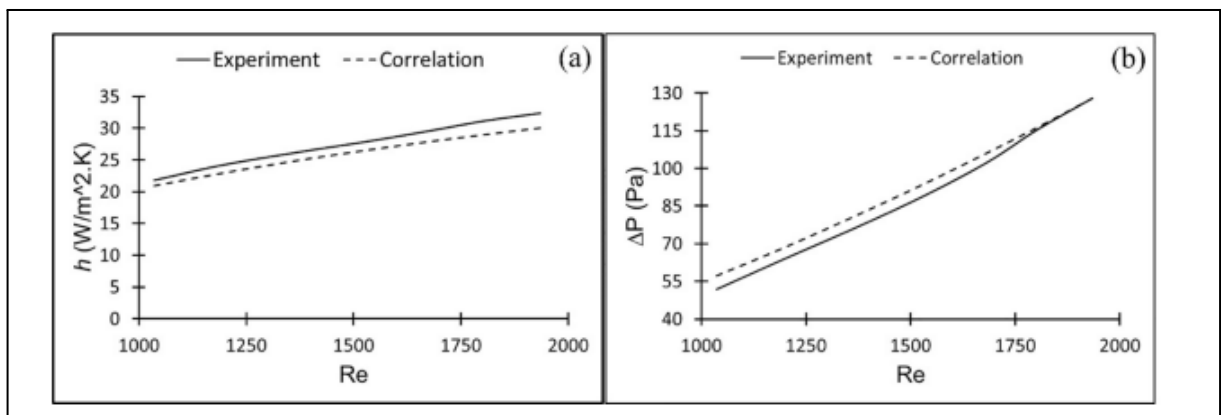


Figure 5.25 : Experimental and Analytical Validation (Shaeri)

The weight of the given heat sink is approximately 372 g. The following optimization procedure would thus, follow a multiobjective approach based on -

- 1) Heat transfer coefficient maximization
- 2) Pressure drop minimization
- 3) Weight minimization

The formulation of the objective function is derived from the correlations for pressure and HTC. The form of optimization is as follows -

$$Z = F(X) \quad \text{Maximize (Heat Transfer Coefficient), \quad Minimize (Pressure Drop, Mass)}$$

Subject to -

$$\begin{aligned} \text{Re} &\leq 2300 \\ 40 \text{ mm} &< W < 70 \text{ mm} \\ 0.8 \text{ mm} &< t_f < 2 \text{ mm} \\ 40 \text{ mm} &< H < 22.86 \text{ mm} \\ 1 \text{ mm} &< W_{ch} < 4 \text{ mm} \\ 10 &< N < 25 \\ 100 \text{ mm} &< L < 203 \text{ mm} \\ 2 \text{ m/s} &< V < 4 \text{ m/s} \end{aligned}$$

Parameters for GA -

Initial Population - 200

Crossover - 0.8

Mutation - Not applied

The above values are taken after a few iterative tests and reference to Ventola's^[20] paper. The suitable offspring (optimal point) is chosen only on the basis of maximized ratio of HTC and pressure drop, however better results could be obtained with other selection algorithms like TOPSIS.

Results and Conclusions -

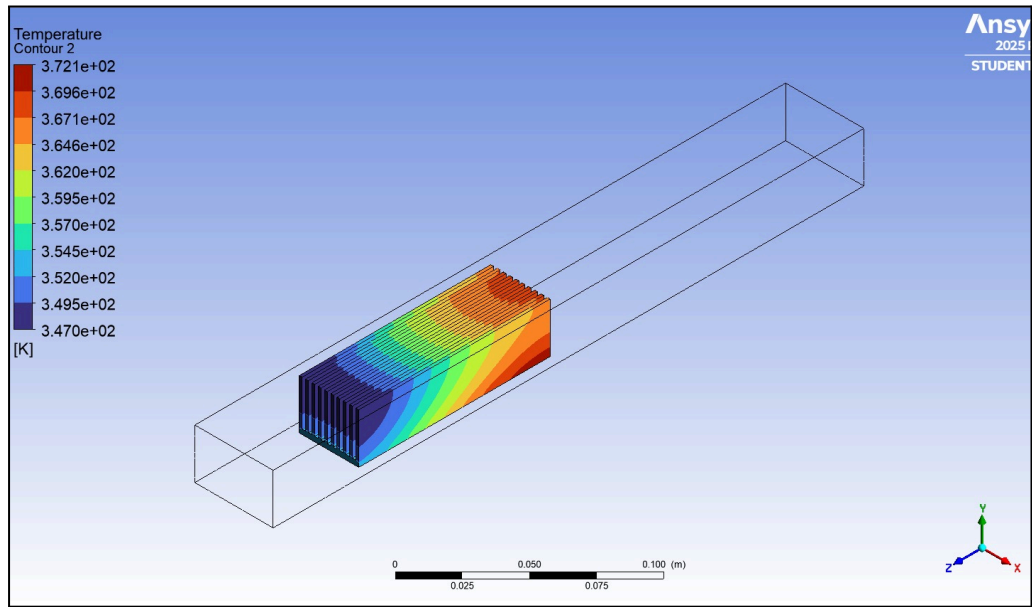


Figure 5.26 : Temperature contour for $Q = 100 \text{ W}$, $v = 2.16 \text{ m/s}$

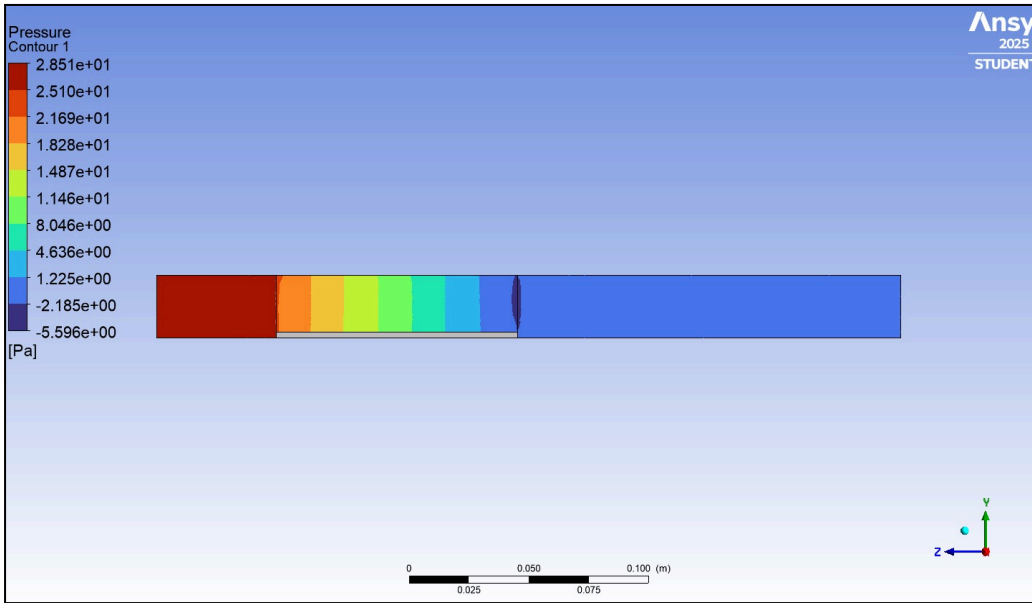


Figure 5.27 : Pressure contour for $Q = 100 \text{ W}$, $v = 2.16 \text{ m/s}$

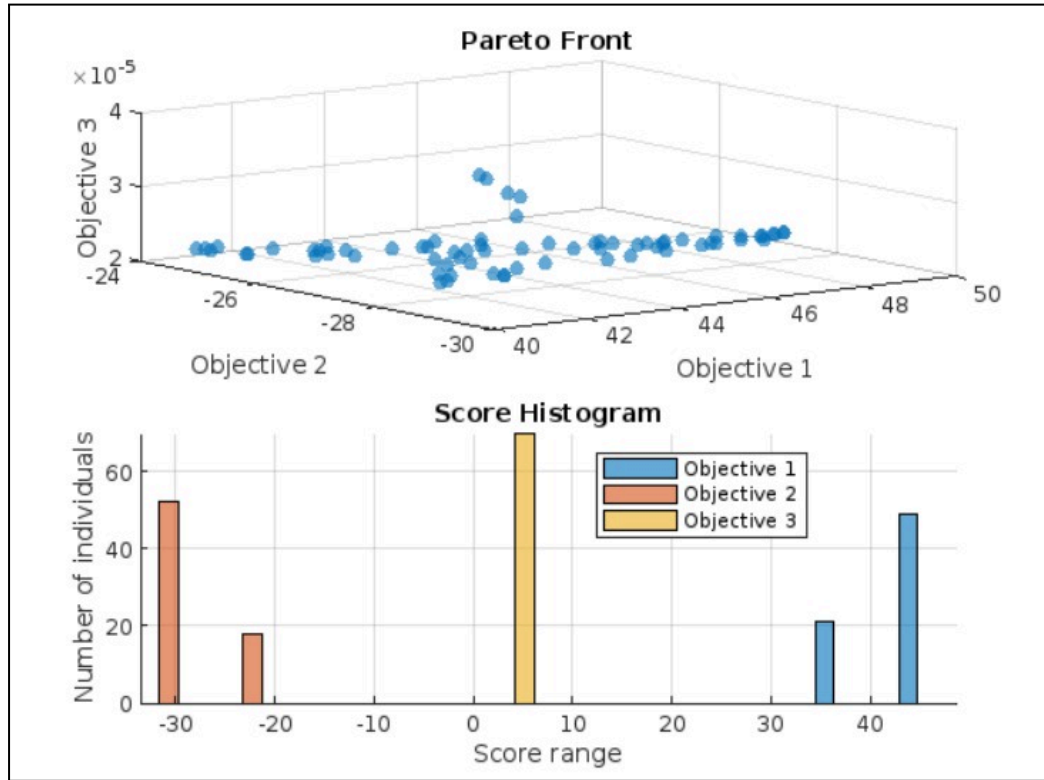


Figure 5.28 : Pareto Plot for multiobjective optimization

The algorithm employs Pareto front based optimization. The Pareto front in optimization refers to the set of all Pareto efficient solutions in a multi-objective optimization problem. A solution is considered Pareto efficient if no other solution can improve one objective without worsening another. The concept is widely used in engineering and economics to identify optimal trade-offs among competing objectives. The multiobjective algorithm returns the following values -

W_{ch} = 1.8 mm , t_f = 1.5 mm, t_b = 2.54 mm, H = 23.6 mm, L = 100.7 mm, N = 9 (number of channels), Pressure drop = 40.1606 Pa, Heat Transfer Coefficient = 29.6323 W/(m².K)

CFD analysis is used to verify the algorithm's result. Based on CFD analysis -

Pressure drop = 34.106 Pa, Error = 17.8%

Heat Transfer Coefficient = 33.7 W/(m².K), Error = 12%

Mass = 118.625 g, Reduction in mass (%) = 68.1183%

The associated errors are due to tradeoffs encountered with the opposing nature of the equations. The errors tend to increase for higher velocities, thus the optimal points of tradeoff in the equations are limited, but can be suitably used to predict the performance for lower velocities. The best solution is to obtain sufficient data points through CFD/ experimental techniques and use a neural network model to determine the closest objective function for a wider range of operating limits, along with genetic algorithms and TOPSIS method, to get an accurate value of optimal performance.

CHAPTER 6 : CONCLUSION AND FUTURE WORK

This study presented a detailed computational analysis of the thermal and hydraulic performance of pin fin heat sinks, focusing on both cylindrical and square pin geometries under forced convection conditions. The objective was to investigate how variations in geometric parameters — such as fin height, fin diameter/edge length, and inlet velocity — impact the key performance indicators, namely the Nusselt number, heat transfer coefficient (HTC), and pressure drop.

The cylindrical pin fin heat sinks were analyzed in both staggered and inline configurations. It was observed that the inline arrangement provided marginally better cooling performance due to a higher number of pins and greater surface area. However, the staggered arrangement offered a better balance between thermal and hydraulic performance, showing a comparatively lower pressure drop due to improved flow redistribution.

The square pin fin heat sinks, also studied under staggered and inline layouts, demonstrated different flow characteristics due to their sharp-edged geometry. The presence of corners led to stronger flow separation and turbulence compared to cylindrical pins, enhancing heat transfer but also causing greater flow resistance. In square fins, the staggered configuration generated better fluid mixing and improved cooling but suffered from a higher pressure drop than the inline configuration, where the airflow remained more streamlined.

Parametric studies revealed that increasing fin height and fin diameter/edge length improved the heat transfer performance by providing more surface area for convection. However, these changes also led to a notable increase in pressure drop, particularly in staggered arrays. Similarly, an increase in inlet velocity improved the convective heat transfer (higher Nusselt number) across all configurations, but with an associated rise in hydraulic losses.

For the machine learning part, a better approach would be with data acquisition as the primary goal, to create a robust model based on the given study, since it demonstrates the capability of neural networks in prediction of data points for which long hours of simulation can be avoided. The neural networks and ELM employed in the study can be used as baseline models for further research based studies.

Genetic algorithms can be used along with TOPSIS algorithm for providing optimal performance outputs, as well as can be used in conjunction with neural networks, to create surrogate models, which would complement the inbuilt optimizer and optimize the weights as per the objective of the user.

REFERENCES

- [1] Ehsan F. Abbas AN OVERVIEW OF HEAT SINK TECHNOLOGY International Journal. of Applied Mechanics and Engineering, 2024, vol.29, No.4, pp.1-23.
- [2] Akshendra Soni, Study of Thermal Performance between Plate-fin, Pin-fin and Elliptical Fin Heat Sinks in Closed Enclosure under Natural Convection.
- [3] Engr. Zulfiqar Khattak, Hafiz Muhammad Ali Plate fin and pin fin heat sink various design aspects.
- [4] R. Sam Sukumar, G. Sriharsha, S. Bala Arun, P. Dilip Kumar, Ch. Sanyasi Naidu, Modelling And Analysis Of Heat Sink With Rectangular Fins Having Through Holes.
- [5] Anil Kumar Rao, Vandana Somukuwar, Heat transfer of a tapered fin heat sink under natural convection.
- [6] Mainak Bhaumik, Anirban Sur, and Kavita Dhanawade, An Experimental Investigation of Aero-Foil-Shaped Pin Fin Arrays.
- [7] Oguzhan OZBALCI, Experimental Study of Thermal and Pressure Performance of Porous Heat Sink Subjected to Al₂O₃-H₂O Nanofluid.
- [8] Amandeep Singh Oberoi, Analysis of Heat Transfer Through Perforated Plate Heat Sink.
- [9] google imag - naam likhna
- [10] Y. S. Touloukian and D. P. DeWitt, Thermophysical Properties of Matter – Volume 1: Thermal Conductivity – Metallic Elements and Alloys. New York: IFI/Plenum, 1970.
- [11] J. Lienhard IV and J. Lienhard V, A Heat Transfer Textbook, 4th ed. Cambridge, MA: Phlogiston Press, 2011.
- [12] M. Bahrami, "Heat Transfer - Thermal Resistance Network," Simon Fraser University
- [13] F. Incropera, D. DeWitt, T. Bergman, and A. Lavine, Fundamentals of Heat and Mass Transfer, 7th ed., Wiley, 2011.
- [14] M. Iyengar and R. Schmidt, "Airflow uniformity through heat sinks," IBM Journal of Research and Development, vol. 46, no. 6, pp. 853–864, Nov. 2002.

- [15] M. Forghan, S. Khoshhal, and M. Farhadi, “Numerical investigation of the effect of pin fin arrangements on the heat transfer enhancement and pressure drop,” *Thermal Science and Engineering Progress*, vol. 10, pp. 208–217, 2019.
- [16] Mohammad Ismail, Ahmad Manasrah, Abdullah Masoud Ali, Mousa Abedalaziz, “Hydrothermal performance of a heat sink using Plate-fins Experimental approach”, 2024.
- [17] Benouis, F. Z., Ould Amer, Y., Arıcı, M., & Meziane, S. (2023). Designing and optimizing a novel heat sink for the enhancement of hydrothermal performances: Modelling and analysis using artificial neural networks.
- [18] Mohammad Reza Shaeri, Ameneh Shadlo, Soroush Sarabi, Andoniaina M. Randriambololona (2022), Machine learning-based optimization of air-cooled heat sinks
- [19] R.C. Adhikari, D.H. Wood , M. Pahlevani (2020), An experimental and numerical study of forced convection heat transfer from rectangular fins at low Reynolds numbers
- [20] Luigi Ventola, Gabriele Curcuruto, Matteo Fasano, Saverio Fotia, Vincenzo Pugliese, Eliodoro Chiavazzo and Pietro Asinari (2016), Unshrouded Plate Fin Heat Sinks for Electronics Cooling: Validation of a Comprehensive Thermal Model and Cost Optimization in Semi-Active Configuration